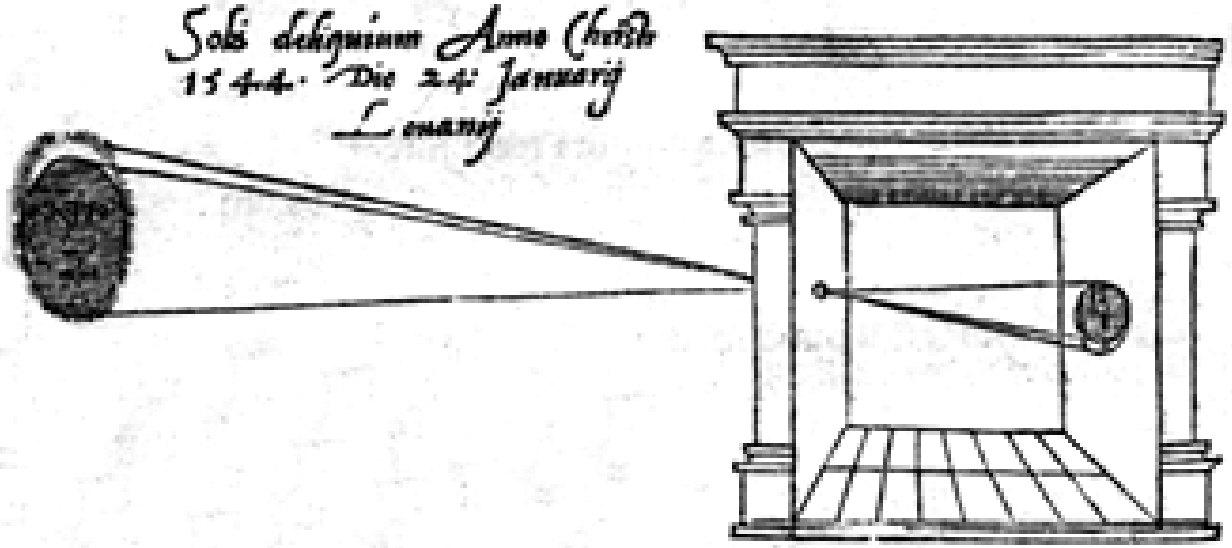
- Introduction
- Planar (2D) Projective Geometry
- Spatial (3D) Projective Geometry
- **Camera Geometry (3D → 2D Projection) and single-view Geometry**

Camera Geometry

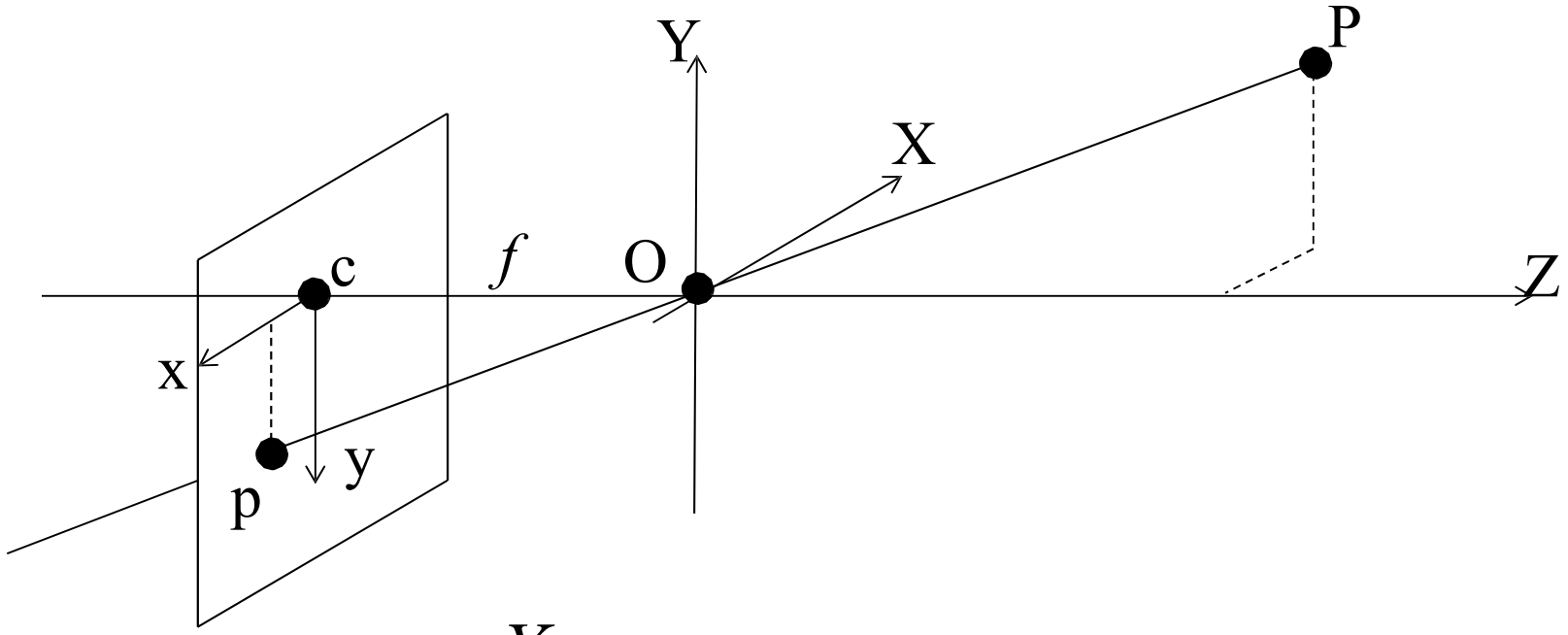
camera projective model

Pinhole camera

illum in tabula per radios Solis, quàm in cœlo contingit: hoc eſt, ſi in cœlo ſuperior pars deliquiũ patiatur, in radiis apparebit inferior deficere, vt ratio exigit optica.



Sic nos exactè Anno .1544. Louanii eclipſim Solis obſeruauimus, inuenimuſq; deficere paulò plus q̃ dex-



$$x = f\frac{X}{Z}$$

$$y = f\frac{Y}{Z}$$

perspective projection

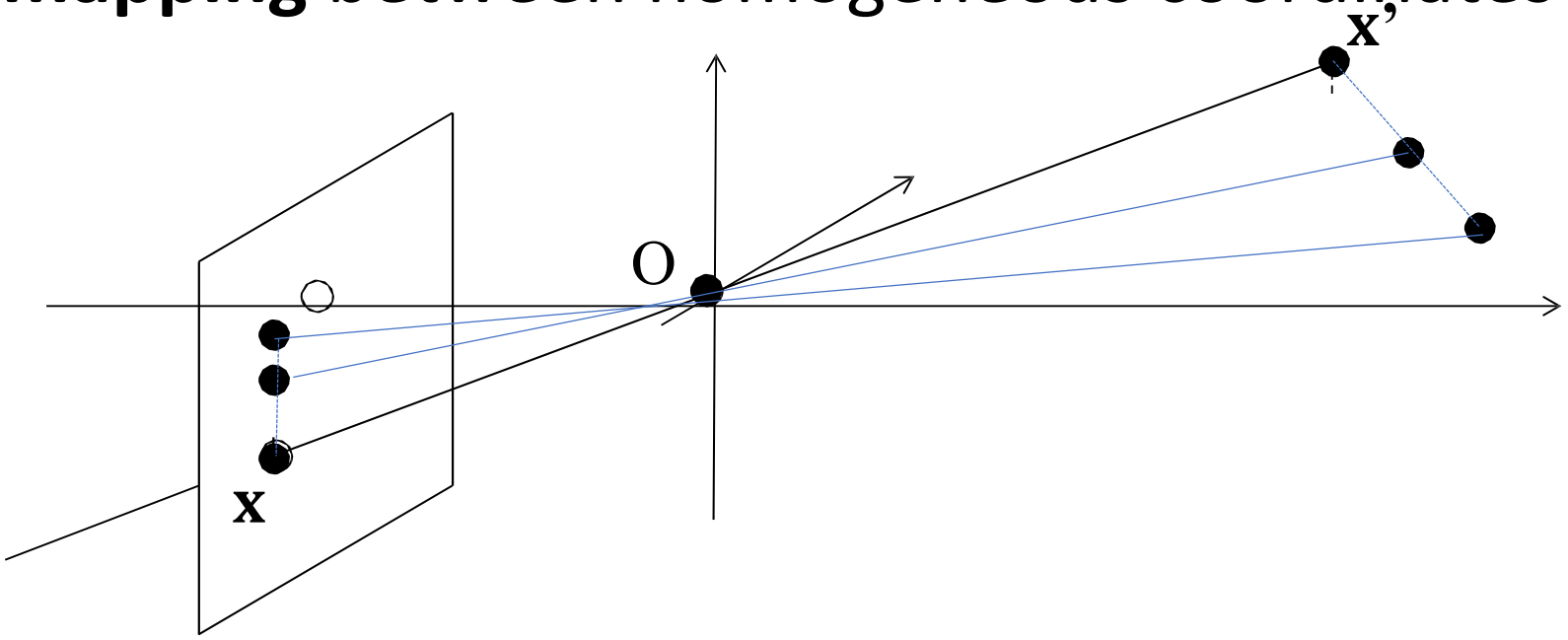
-nonlinear
-not shape-preserving
-not length-ratio preserving

Scene-to-image projection

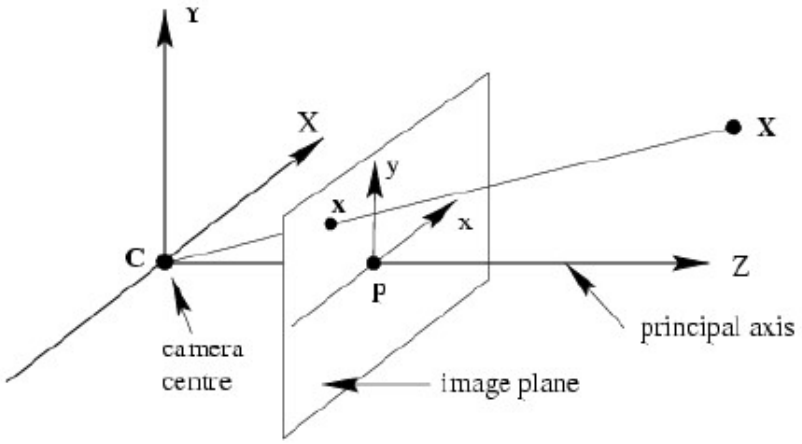
Colinear points are projected onto colinear image points

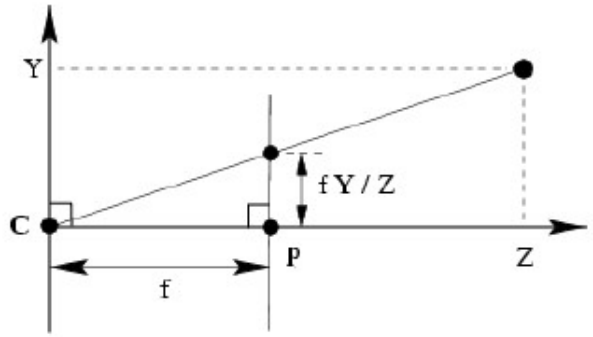
→ colinearity is preserved

→ **linear mapping** between homogeneous coordinates



CAMERA GEOMETRY





colinearity is preserved → linear relation among homogeneous coords

$$\mathbf{X} = \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \mapsto \mathbf{u} = \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \mathbf{P}_{3\text{x}4} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} = \mathbf{P}_{3\text{x}4}\mathbf{X} = \left| \mathbf{M}_{3\text{x}3} \quad \mathbf{m}_{3\text{x}1} \right| \mathbf{X}$$

camera projection matrix

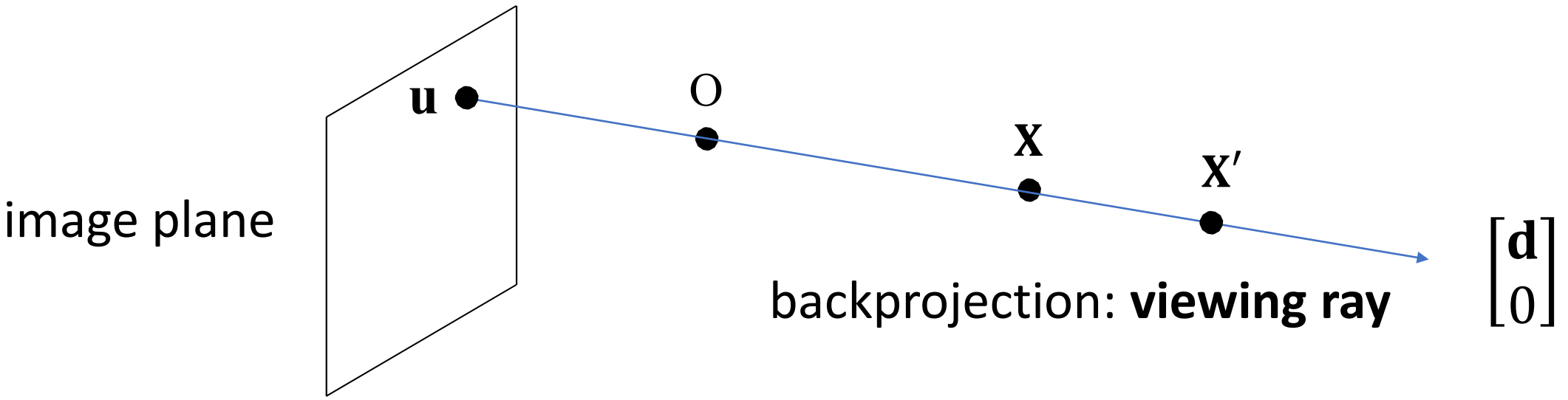
point in 3D space

image

invertible

- SCENE
- CAMERA

The viewing ray associated to an image point $\mathbf{u}$



The backprojection of image point $\mathbf{u}$ for the camera $\mathbf{P} = [\mathbf{M} \quad \mathbf{m}]$, is a straight line

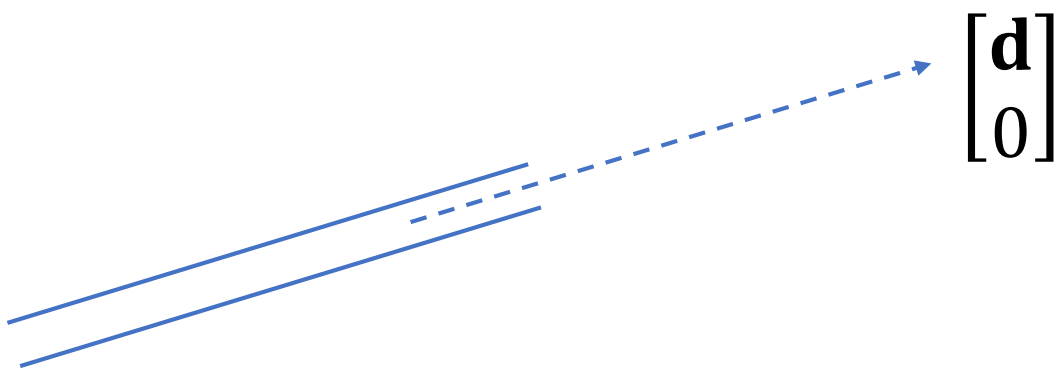
- through $\mathrm{O} = RNS(\mathbf{P})$ *Proof*: any $\mathbf{X}' = \mathrm{O} + \lambda\mathbf{X}$ projects to $\mathbf{P}\mathbf{X}' = \mathbf{P}\mathrm{O} + \mathbf{P}\mathbf{X} = \mathbf{P}\mathbf{X}$ (0)
- whose direction is $\mathbf{d} = \mathbf{M}^{-1}\mathbf{u}$ *Proof*: $\mathbf{u} = [\mathbf{M} \quad \mathbf{m}]\begin{bmatrix}\mathbf{d}\\0\end{bmatrix} = \mathbf{M}\mathbf{d}$ → $\boxed{\mathbf{d} = \mathbf{M}^{-1}\mathbf{u}}$

Where is O ? from $\mathbf{P}\mathrm{O} = [\mathbf{M} \quad \mathbf{m}]\begin{bmatrix}\mathbf{o}\\1\end{bmatrix} = 0$ → $\mathbf{o} = -\mathbf{M}^{-1}\mathbf{m}$ cartesian coordinates

VANISHING POINTS

parallel lines concur at the point at the infinity $\begin{bmatrix}\mathbf{d}\\0\end{bmatrix}$ of their common direction $\mathbf{d}$

images of parallel lines concur at the image V of the point at the infinity of direction $\mathbf{d}$: $V = [\mathbf{M} \quad \mathbf{m}]\begin{bmatrix}\mathbf{d}\\0\end{bmatrix} = \mathbf{M}\mathbf{d}$





VANISHING POINTS

Remember: the direction of the viewing ray of image point $\mathbf{V}$ is

$$\mathbf{M^{-1}V = M^{-1}\,Md = d}$$

→

Vanishing Point Theorem:

The viewing ray associated to the vanishing point $\mathbf{V}$ of a direction $\mathbf{d}$ is parallel to $\mathbf{d}$ (i.e., its direction is $\mathbf{d}$)

Application of vanishing points: navigation along a desired direction

mobile robot navigation



vehicle driving



Find $\mathbf{V}$ and, knowing $\mathbf{M}$, follow the direction $\mathbf{d} = \mathbf{M}^{-1}\mathbf{V}$ of the corridor

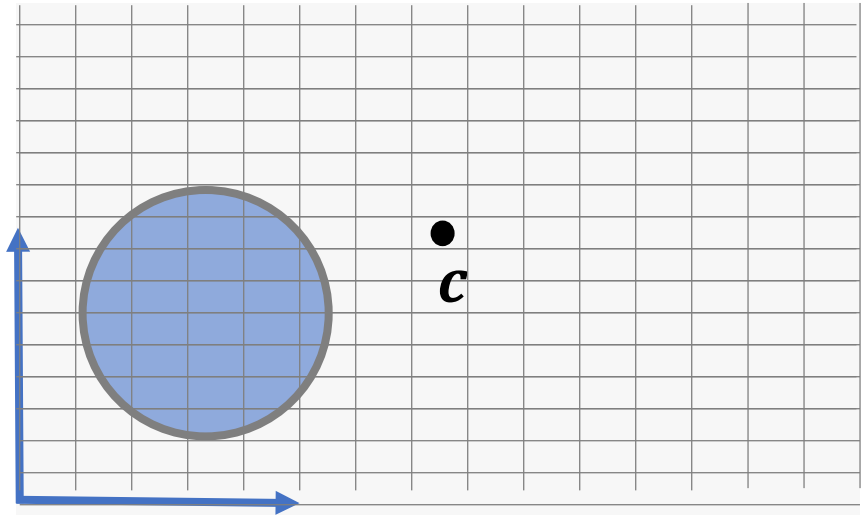
$\mathbf{M}$ can be estimated through extrinsic + intrinsic calibration

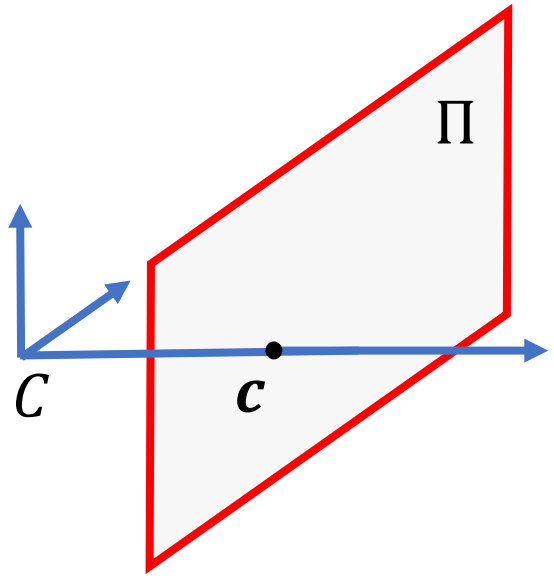
- SCENE
- CAMERA (intrinsic camera parameters)

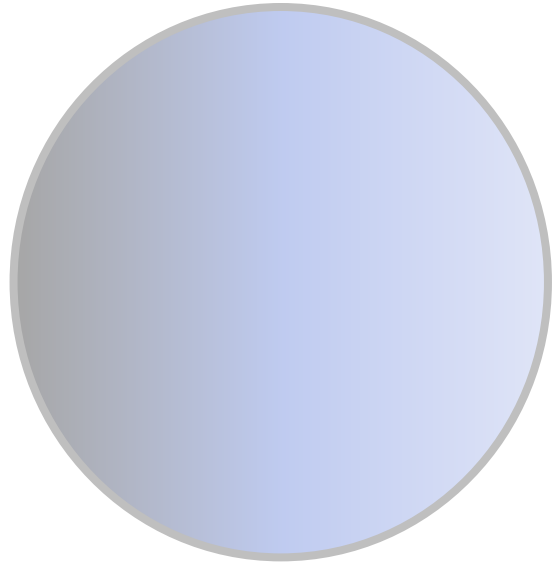
Camera Projection to Images

- Points in images:
- Might refer to a different reference system
- Images are discretized in pixels (which might be non-square)
- The physical sensor might introduce nonlinearities in the projection

Digital image

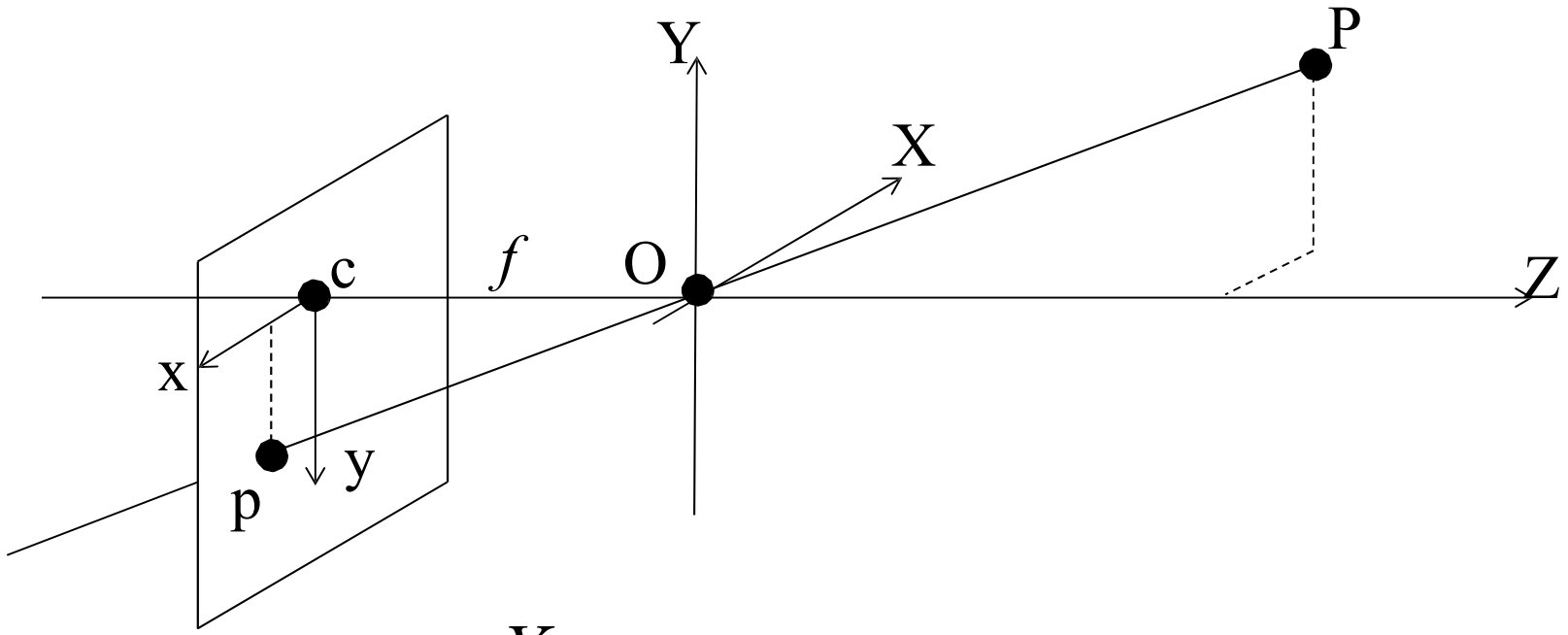






3D world

wrt to this description …



$$x = f\,\frac{X}{Z}$$

$$y = f\,\frac{Y}{Z}$$

all quantities, namely
$x\,, y\,, f\,, X\,, Y\,, Z$
are expressed in cm

… compensate Offset in Digital images

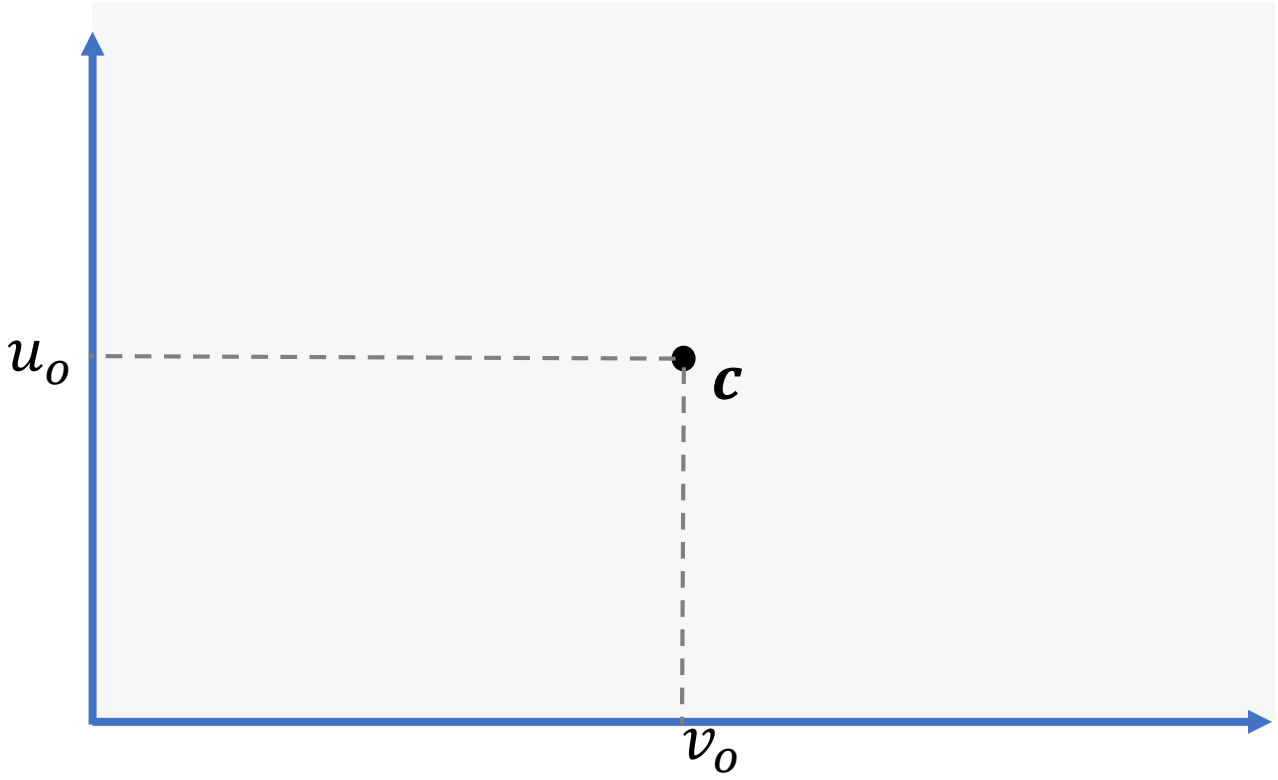
- 2D points in the image plane and 2D point in image coordinates differ by an offset $\boldsymbol{c} = [u_o, v_o]$ with $\boldsymbol{c}$: orthogonal projection of O onto the image plane
- We can easily accomodate this in the camera projection equations

Camera Projection

$$P = [X_P, Y_P, Z_P] \rightarrow \mathbf{p} = [x_p, y_p]$$

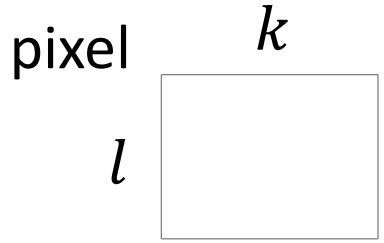
where

$$\begin{cases} x_p = f * \dfrac{X_P}{Z_P} + u_o \\ y_p = f * \dfrac{Y_P}{Z_P} + v_o \end{cases}$$



… compensate for different units

- Coordinates in images are expressed in terms of pixels, while on the image plane Π they are still expressed in cm as in the 3D world
- Pixels might have aspect ratio $\neq 1$
- Let l,k be the size of the pixel in cm (expressed in cm/pixel)

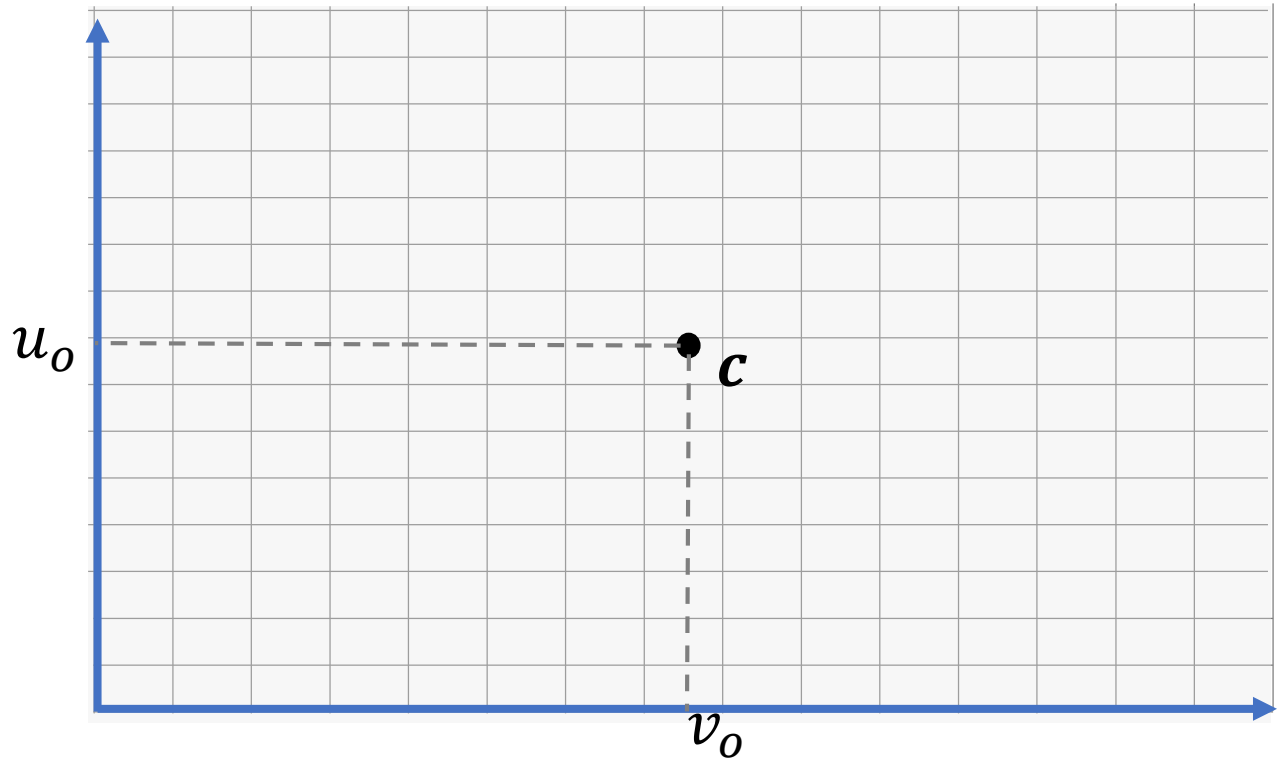


Camera Projection

$$P = [X_P, Y_P, Z_P] \rightarrow \mathbf{p} = [x_p, y_p]$$

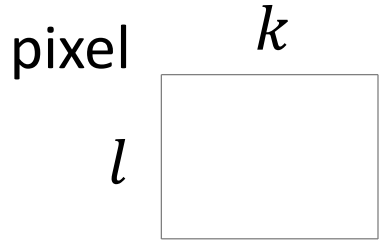
where

$$\begin{cases} x_p = \overbrace{f/k}^{f_x} * \dfrac{X_P}{Z_P} + u_o \\ y_p = \underbrace{f/l}_{f_y} * \dfrac{Y_P}{Z_P} + v_o \end{cases}$$



… compensate for different units

- Coordinates in images are expressed in terms of pixels, while on the image plane Π they are still expressed in cm as in the 3D world
- Pixels might have aspect ratio $\neq 1$
- Let l,k be the size of the pixel in cm (expressed in cm/pixel)

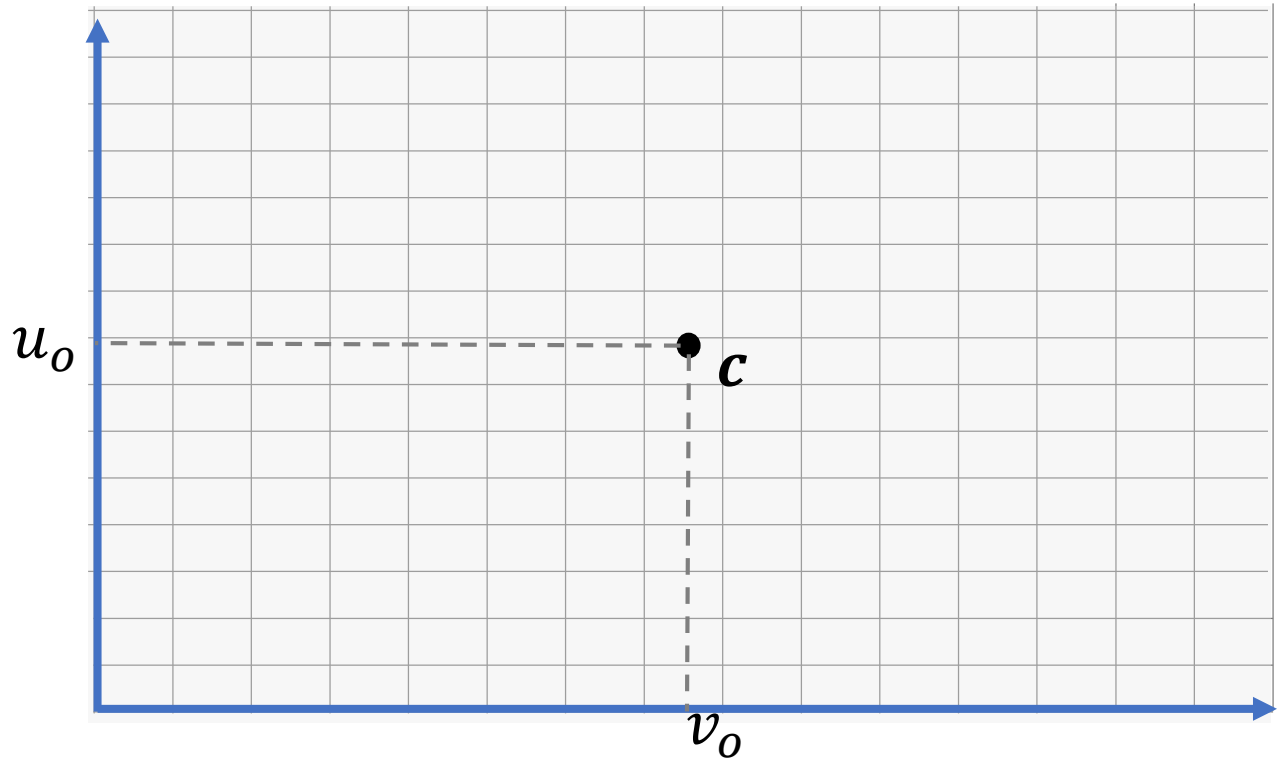


Camera Projection

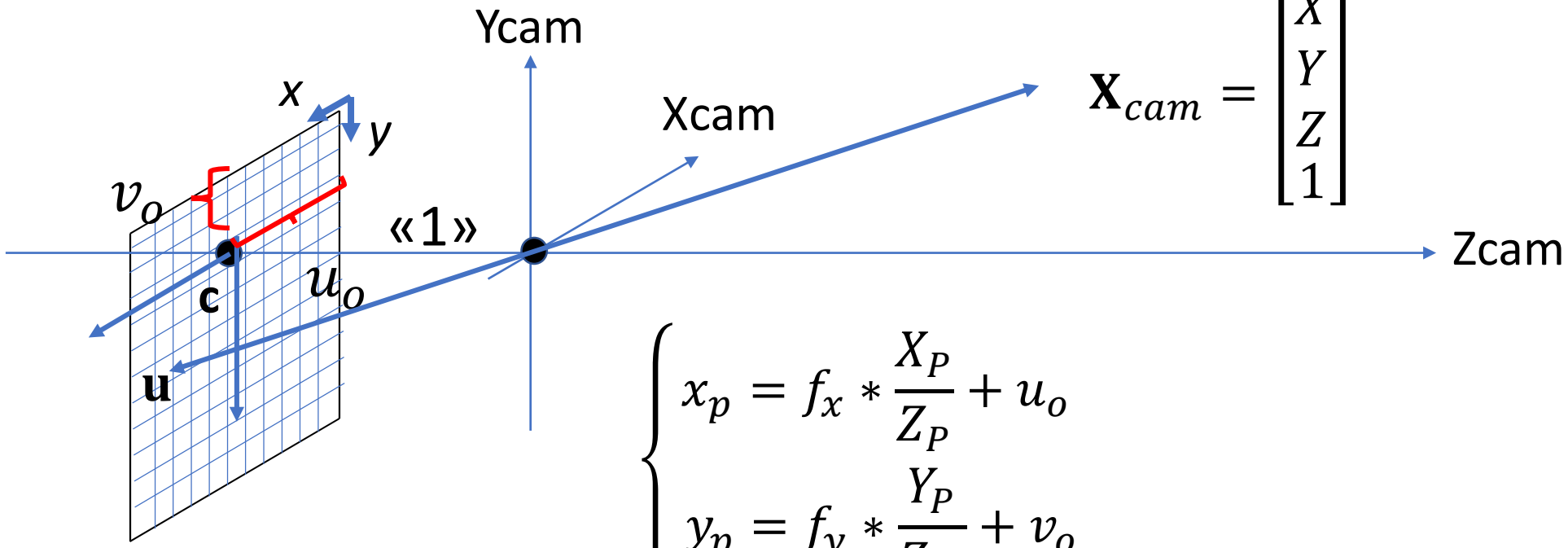
$$P = [X_P, Y_P, Z_P] \rightarrow \mathbf{p} = [x_p, y_p]$$

where

$$\begin{cases} x_p = f_x * \dfrac{X_P}{Z_P} + u_o \\ y_p = f_y * \dfrac{Y_P}{Z_P} + v_o \end{cases}$$



homogeneous **pixel** coordinates of image point: $\mathbf{u} = [x, y, w]^T$



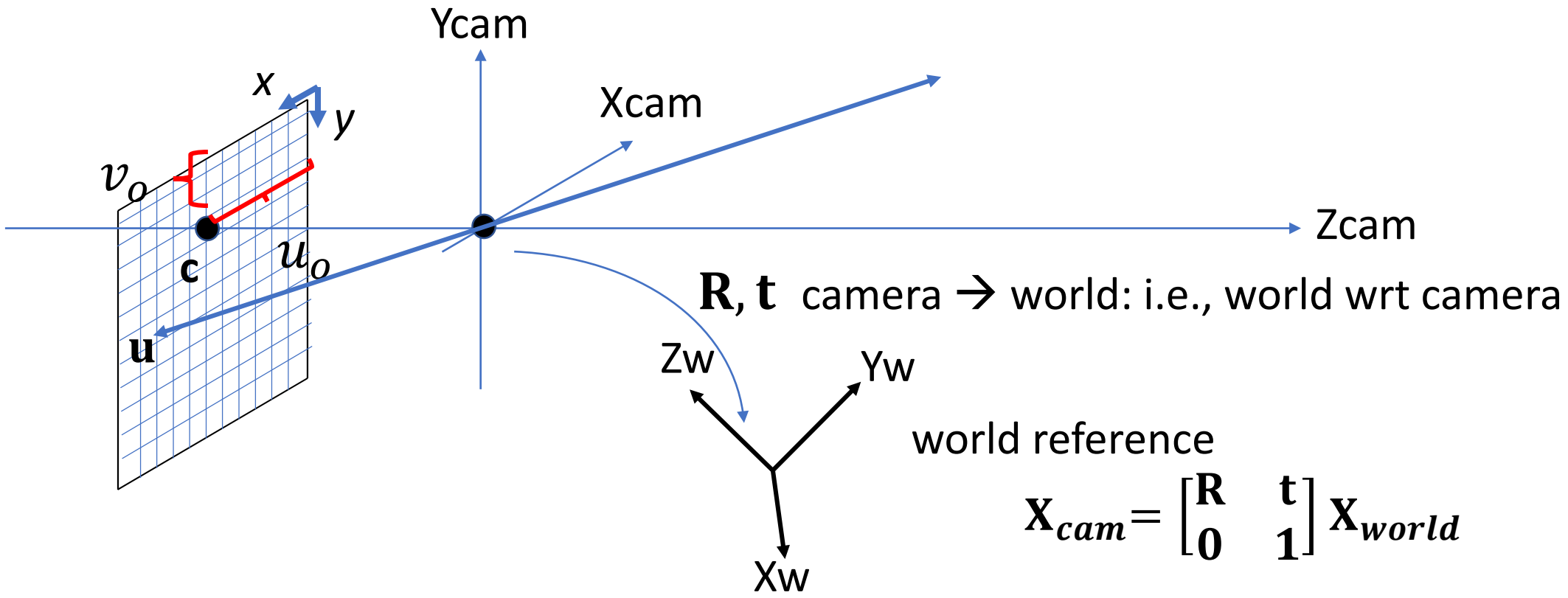
$$\mathbf{X}_{cam} = \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\begin{cases} x_p = f_x * \dfrac{X_P}{Z_P} + u_o \\ y_p = f_y * \dfrac{Y_P}{Z_P} + v_o \end{cases}$$

$$\mathbf{u} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_o \\ 0 & f_y & v_o \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_o \\ 0 & f_y & y_o \\ 0 & 0 & 1 \end{bmatrix} P = [\mathbf{K} \quad \mathbf{0}] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = [\mathbf{K} \quad \mathbf{0}]\mathbf{X}_{cam}$$

K : called **calibration matrix**

In general, world reference is ≠ camera reference



$\mathbf{R}, \mathbf{t}$ camera → world: i.e., world wrt camera

world reference

$$\mathbf{X}_{cam} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \mathbf{X}_{world}$$

$$\mathbf{u} = [\mathbf{K} \quad \mathbf{0}]\mathbf{X}_{cam} = [\mathbf{K} \quad \mathbf{0}] \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & \mathbf{1} \end{bmatrix} \mathbf{X}_{world} = [\mathbf{KR} \quad \mathbf{Kt}]\, \mathbf{X}_{world}$$

remember $\mathbf{u} = \mathbf{P}\, \mathbf{X}_{world} = [\mathbf{M} \quad \mathbf{m}]\, \mathbf{X}_{world}$ → $\boxed{\mathbf{M} = \mathbf{KR}}$ and $\boxed{\mathbf{m} = \mathbf{Kt}}$

Calibration matrix

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & U_o \\ 0 & f_y & V_o \\ 0 & 0 & 1 \end{bmatrix}$$

Intrinsic parameters: don't vary under camera displacement

- relative position of camera center and image plane
- pixel aspect ratio $a = \frac{f_x}{f_y}$ (for ***natural*** *camera*, square pixels, is $a = 1$)

Camera projection matrix

$$P = [\mathbf{KR} \quad \mathbf{Kt}] = \mathbf{K}[\mathbf{R} \quad \mathbf{t}]$$

Intrinsic parameters + **Extrinsic parameters:**

- relative pose (= position & orientation) of camera reference and world reference

Camera calibration

Intrinsic camera **calibration**:
estimation of matrix $\mathbf{K}$
- focal distance f_X
- focal distance f_Y
- principal point (U_o, V_o)
- skew factor (in old cameras)
aspect ratio $a = f_X/f_Y = l/k$:
ratio between pixel height and lenght

intrinsic camera parameters don't vary
under camera displacement

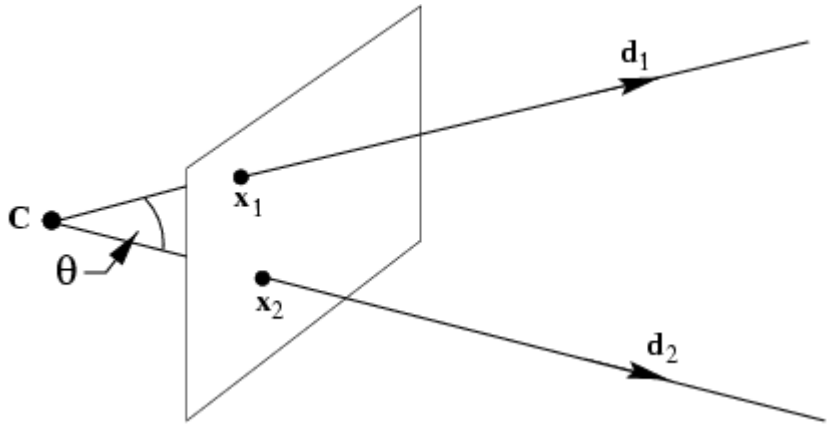
Extrinsic camera **calibration**:
estimation of matrix $\mathbf{R}$ and vector $\boldsymbol{t}$
- camera→world rotation $\mathbf{R}$
- camera→world translation $\boldsymbol{t}$

extrinsic camera parameters vary under camera displacement

What does INTRINSIC calibration give?

$$x = K[I \mid 0]\begin{bmatrix} d \\ 0 \end{bmatrix}$$

$$d = K^{-1}x$$



All the viewing rays are constrained to go through camera center, the direction of the viewing ray associated to an image point x is

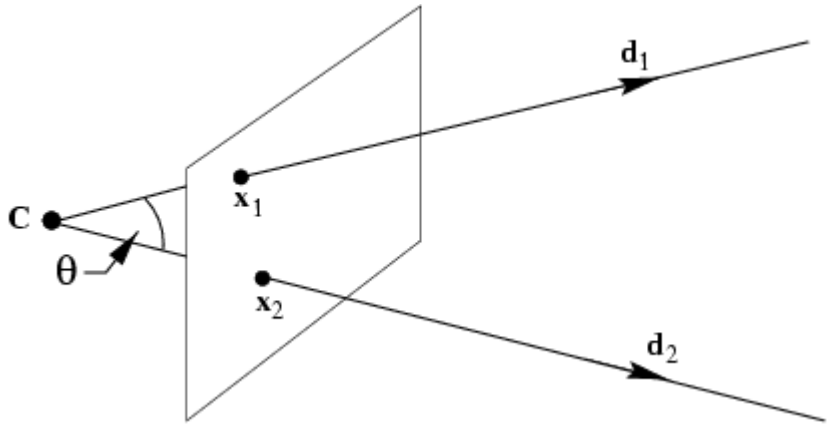
$$d = K^{-1}x \text{ (relative to the } \textbf{camera}\text{)}$$

→ **Intrinsic** camera calibration gives the mapping btw any image point and the 3D position of its associated viewing ray, relative to **camera**

What does EXTRINSIC calibration give?

$$x = K[R|t]\begin{bmatrix} d \\ 0 \end{bmatrix}$$

$$d = (KR)^{-1}x$$



All the viewing rays are constrained to go through camera center, the direction of the viewing ray associated to an image point x is

$$d = (KR)^{-1}x \text{ (relative to the } \textbf{world}\text{)}$$

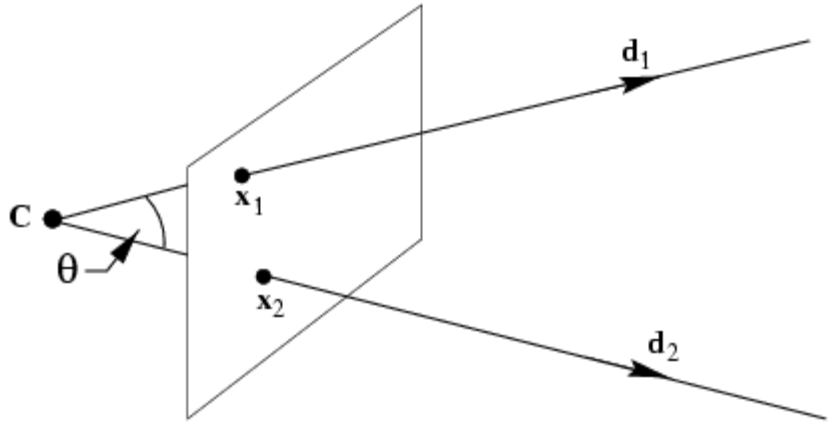
→ **Extrinsic** camera calibration gives the mapping btw any image point and the 3D position of its associated viewing ray, relative to **world**

Angle between two viewing rays

$$u = [KR|Kt]\begin{bmatrix} d \\ 0 \end{bmatrix} = KRd$$

$$d = (KR)^{-1}u = R^{-1}K^{-1}u$$

$$d^T = u^T(KR)^{-T} = u^TK^{-T}R^{-T} = u^TK^{-T}R$$



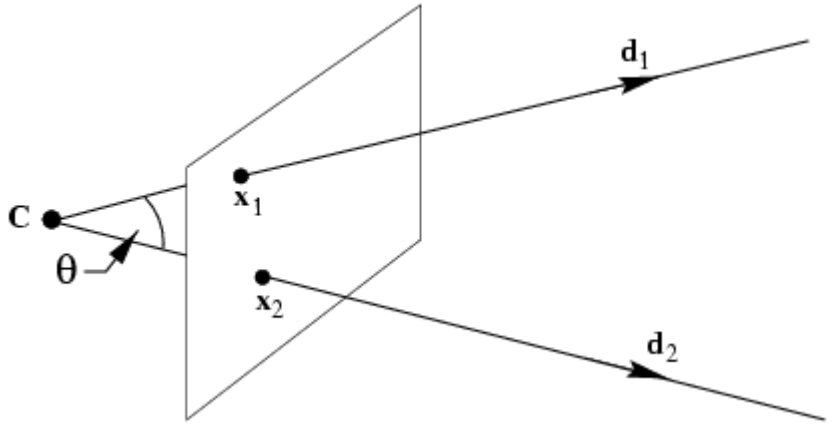
$$\cos\theta = \frac{{d_1}^T d_2}{\sqrt{({d_1}^T d_1)({d_2}^T d_2)}} = \frac{{u_1}^T(K^{-T}\cancel{RR^{-1}}K^{-1})u_2}{\sqrt{({u_1}^T(K^{-T}K^{-1})u_1)({u_2}^T(K^{-T}K^{-1})u_2)}}$$

These angles don't depend on the absolute position of the camera

→ relative position of viewing rays associated to different image points

The ω matrix (also called the IAC)

Define $\omega \triangleq (\mathbf{K}\mathbf{K}^{\boldsymbol{T}})^{-1}$ (ω is a symmetric matrix)



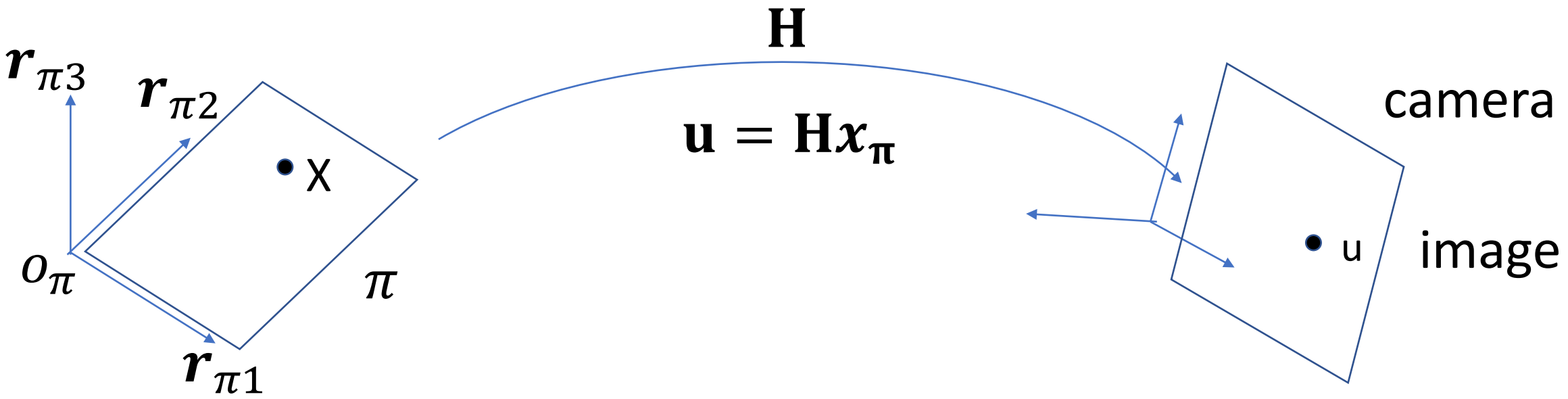
Property: The directions d_1 and d_2 of the viewing rays of two image points u_1 and u_2 form an angle given by

$$\cos\theta = \frac{u_1{}^{T}\omega\, u_2}{\sqrt{(u_1{}^{T}\omega\, u_1)(u_2{}^{T}\omega u_2)}}$$

Useful properties

The image of a planar scene:
Homography between a plane π and its image

Homography between a plane π and its image

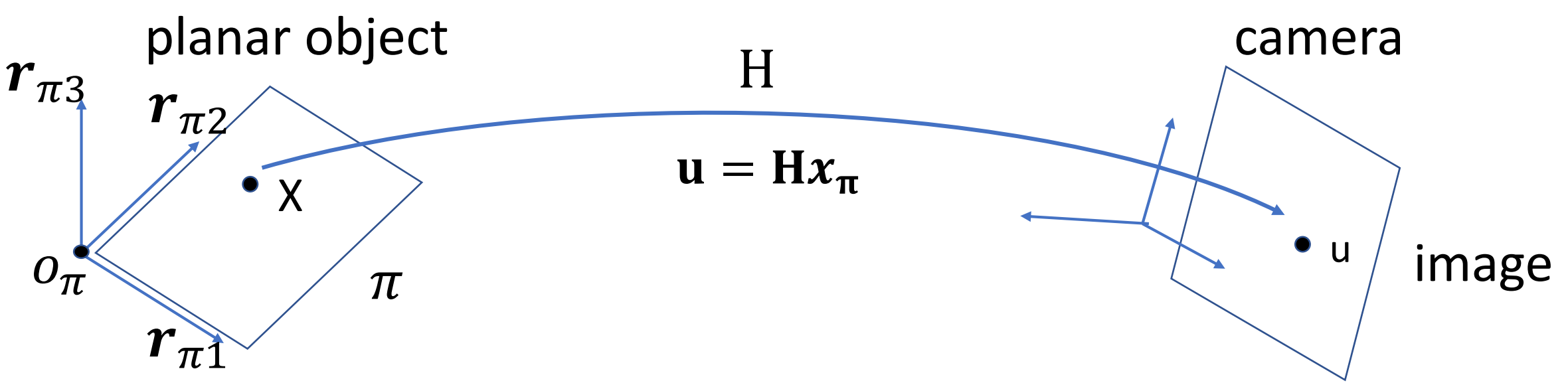


plane π reference: relative pose plane wrt camera → rototranslation $\mathbf{R}_\pi$, $\boldsymbol{o}_\pi$

world reference ≡ camera reference

$\mathrm{P} = [\mathbf{KR} \quad \mathbf{Kt}]$ →

$\mathrm{P} = [\mathbf{K} \quad \mathbf{0}]$



plane reference

world reference ≡ camera reference → $\mathrm{P} = [\mathbf{K} \quad \mathbf{0}]$

$$\mathbf{X}_{\boldsymbol{\pi}} = \begin{bmatrix} x \\ y \\ 0 \\ w \end{bmatrix} \qquad \mathbf{X}_{\mathbf{w}} = \begin{bmatrix} \boldsymbol{r}_{\boldsymbol{\pi 1}} & \boldsymbol{r}_{\boldsymbol{\pi 2}} & \boldsymbol{r}_{\boldsymbol{\pi 3}} & \mathbf{o}_{\boldsymbol{\pi}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 0 \\ w \end{bmatrix} = \begin{bmatrix} \boldsymbol{r}_{\pi 1} & \boldsymbol{r}_{\pi 2} & \mathbf{o}_{\pi} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix} = \boldsymbol{x}_{\boldsymbol{\pi}}$$

$$\mathbf{u} = \mathbf{P}\mathbf{X}_{\mathbf{w}} = [\mathbf{K} \quad \mathbf{0}] \begin{bmatrix} \boldsymbol{r}_{\pi 1} & \boldsymbol{r}_{\pi 2} & \mathbf{o}_{\pi} \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{x}_{\boldsymbol{\pi}} = \mathbf{K}[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_{\pi}]\boldsymbol{x}_{\boldsymbol{\pi}}$$

plane-to-image homography $\mathbf{H} = \mathbf{K}[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_{\pi}]$

The image of a planar scene: Homography between the plane π_∞ and its image

Particular case:

plane $\pi = \pi_\infty$ is the plane at the infinity

vanishing points = image of points at the ∞: $\mathbf{v_d} = \mathbf{P}\mathrm{X}_\infty = \mathbf{K}\,[\mathbf{R} \quad \mathbf{t}] \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} = \mathbf{KR}\,\mathbf{d}$

→

Plane-at-the-infinity – to – image homography $\mathbf{H} = \mathbf{KR}$

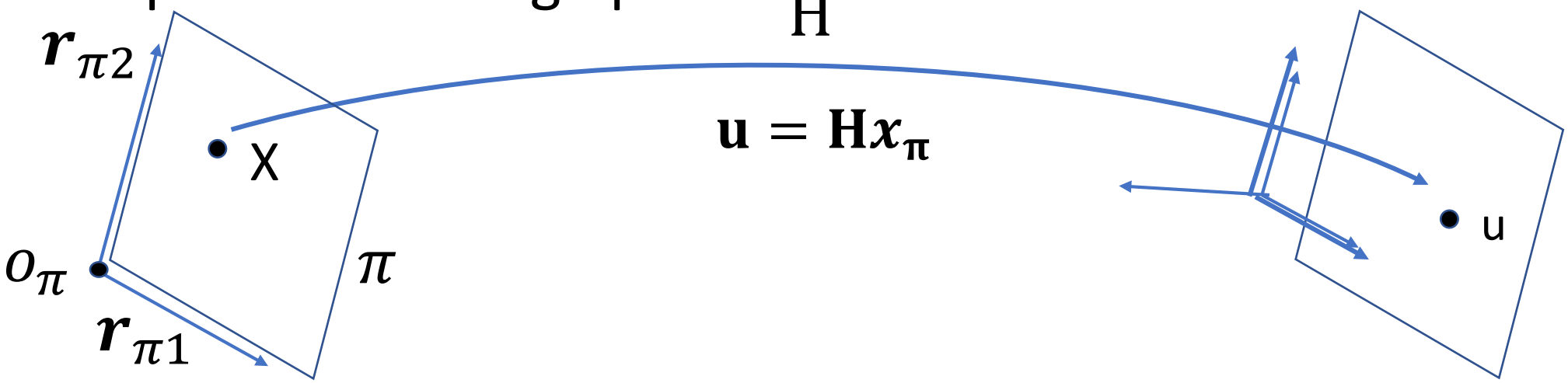
(inverse: $\mathbf{d} = (\mathbf{KR})^{-1}\mathbf{v_d}$)

The image of a planar scene:
Homography between a plane π and its image

Particular case:

- plane π **parallel to the image plane**

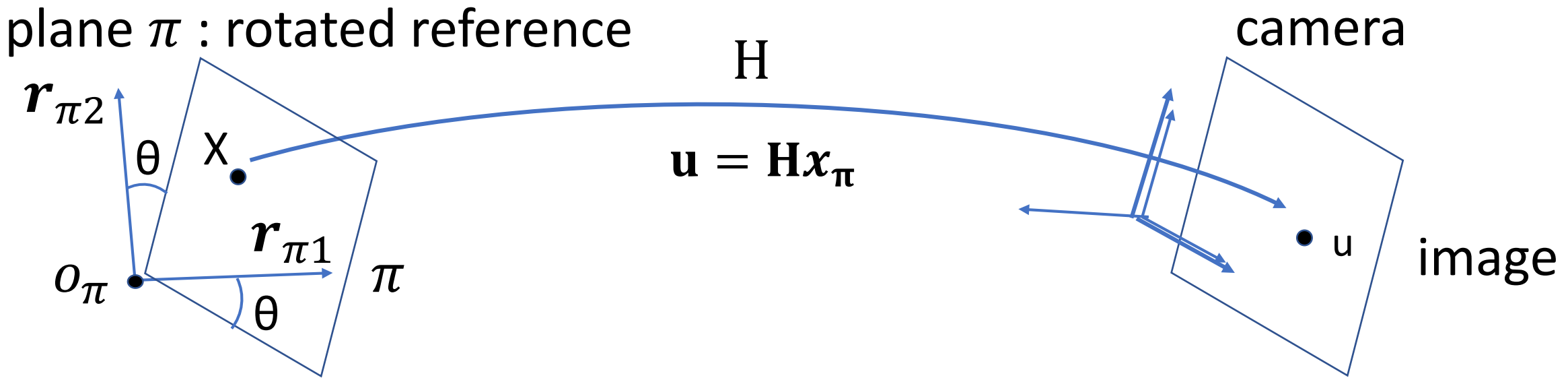
plane π parallel to image plane



$$\boldsymbol{r}_{\boldsymbol{\pi 1}} = \begin{bmatrix}1\\0\\0\end{bmatrix},\ \boldsymbol{r}_{\boldsymbol{\pi 2}} = \begin{bmatrix}0\\1\\0\end{bmatrix} \rightarrow [\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad o_\pi] = \begin{bmatrix}1 & 0 & X_o\\0 & 1 & Y_o\\0 & 0 & 1\end{bmatrix}$$

$$\mathbf{u} = \mathbf{K}[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_\pi]\boldsymbol{x}_{\boldsymbol{\pi}} = \mathbf{K}\begin{bmatrix}1 & 0 & X_o\\0 & 1 & Y_o\\0 & 0 & 1\end{bmatrix}\boldsymbol{x}_{\boldsymbol{\pi}} = \begin{bmatrix}f_x & 0 & U_o\\0 & f_y & V_o\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & X_o\\0 & 1 & Y_o\\0 & 0 & 1\end{bmatrix}\boldsymbol{x}_{\boldsymbol{\pi}}$$

plane-to-image homography $\mathbf{H}$ = 2D-affine * 2D-isometry = **2-D affinity**



$$\boldsymbol{r}_{\pi 1} = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix},\ \boldsymbol{r}_{\pi 2} = \begin{bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{bmatrix} \rightarrow [\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad o_\pi] = \begin{bmatrix} R_2 & \begin{matrix} X_o \\ Y_o \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 1 \end{bmatrix}$$

$$\mathbf{u} = \mathbf{K}[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_\pi]\boldsymbol{x}_\pi = \mathbf{K}\begin{bmatrix} R_2 & \begin{matrix} X_o \\ Y_o \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 1 \end{bmatrix}\boldsymbol{x}_\pi = \begin{bmatrix} f_x & 0 & U_o \\ 0 & f_y & V_o \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} R_2 & \begin{matrix} X_o \\ Y_o \end{matrix} \\ \begin{matrix} 0 & 0 \end{matrix} & 1 \end{bmatrix}\boldsymbol{x}_\pi$$

plane-to-image homography $\mathbf{H}$ = 2D-affine * 2D-isometry = **2-D affinity**

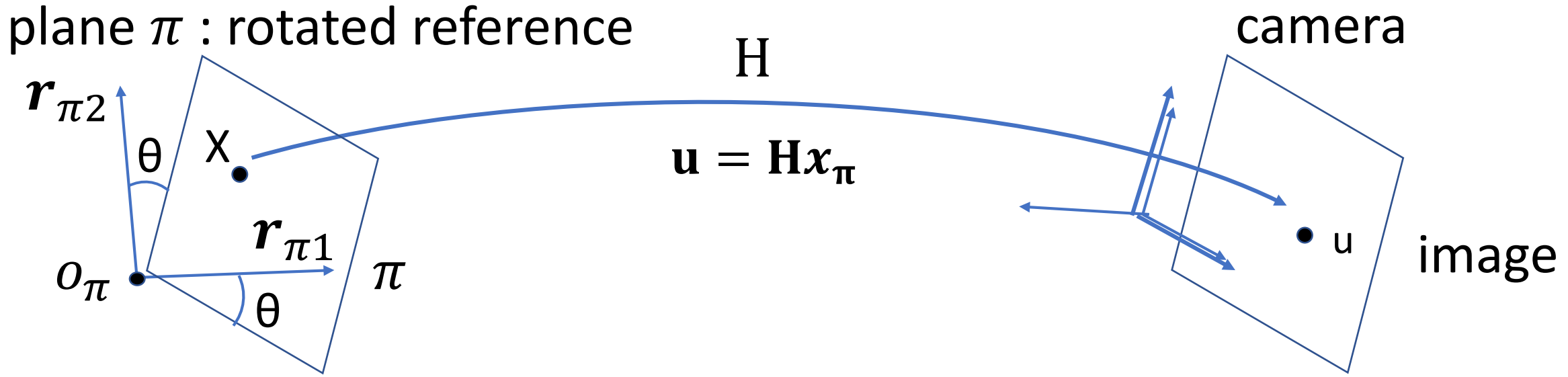
The image of a planar scene:
Homography between a plane π and its image

Particular case:

- plane π parallel to the image plane

AND

- **natural camera**, i.e., square pixels $f_x = f_y = f$



$$\boldsymbol{r}_{\boldsymbol{\pi 1}} = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix},\ \boldsymbol{r}_{\boldsymbol{\pi 2}} = \begin{bmatrix} -\sin\theta \\ \cos\theta \\ 0 \end{bmatrix} \rightarrow [\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad o_\pi] = \begin{bmatrix} R & & X_o \\ & & Y_o \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{u} = \mathbf{K}[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_\pi]\boldsymbol{x}_{\boldsymbol{\pi}} = \mathbf{K}\begin{bmatrix} R & & X_o \\ & & Y_o \\ 0 & 0 & 1 \end{bmatrix}\boldsymbol{x}_{\boldsymbol{\pi}} = \begin{bmatrix} f & 0 & U_o \\ 0 & f & V_o \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} R & & X_o \\ & & Y_o \\ 0 & 0 & 1 \end{bmatrix}\boldsymbol{x}_{\boldsymbol{\pi}}$$

plane-to-image homography $\mathbf{H}$ = 2D-similarity * 2D-isometry = **2D-similarity**

From projection matrix $\mathbf{P}$ to calibration matrix $\mathbf{K}$

remember

$$\mathbf{P} = [\mathbf{M} \quad \boldsymbol{m}] = [\mathbf{KR} \quad \mathbf{K}\boldsymbol{t}]$$

from $\omega \triangleq \left(\mathbf{K}\mathbf{K}^{\boldsymbol{T}}\right)^{-1}$

$$\mathbf{M}\mathbf{M}^{\mathrm{T}} = \mathbf{K}\mathbf{R}(\mathbf{K}\mathbf{R})^{\mathrm{T}} = \mathbf{K}\mathbf{K}^{\mathrm{T}} = \boldsymbol{\omega}^{-1}$$

From $\mathrm{P} = [\mathbf{M} \quad \boldsymbol{m}]$:

- take $\omega^{-1} = \mathbf{M}\mathbf{M}^{\mathrm{T}}$
- $\mathbf{K}$ = Choleski factorization of $\omega^{-1} = \mathbf{M}\mathbf{M}^{\mathrm{T}} = \mathbf{K}\mathbf{K}^{\mathrm{T}}$

Camera calibration through the ω matrix

USEFUL FACTS

vanishing points = image of points at the ∞: $\mathbf{v_d} = \mathbf{P}\mathrm{X}_\infty = \mathbf{K}\,[\mathbf{R} \quad \mathbf{t}] \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} = \mathbf{KR}\,\mathbf{d}$

→ Homography from $\boldsymbol{\pi_\infty}$ to image plane: $\mathbf{H} = \mathbf{KR}$ → inverse: $\mathbf{d} = (\mathbf{KR})^{-1}\mathbf{v_d}$

From $\boldsymbol{\omega}$ to calibration matrix $\mathbf{K}$: the Cholesky factorisation of inverse $\boldsymbol{\omega}^{-1} = \mathbf{K}\mathbf{K}^{\mathrm{T}}$ provides an upper triangular matrix: i.e., $\mathbf{K}$

Collect constraints on $\boldsymbol{\omega}$ from:

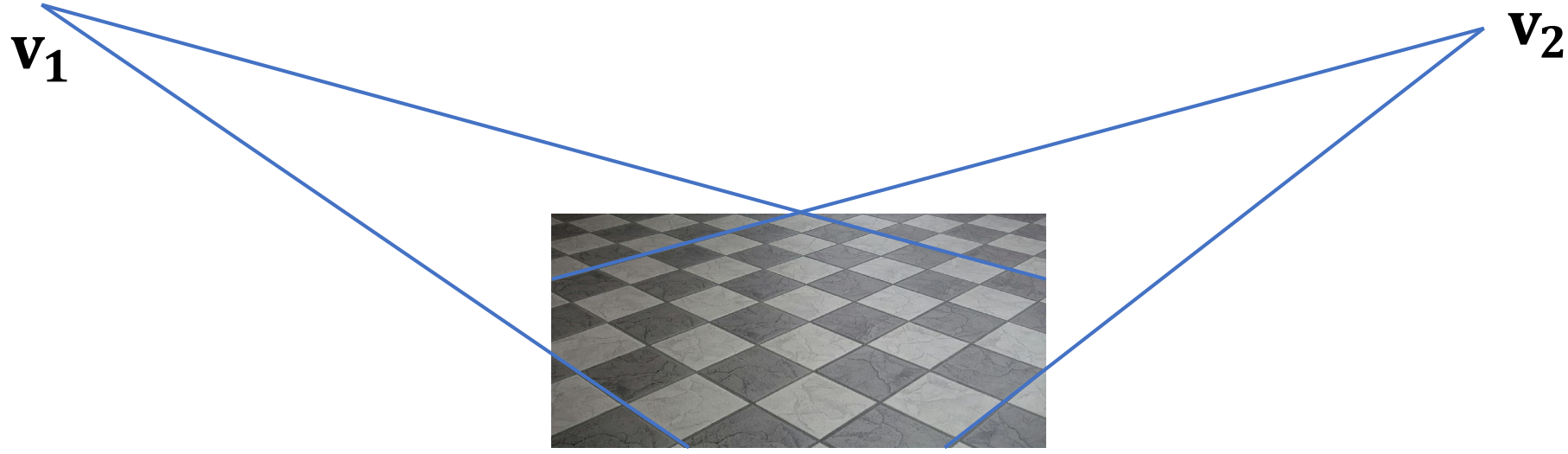
- known angles btw directions, observing: their vanishing points
- (includes) self-orthogonality, observing: images of circular points
- known planar shapes, observing: their images

Constraints on the ω matrix from angles θ

- **Known: angles btw directions $\mathbf{d}_i$**, and **their vanishing points $\mathbf{v}_i = \mathbf{KRd}_i$**

$$\cos\theta = \frac{\mathbf{d_1}^{\mathrm{T}}\mathbf{d_2}}{\sqrt{(\mathbf{d_1}^{\mathrm{T}}\mathbf{d_1})(\mathbf{d_2}^{\mathrm{T}}\mathbf{d_2})}} = \frac{\mathbf{v_1}^{\mathrm{T}}(\mathbf{KR})^{-\mathrm{T}}(\mathbf{KR})^{-1}\mathbf{v_2}}{\sqrt{(\mathbf{v_1}^{\mathrm{T}}..\mathbf{v_1})(\mathbf{v_2}^{\mathrm{T}}..\mathbf{v_2})}} = \frac{\mathbf{v_1}^{\mathrm{T}}\omega\mathbf{v_2}}{\sqrt{(\mathbf{v_1}^{\mathrm{T}}\omega\mathbf{v_1})(\mathbf{v_2}^{\mathrm{T}}\omega\mathbf{v_2})}}$$

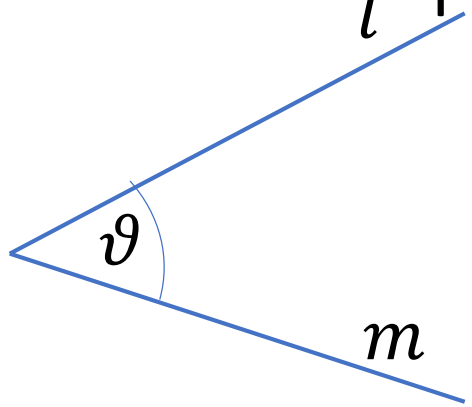
if directions are orthogonal $\theta = 90°$ → **linear** constraint: $\mathbf{v_1}^{\mathrm{T}}\omega\mathbf{v_2} = \mathbf{0}$



ATTENTION: DO NOT MIX UP WITH 2D GEOMETRY !

$l', m', C_\infty^{*\prime}$ image lines and conic dual to the circular pts

$$\cos\vartheta = \frac{l'^T C_\infty^{*\prime} m'}{\sqrt{(l'^T C_\infty^{*\prime} l')(m'^T C_\infty^{*\prime} m')}}$$

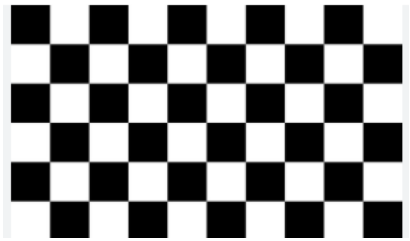
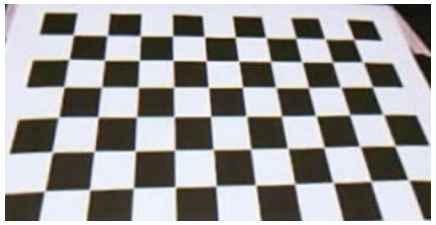


$$\cos\theta = \frac{\mathbf{v_1}^{\mathrm{T}} \omega \mathbf{v_2}}{\sqrt{(\mathbf{v_1}^{\mathrm{T}} \omega \mathbf{v_1})(\mathbf{v_2}^{\mathrm{T}} \omega \mathbf{v_2})}}$$

HERE

$\mathbf{v_1}, \mathbf{v_2}, \omega$ vanishing points and ω matrix

Constraints on the ω matrix from homographies H

- **known planar shapes and**  $\mathbf{H}$  **their images**

plane-to-image homography $\mathbf{H} = \mathbf{K}[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_{\pi}]$

$$\rightarrow [\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_{\pi}] = \mathbf{K}^{-1}\mathbf{H} = \left[\mathbf{K}^{-1}\boldsymbol{h}_1 \quad \mathbf{K}^{-1}\boldsymbol{h}_2 \quad \mathbf{K}^{-1}\boldsymbol{h}_3\right]$$

vectors $\mathbf{K}^{-1}\boldsymbol{h}_1, \mathbf{K}^{-1}\boldsymbol{h}_2$ are orthogonal and have the same module

$$\rightarrow$$

$$\mathbf{0} = \boldsymbol{h}_1^T\mathbf{K}^{-T}\mathbf{K}^{-1}\boldsymbol{h}_2 = \boldsymbol{h}_1^T\,\omega\,\boldsymbol{h}_2$$

and

$$\mathbf{0} = \boldsymbol{h}_1^T\mathbf{K}^{-T}\mathbf{K}^{-1}\boldsymbol{h}_1 - \boldsymbol{h}_2^T\mathbf{K}^{-T}\mathbf{K}^{-1}\boldsymbol{h}_2 = \boldsymbol{h}_1^T\,\omega\,\boldsymbol{h}_1 - \boldsymbol{h}_2^T\,\omega\,\boldsymbol{h}_2$$

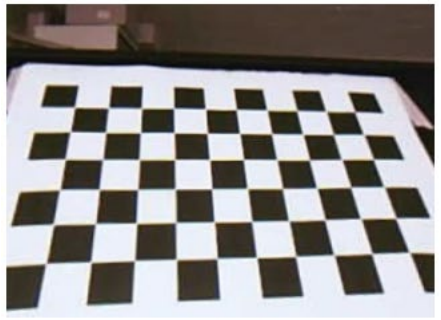
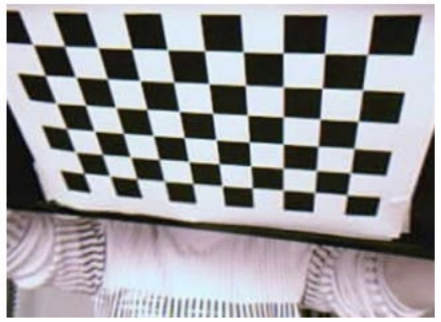
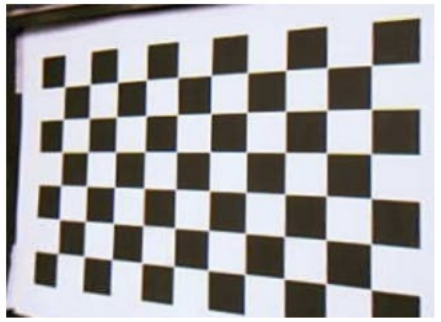
Camera calibration from images of known planar shapes → H (Zhang constraints)

- Estimate homographies from known planar shapes to their image
- For each homography $\mathbf{H} = [\boldsymbol{h}_1 \quad \boldsymbol{h}_2 \quad \boldsymbol{h}_3]$

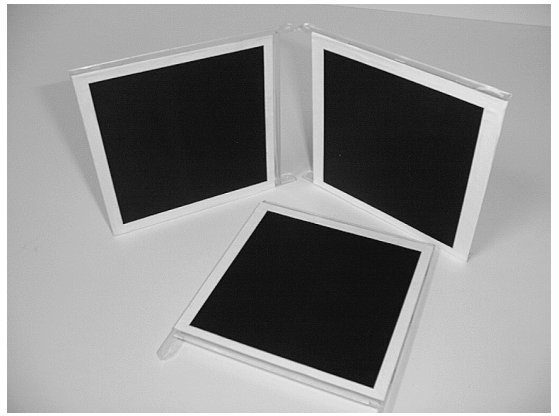
$$\boldsymbol{h_1}^T \omega \boldsymbol{h_2} = 0$$
$$\boldsymbol{h_1}^T \omega \boldsymbol{h_1} - \boldsymbol{h_2}^T \omega \boldsymbol{h_2} = 0$$

2 homogeneous equations in ω → at least 3 homographies needed

e.g.

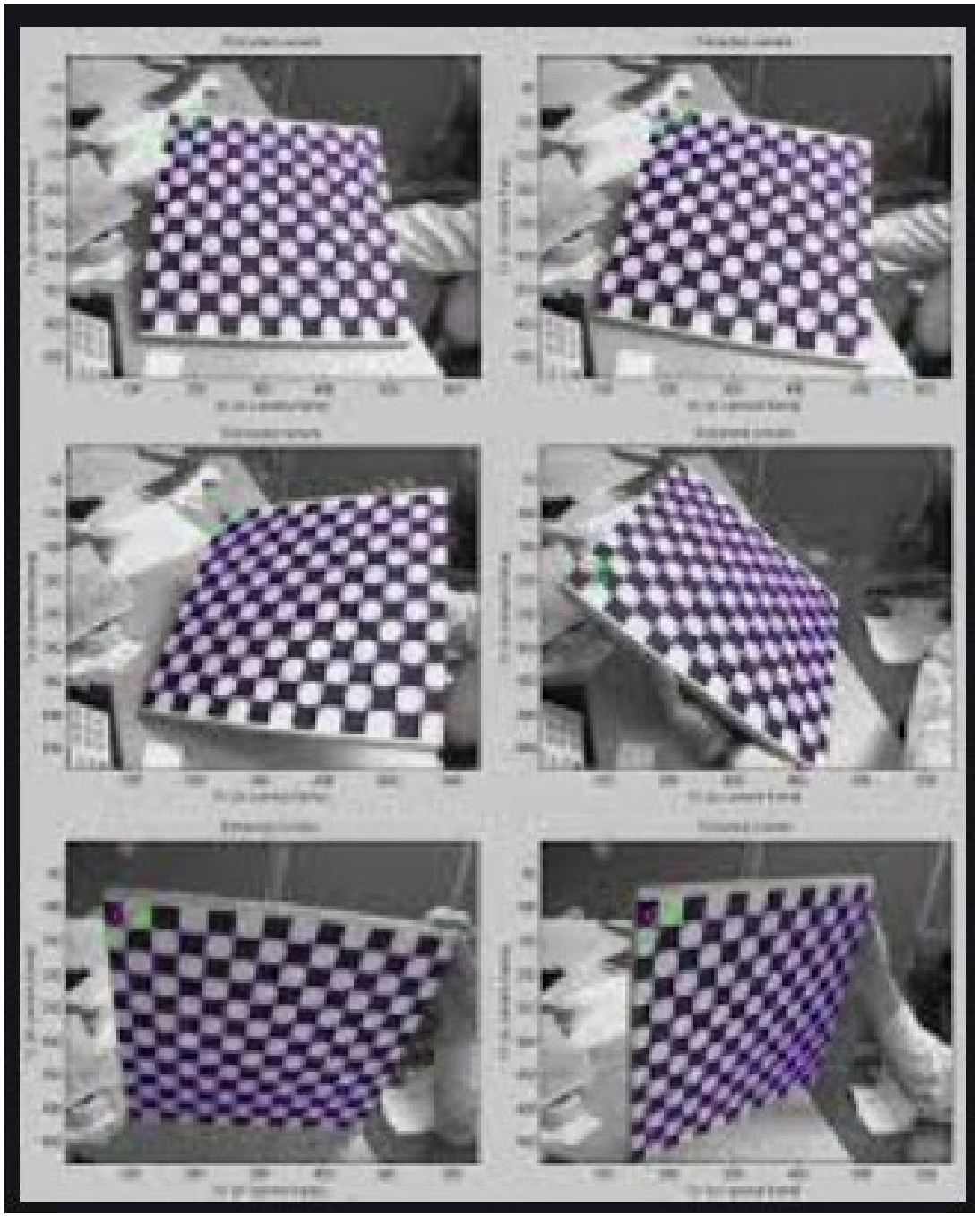
or



from Cholesky factorisation of ω^{-1} → $\mathbf{K}$

OpenCV camera calibration toolbox

- implements Zhang method
- planar target (easily printable)
- several images (~ 20) to cope with noise
- also estimates distortion param.
- provides accurate calibration





OpenCV and Matlab Calibration Toolbox also contains estimation of **distortion** parameters

$$x = x_o + (x_o - c_x)(K_1 r^2 + K_2 r^4 + \ldots)$$
$$y = y_o + (y_o - c_y)(K_1 r^2 + K_2 r^4 + \ldots)$$

$$r = (x_o - c_x)^2 + (y_o - c_y)^2 \ .$$

Scenario n. 1

Known: vanishing points and angle θ between directions

$$\cos\theta = \frac{\mathbf{d_1}^{\mathrm{T}}\mathbf{d_2}}{\sqrt{(\mathbf{d_1}^{\mathrm{T}}\mathbf{d_1})(\mathbf{d_2}^{\mathrm{T}}\mathbf{d_2})}} = \frac{\mathbf{v_1}^{\mathrm{T}}\omega\mathbf{v_2}}{\sqrt{(\mathbf{v_1}^{\mathrm{T}}\omega\mathbf{v_1})(\mathbf{v_2}^{\mathrm{T}}\omega\mathbf{v_2})}}$$

→

constraint on the ω matrix

(linear if directions are orthogonal)

→ camera calibration

Scenario n. 2

Known: vanishing points and ω matrix

$$\cos\,\theta = \frac{\mathbf{d_1}^{\mathrm{T}}\mathbf{d_2}}{\sqrt{(\mathbf{d_1}^{\mathrm{T}}\mathbf{d_1})(\mathbf{d_2}^{\mathrm{T}}\mathbf{d_2})}} = \frac{\mathbf{v_1}^{\mathrm{T}}\omega\mathbf{v_2}}{\sqrt{(\mathbf{v_1}^{\mathrm{T}}\omega\mathbf{v_1})(\mathbf{v_2}^{\mathrm{T}}\omega\mathbf{v_2})}}$$

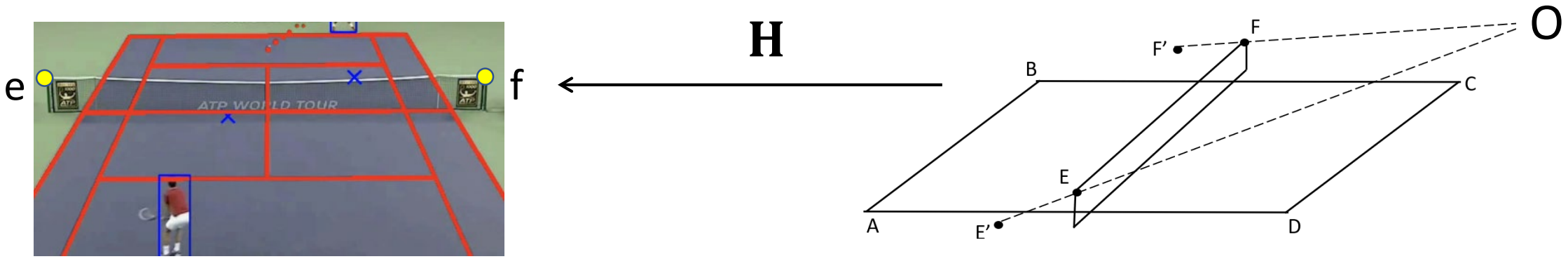
→

compute angle θ between directions

→ Localization (see later)

→ Reconstruction of the shape (see Appendix 2)

Extrinsic + intrinsic camera calibration from a single image of a known planar shape and known camera center position O



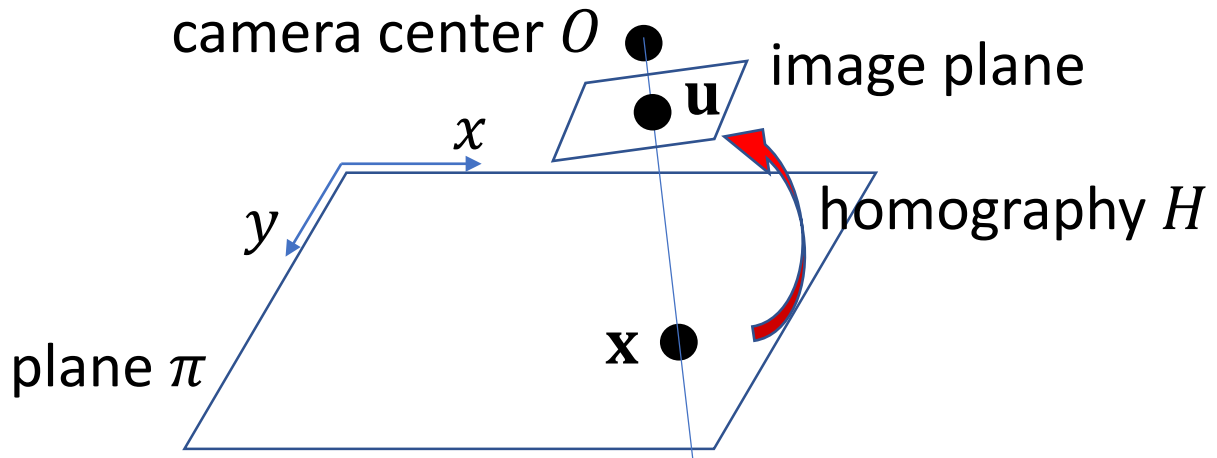
- Estimate H from ground plane to image
- Compute 3D position of O wrt the tennis court:
- Find coordinates of E' and F' on the ground plane by $\mathrm{E}' = \mathrm{H}^{-1}\mathbf{e}$ and $\mathrm{F}' = \mathrm{H}^{-1}\mathbf{f}$
- From known position of E and F , O is at the intersection of lines E'E and F'F

Calibration from known plane π and camera center O

Remember what calibration gives: viewing ray associated to any image point $\mathbf{u}$., i.e., the mapping between $\mathbf{u}$ and the viewing ray direction $\mathbf{d}$

Take any $\mathbf{u}$: Knowing H, find backprojection point $\mathbf{x}$ onto π as $\mathbf{x} = H^{-1}\mathbf{u}$

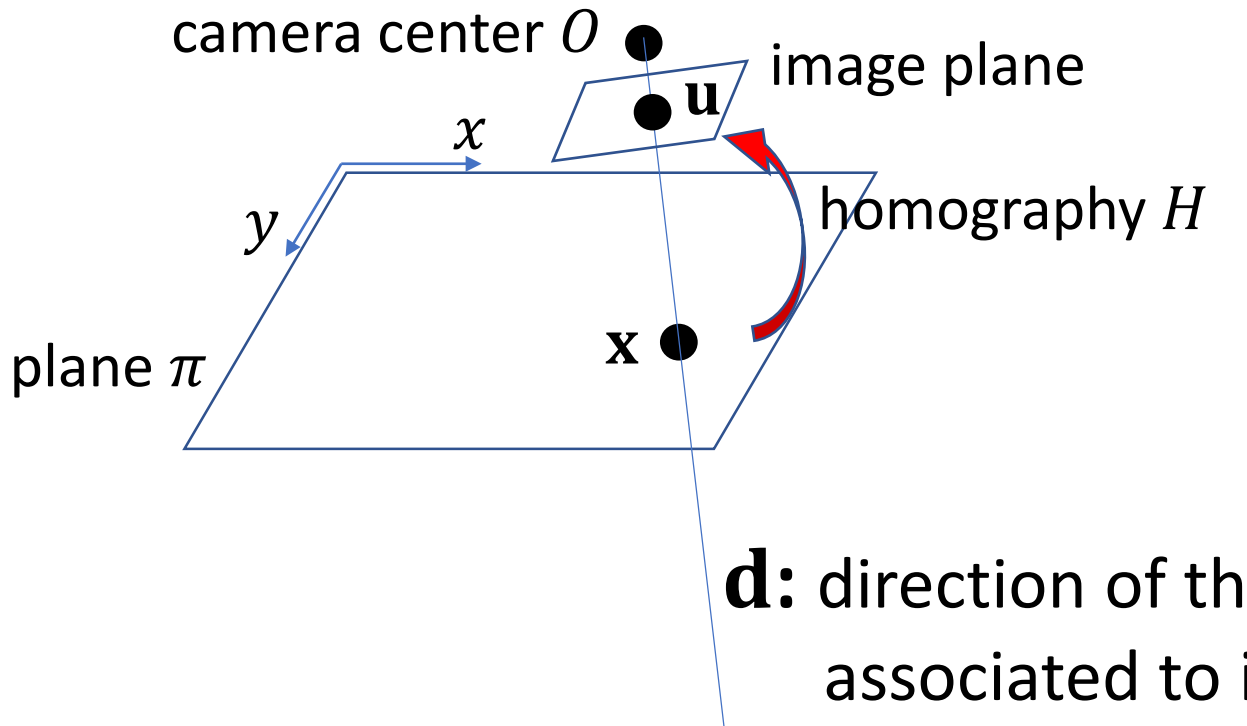
Knowing O, find direction $\mathbf{d}$ from O to $\mathbf{x}$ (express $\mathbf{d}$ in terms of $\mathbf{x}$)



$\mathbf{d}$: direction of the viewing ray associated to image point $\mathbf{u}$

Calibration from known plane π and camera center O

Let us refer the coordinates to a reference attached to a known plane π : a generic point $\mathbf{x}$ on this plane has **homogeneous** coordinates $\mathbf{x} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$, while $O = \begin{bmatrix} \mathrm{x}_o \\ \mathrm{y}_o \\ \mathrm{z}_o \end{bmatrix}$ are the known **cartesian** coordinates of the camera center,



$\mathbf{d}$: direction of the viewing ray associated to image point $\mathbf{u}$

Calibration from known plane π and camera center O

1) Estimate homography H from known points on the plane π and their images

2) Call M_o the matrix relating any point $\mathbf{x}$ on π to the direction $\mathbf{d}$ of a ray from O to $\mathbf{x}$:

$$\mathbf{d} = \begin{bmatrix} x - \mathrm{x}_o \\ y - \mathrm{y}_o \\ 0 - \mathrm{z}_o \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\mathrm{x}_o \\ 0 & 1 & -\mathrm{y}_o \\ 0 & 0 & -\mathrm{z}_o \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = {M_o}^{-1} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \text{ with } {M_o}^{-1} = \begin{bmatrix} 1 & 0 & -\mathrm{x}_o \\ 0 & 1 & -\mathrm{y}_o \\ 0 & 0 & -\mathrm{z}_o \end{bmatrix}$$

3) Compute matrix $\mathbf{M}$ relating any image point $\mathbf{u}$ to the direction $\mathbf{d} = \mathbf{M}^{-1}\mathbf{u}$ of its viewing ray from $\mathbf{u} = H\,\mathbf{x}$, is $\mathbf{d} = {M_o}^{-1}\mathbf{x} = {M_o}^{-1}H^{-1}\mathbf{u}$ → $\mathbf{M}^{-1} = {M_o}^{-1}H^{-1}$

4) **Q-R decompose** matrix $\mathbf{M}^{-1}$ as $\mathbf{M}^{-1} = \mathbf{R}^{-1}\mathbf{K}^{-1}$, where $\mathbf{K}$ is the camera intrinsic calibration matrix and $\mathbf{R}^{-1}$ is the rotation matrix from the world (reference attached to π) to the camera

Camera calibration from

- a planar face of known (reconstructed) shape and
- a vanishing point of the normal to the plane

calibration after reconstruction of a planar face



Shape of the face:
known after metric rectificaton



How much additional information is needed to calibrate the camera?

- Vanishing point of the direction normal to the reconstructed face



Vanishing point of vertical direction

V

Camera calibration (assume skew = 0)

- Only four unknowns:

$$\omega = (KK^T)^{-1} = \begin{vmatrix} a^2 & 0 & -u_0 a^2 \\ * & 1 & -v_0 \\ * & * & f_Y^2 + a^2 u_0^2 + v_0^2 \end{vmatrix} \qquad a \triangleq \frac{f_X}{f_Y}$$

→ only four equations are needed


h1
h3
h2
h4
V

Calibration: rectified planar face plus vanishing point of the direction normal to the face

$$h_1^T \omega h_2 = 0$$
$$h_3^T \omega h_4 = 0$$
$$v^T \omega h_1 = 0$$
$$v^T \omega h_2 = 0$$

3° and 4° equations are linearly independent,

but there are no further ones (why?)

- → solve for ω
- → find $\mathbf{K}$ (by Cholesky factorisation of $\omega^{-1} = \mathbf{K}\mathbf{K}^{\mathrm{T}}$)

Calibration from rectified face plus orthogonal vanishing point

direct method:

from

- the reconstructing homography H_R from given img to rectified img
- the image of the line at the infinity $\boldsymbol{l}'_{\infty}$ of the planar face
- the vanishing point $\mathbf{v}$ along the direction orthogonal to the face

Calibration from rectified face plus orthogonal vanishing point

- From plane-to-image homography $\mathbf{H} = [h_1 \quad h_2 \quad h_3]$ inverse of the rectifying homography $\mathbf{H}_R$: $\mathbf{H} = \mathbf{H}_R^{-1}$

→

$$h_1{}^T \omega h_2 = 0$$
$$h_1{}^T \omega h_1 - h_2{}^T \omega h_2 = 0$$

(Zhang constraints)

- from $\mathbf{v}_1^T \omega \mathbf{v} = 0$ and $\mathbf{v}_2^T \omega \mathbf{v} = 0$, and $\boldsymbol{l}'_\infty = \mathbf{v}_1 \times \mathbf{v}_2$ →

$$\boldsymbol{l}'_\infty = \omega \mathbf{v} \text{ (2 eqns)}$$

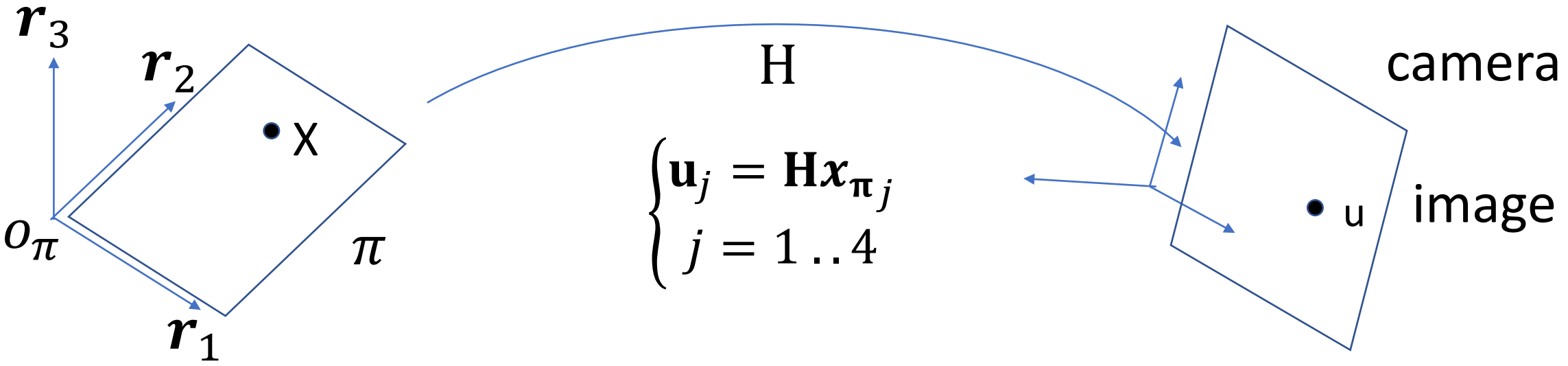
- solve for ω
- find $\mathbf{K}$ (by Cholesky factorisation of $\omega^{-1} = \mathbf{K}\mathbf{K}^{\mathrm{T}}$)

Localization

of a known planar shape
from a calibrated image

Localization of a known planar shape from a calibrated image

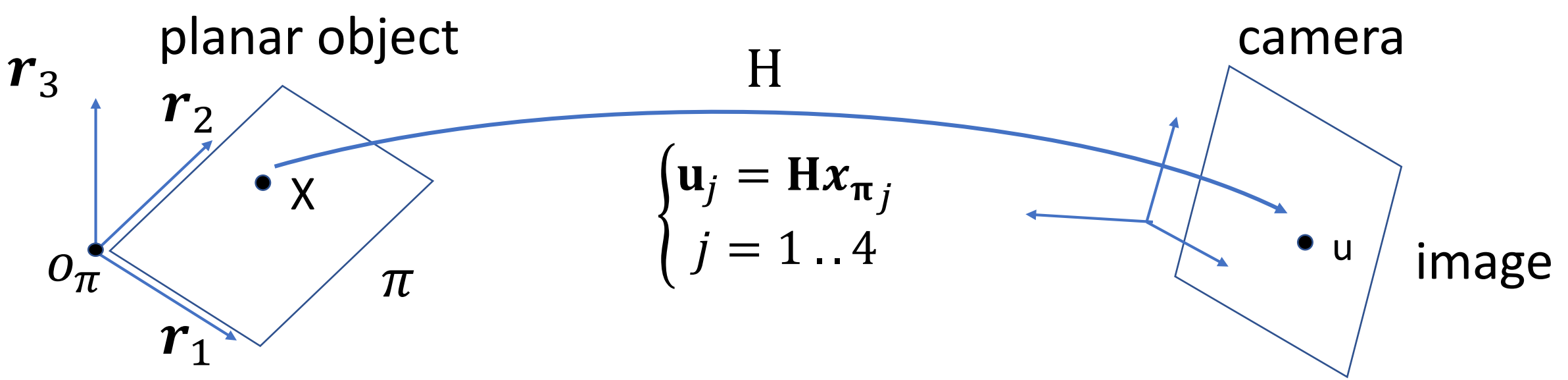
Find homography H between the plane π and its image by DLT



plane π reference: relative pose plane wrt camera → rototranslation $\mathbf{R}_\pi$, $\boldsymbol{o}_\pi$

world reference ≡ camera reference

$$\mathrm{P} = [\mathbf{K} \quad \mathbf{0}]$$



$$\begin{cases} \mathbf{u}_j = \mathbf{H}\boldsymbol{x}_{\boldsymbol{\pi}\,j} \\ j = 1\,..\,4 \end{cases}$$

plane reference

world reference ≡ camera reference → $\mathbf{P} = [\mathbf{K} \quad \mathbf{0}]$

$$\mathbf{X}_{\boldsymbol{\pi}} = \begin{bmatrix} x \\ y \\ 0 \\ w \end{bmatrix} \qquad \mathbf{X}_{\mathbf{w}} = \begin{bmatrix} \boldsymbol{r}_1 & \boldsymbol{r}_2 & \boldsymbol{r}_3 & \mathbf{o}_\pi \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 0 \\ w \end{bmatrix} = \begin{bmatrix} \boldsymbol{r}_1 & \boldsymbol{r}_2 & \mathbf{o}_\pi \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix} \rightarrow \boldsymbol{x}_\pi$$

$$\mathbf{u} = \mathbf{P}\mathbf{X}_{\mathbf{w}} = [\mathbf{K} \quad \mathbf{0}] \begin{bmatrix} \boldsymbol{r}_1 & \boldsymbol{r}_2 & \mathbf{o}_\pi \\ 0 & 0 & 1 \end{bmatrix} \boldsymbol{x}_{\boldsymbol{\pi}} = \mathbf{K}[\boldsymbol{r}_1 \quad \boldsymbol{r}_2 \quad \mathbf{o}_\pi]\boldsymbol{x}_{\boldsymbol{\pi}}$$

homography $\mathbf{H} = \mathbf{K}[\boldsymbol{r}_1 \quad \boldsymbol{r}_2 \quad \mathbf{o}_\pi]$ → obj. pose wrt camera $[\boldsymbol{r}_1 \quad \boldsymbol{r}_2 \quad \mathbf{o}_\pi] = \mathbf{K}^{-1}\mathbf{H}$

homography $\mathbf{H} = \mathbf{K}[\boldsymbol{r}_1 \quad \boldsymbol{r}_2 \quad \mathbf{o}_\pi]$ → obj. pose wrt camera $[\boldsymbol{r}_1 \quad \boldsymbol{r}_2 \quad \mathbf{o}_\pi] = \mathbf{K}^{-1}\mathrm{H}$

and

$$\boldsymbol{r}_3 = \boldsymbol{r}_1 \times \boldsymbol{r}_2$$

NOTE: if the coordinates of the points $\boldsymbol{x}_{\boldsymbol{\pi}_j}$ on the plane are known modulo scale, then the translation $\mathbf{o}_\pi$ is only known modulo scale
(i.e., its direction is known but its module is unknown)

→

- Shape and image → orientation and viewing direction (5 dof)
- Shape + size and image → position and orientation (6 dof)

↑

Localization of a planar set of points

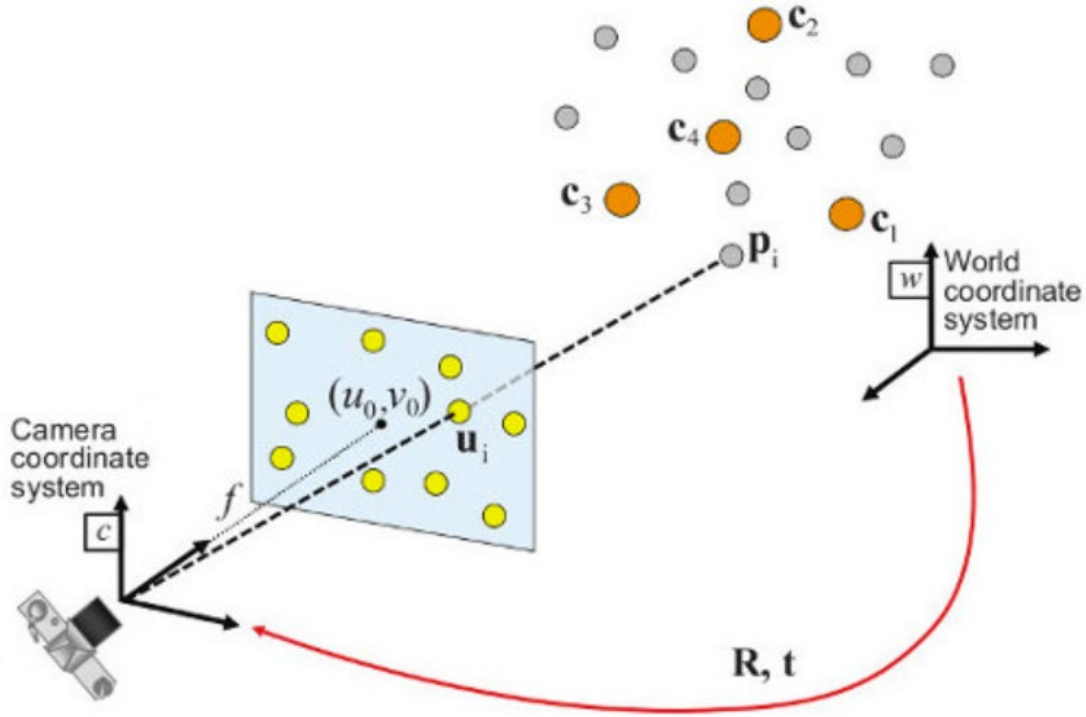
Localization of a nonplanar set of points

↓

Localization of 3D objects: the Perspective n points (PnP) problem

Given the images of N points of known 3D coordinates, find camera position and orientation $[\mathbf{R} \quad \mathbf{t}]$ from $\mathbf{P}\mathbf{X}_i = \mathbf{u}_i$

$$\mathbf{P}\mathbf{X}_i = [\mathbf{K}\mathbf{R} \quad \mathbf{K}\mathbf{t}]\mathbf{X}_i = \mathbf{K}[\mathbf{R} \quad \mathbf{t}]\mathbf{X}_i = \mathbf{u}_i$$



If camera is calibrated (**K** known)

$$[\mathbf{R} \quad \mathbf{t}]\mathbf{X}_i = \mathbf{K}^{-1}\mathbf{u}_i \,, i = 1..n$$

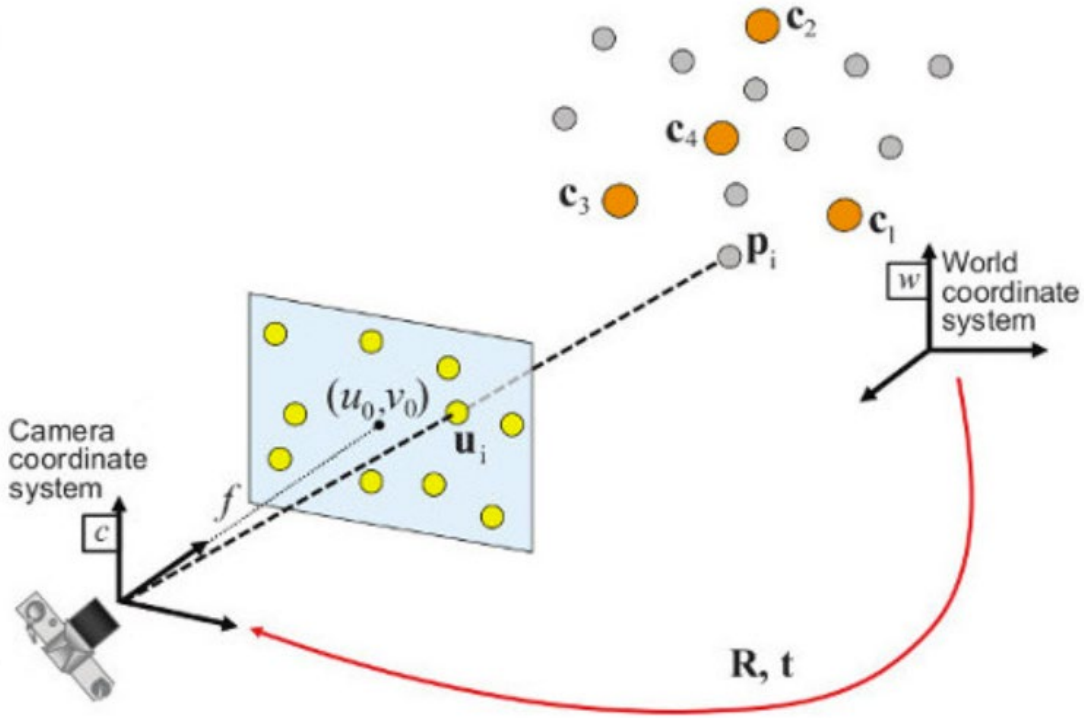
$\mathbf{R} \quad \mathbf{t}$ 6 dof provided that orthogonality constraints for $\mathbf{R}$ (which are nonlinear) are included; each equality yields 2 equations → $\mathrm{n} \geq 3$

PnP is included in main libraries, such as, e,g, OpenCV

Localization of 3D objects: the Perspective n points (PnP) problem

Given the images of N points of known 3D coordinates, find camera position and orientation $[\mathbf{R} \quad \mathbf{t}]$ from $\mathbf{PX}_i = \mathbf{u}_i$

$$\mathbf{PX}_i = [\mathbf{KR} \quad \mathbf{Kt}]\mathbf{X}_i = \mathbf{K}[\mathbf{R} \quad \mathbf{t}]\mathbf{X}_i = \mathbf{u}_i$$



If camera is uncalibrated (**K** unknown)

$$\mathbf{PX}_i = \mathbf{u}_i \,, i \;=\; 1 \,..\, n$$

P has 11 dof; each equality yields 2 linear equations → $\mathrm{n} \geq 6$

Use DLT

Rectification of a single *calibrated* image from vanishing points *(Scenario n.2)*

Shape reconstruction of a planar face given a calibrated image from vanishing points



From vanishing points to vanishing line l'_∞

$\mathbf{v}_1 = l_1 \times l_2$

$l'_\infty = \mathbf{v}_1 \times \mathbf{v}_2$

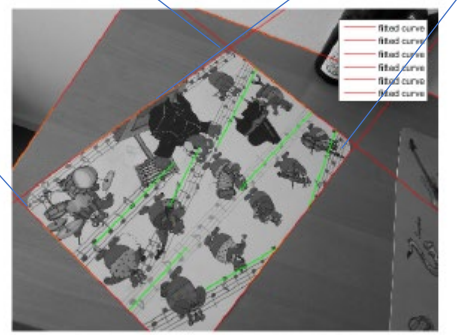
$\mathbf{v}_2 = l_3 \times l_4$

l_1

l_2

l_3

l_4



Unknown shape, known image, known K (calibrated camera), known vanishing points: find $\mathbf{H}_R$ image→plane

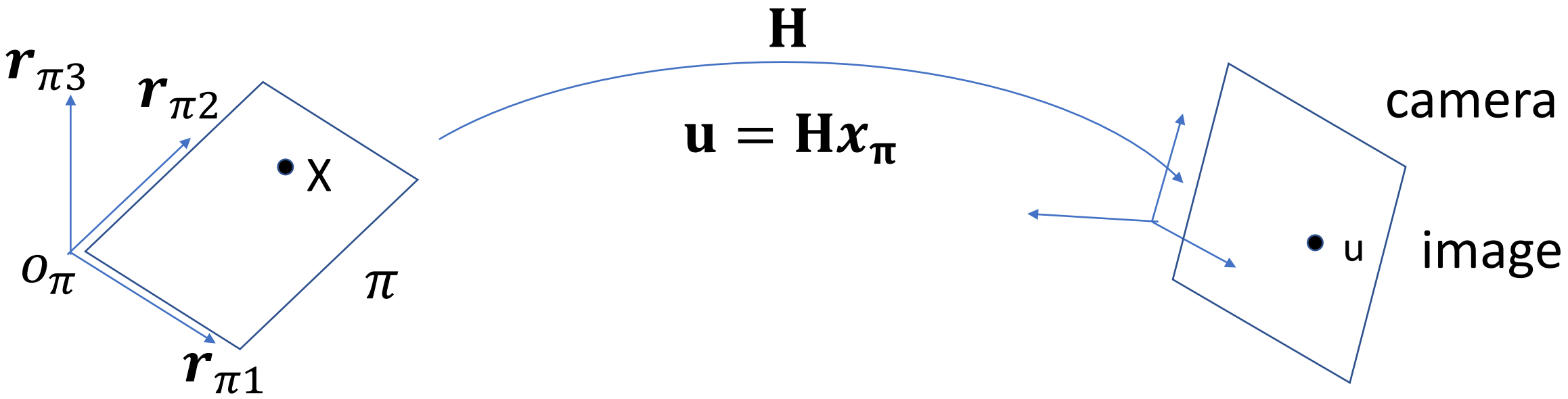


$\mathbf{H}_R$?



Solution (shape): $\mathbf{H}_R = \mathbf{H}^{-1}$ where $\mathbf{H}$ is the plane-to-image Homography (modulo reference frame on the plane and scale factor)

remember:
homography between a plane π and its image



relative pose plane wrt camera →rototranslation $\mathbf{R}_\pi$, $\boldsymbol{o}_\pi$

Goal: find plane-to-image homography $\mathbf{H} = \mathbf{K}[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_\pi]$

Reconstructing homography $\mathbf{H}_R = \mathbf{H}^{-1}$

plane-to-image homography $\mathbf{H} = \mathbf{K}[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_{\pi}]$:

$\mathbf{H}$ is related to $\mathbf{K}$ and to the relative pose of plane π wrt camera

→

free to choose a «comfortable» reference frame on the plane π:
the only constrained element is the direction $\mathbf{n}_{\pi}$ normal to plane π
$\mathbf{n}_{\pi} = \mathbf{K}^T l'_{\infty}$ (*for proof see next slide, with* $\mathbf{M} = \mathbf{K}$)

→

Choose any $\boldsymbol{r}_{\pi 1} \perp \boldsymbol{r}_{\pi 2}$ both orthogonal to $\mathbf{n}_{\pi}$ and normalize all of them;
then, take $\mathbf{o}_{\pi} = \mathbf{n}_{\pi}$ → in this way $[\boldsymbol{r}_{\pi 1} \quad \boldsymbol{r}_{\pi 2} \quad \mathbf{o}_{\pi}] = \mathbf{R}_{\pi}$

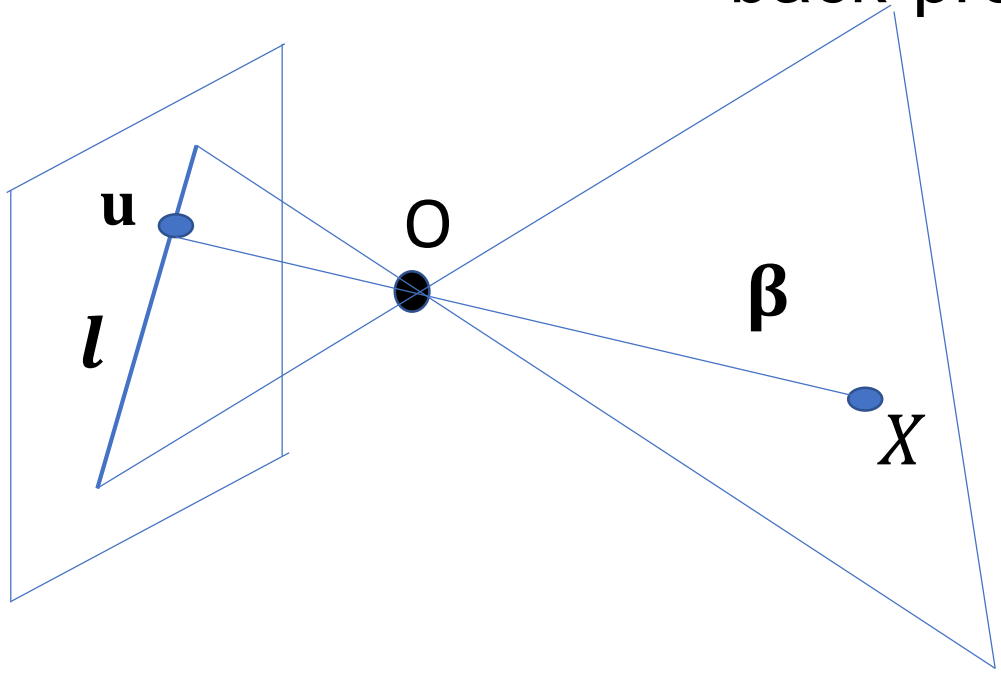
→

Solution: $\mathbf{H}_R = \mathbf{H}^{-1} = (\mathbf{K}\mathbf{R}_{\pi})^{-1} = \mathbf{R}_{\pi}{}^T \mathbf{K}^{-1}$

DIGRESSION: Back-projection of an image line

set of space points X, whose image projection $\mathbf{u} = \mathbf{P}\mathrm{X}$ is on image line $\boldsymbol{l}$

back-projection



$$l^{\mathrm{T}}\mathbf{u} = l^{\mathrm{T}}\mathbf{P}\mathrm{X} = 0$$

$$\boldsymbol{\beta}^{T}\mathrm{X} = 0$$

back-projection of $\boldsymbol{l}$: plane $\boldsymbol{\beta} = \mathbf{P}^{\mathbf{T}}\boldsymbol{l}$ through O, in fact, since $\mathrm{O} = RNS(\mathbf{P})$

$$\boldsymbol{\beta}^{T}\mathrm{O} = \boldsymbol{l}^{\mathrm{T}}\mathbf{P}\,\mathrm{O} = \boldsymbol{l}^{\mathrm{T}}0 = 0$$

back-projection of $\boldsymbol{l}$: plane $\boldsymbol{\pi} = \mathbf{P}^{\mathbf{T}}\boldsymbol{l} = \begin{bmatrix}\mathbf{M}^{T} \\ \boldsymbol{m}^{T}\end{bmatrix}\boldsymbol{l}$ → normal: $\mathbf{n}_{\beta} = \mathbf{M}^{T}\boldsymbol{l}$

calibrated rectification from vanishing points



$$\mathbf{H}_R = \mathbf{R}_\pi{}^T \mathbf{K}^{-1}$$



rectified image

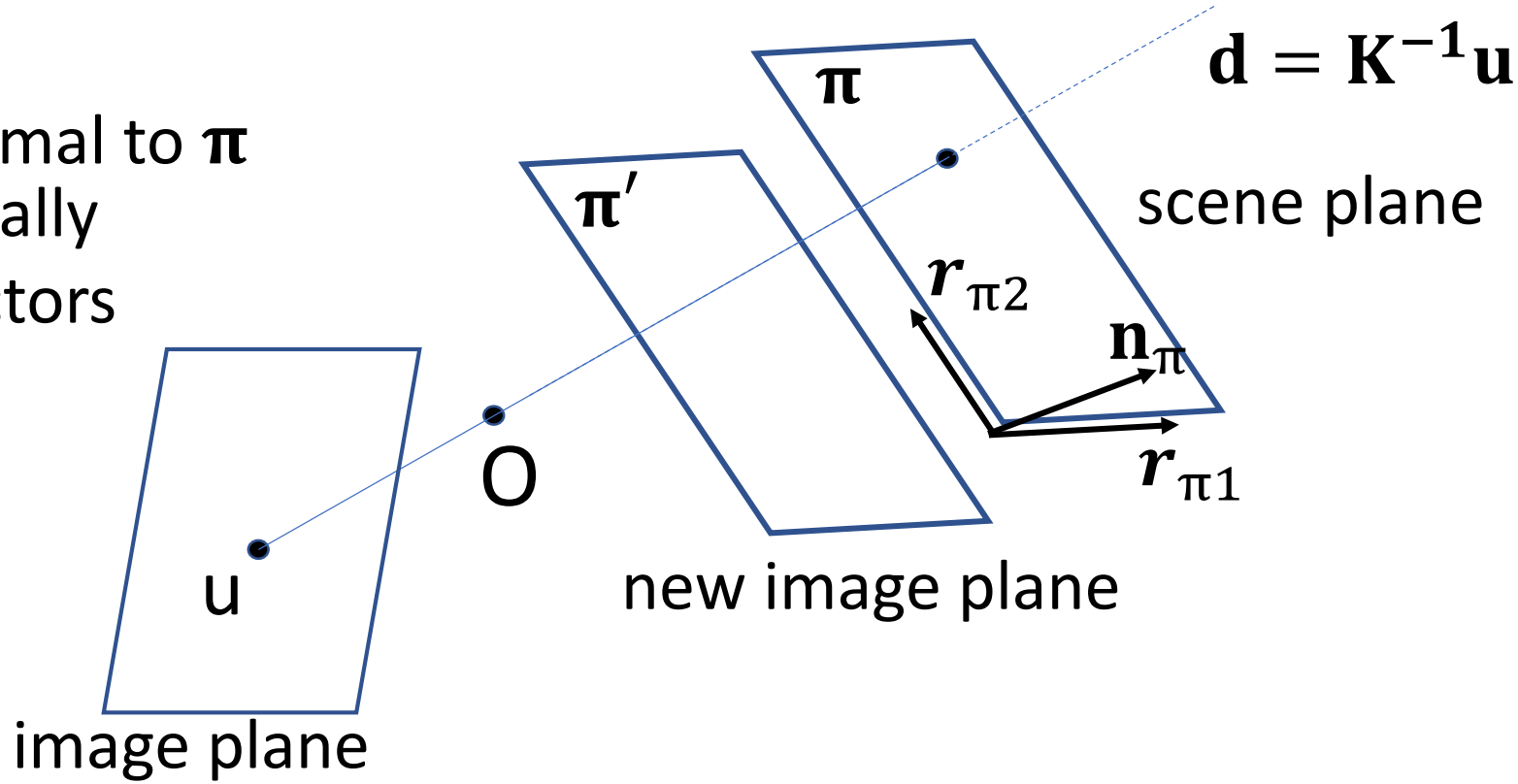
Rectification of a single calibrated image from vanishing points

Geometric argument

Rectification (reconstruction):

reprojection of the image onto a new image plane $\boldsymbol{\pi}'$
parallel to the scene plane $\boldsymbol{\pi}$ by a **natural** camera:
Scene plane || image plane and natural camera
→ new (rectified) image **similar** to planar scene

$\mathbf{n}_\pi$: unit vector normal to $\boldsymbol{\pi}$
$\boldsymbol{r}_{\pi 1}$ and $\boldsymbol{r}_{\pi 2}$: mutually orthogonal unit vectors orthogonal to $\mathbf{n}_\pi$



Rectification (reconstruction):

reprojection of the image onto a new image plane $\boldsymbol{\pi}'$

parallel to the scene plane $\boldsymbol{\pi}$ by a **natural** camera:

Scene plane || image plane AND natural camera

→ new (rectified) image **similar** to planar scene

coord.s of the image on the new plane:

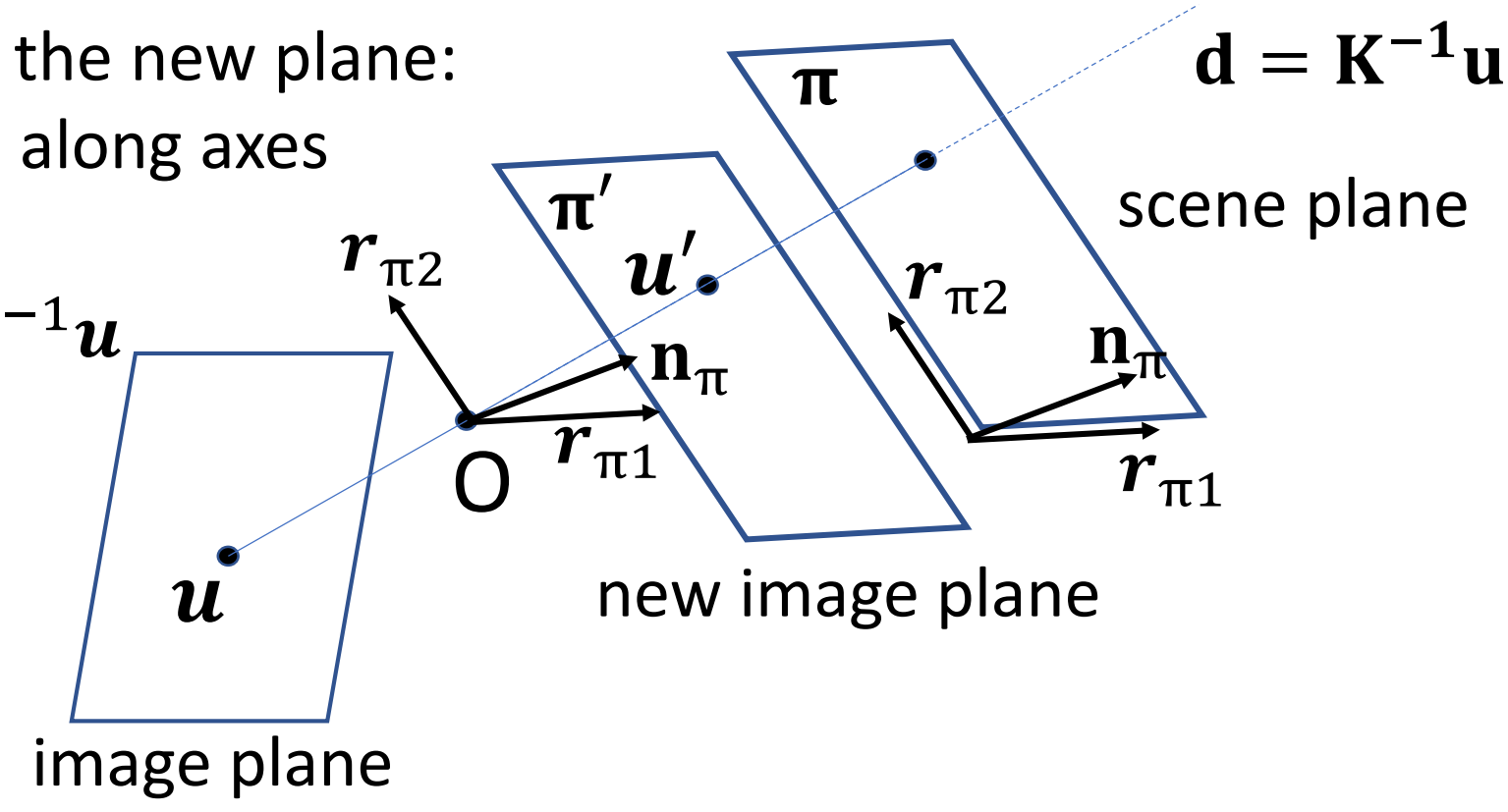
components of vector **d** along axes

$$\boldsymbol{u}' = \begin{bmatrix} \boldsymbol{r}_{\pi 1}{}^T \mathbf{d} \\ \boldsymbol{r}_{\pi 2}{}^T \mathbf{d} \\ \boldsymbol{n}_{\pi}{}^T \mathbf{d} \end{bmatrix} = \mathbf{R}_{\pi}{}^T \mathbf{K}^{-1} \boldsymbol{u}$$

→

$$\mathbf{H}_R = \mathbf{R}_{\pi}{}^T \mathbf{K}^{-1}$$

$$\mathbf{d} = \mathbf{K}^{-1}\mathbf{u}$$



HOMEWORK EXAMPLE (2024-25)

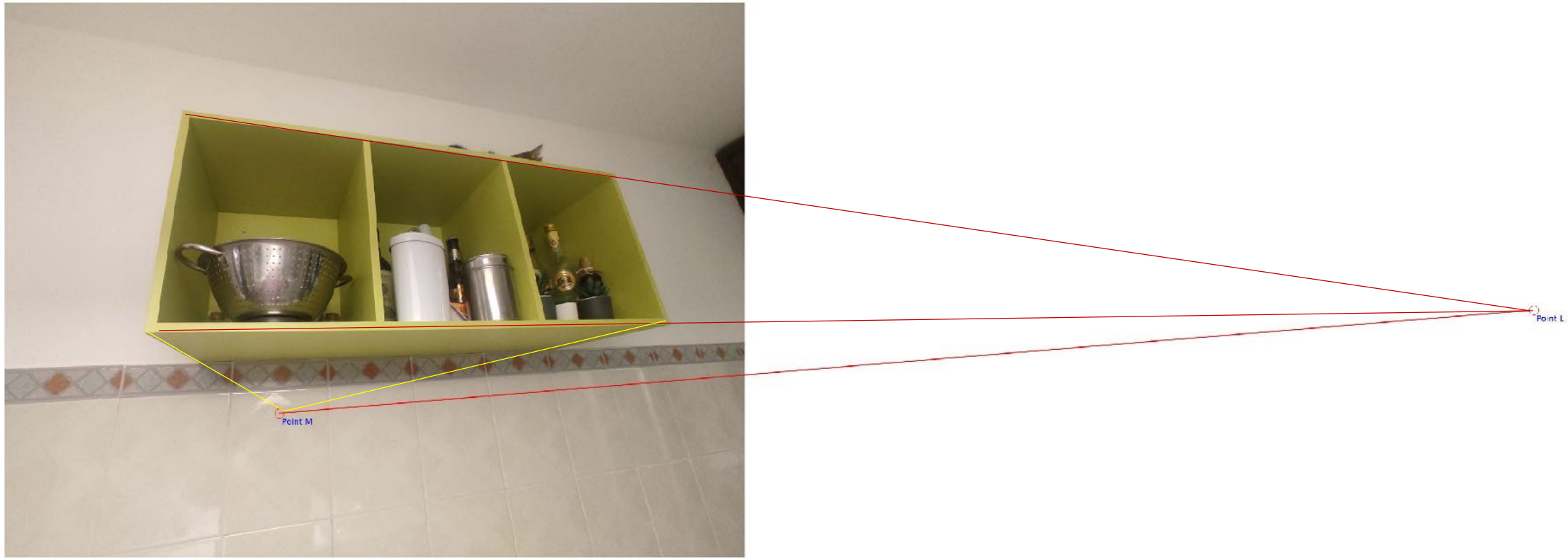


Given image

A horizontal circumference



1. Vanishing line l'_∞ : line joining vanishing points



2. Euclidean rectification of the horizontal face(s): direct method

- Intersect vanishing line l'_∞ and image C of horizontal circumference → $\boldsymbol{I'}$ and $\boldsymbol{J'}$
- Image of the circular points → image of the conic dual to the circular points

$$C^{*\prime}_\infty = \boldsymbol{I'J'^T} + \boldsymbol{J'I'^T}$$

- Singular value decomposition

$$\text{svd}(C^{*\prime}_\infty) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{SR}^{-1} C^*_\infty H_{SR}^{-T}$$

- Rectifying transformation (from svd output U)

$$H_{SR} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

- Euclidean reconstructed model $M_S = H_{SR} * \text{Image}$

2. reconstruction of the horizontal faces: stratified method

2.1 Affine rectification

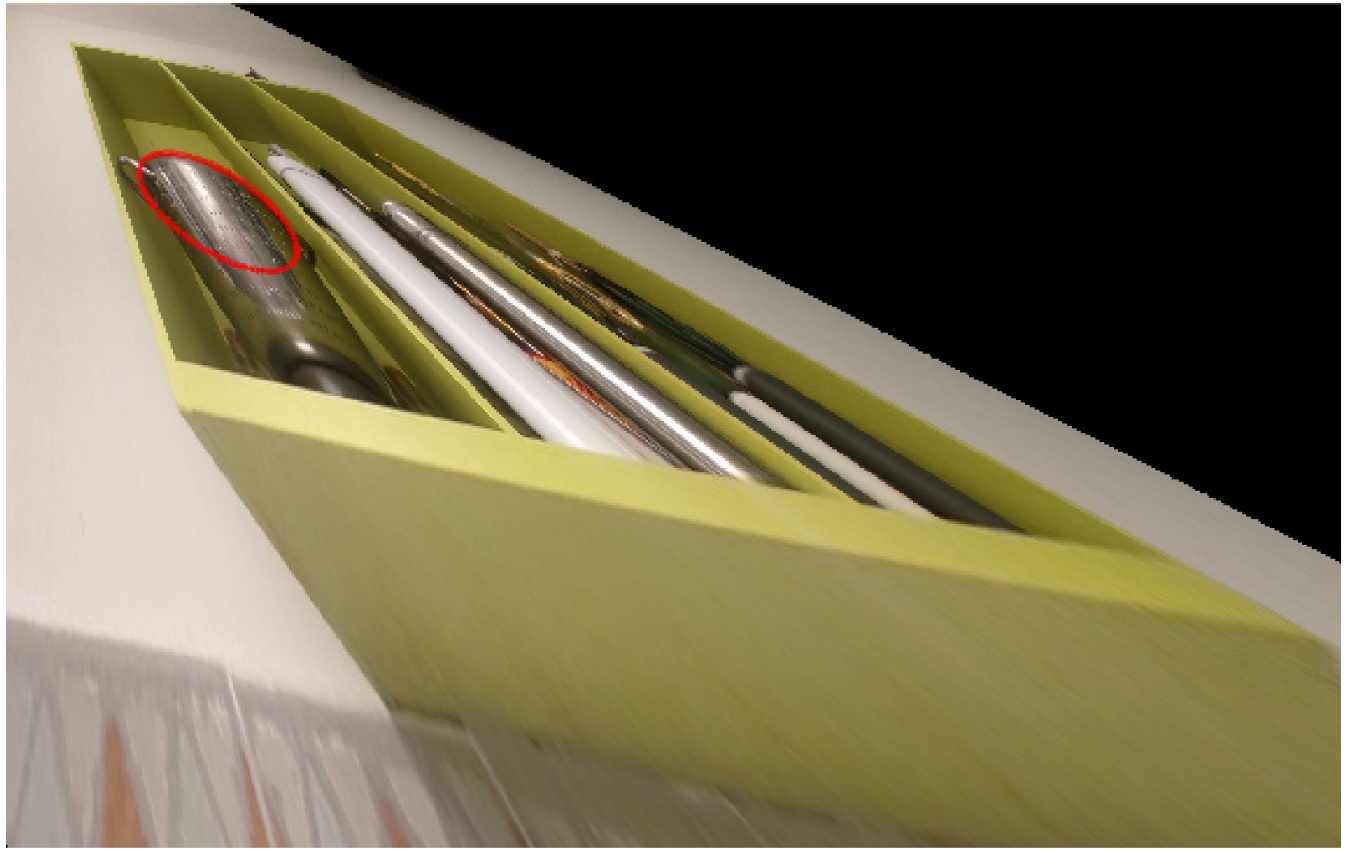
using the vanishing line

2.2 Euclidean rectification applied to affinely rectified image

using the image of the horizontal circumference

2.1. Affine reconstruction of horizontal face

$$H_a = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ & l'^T_\infty & \end{bmatrix}$$

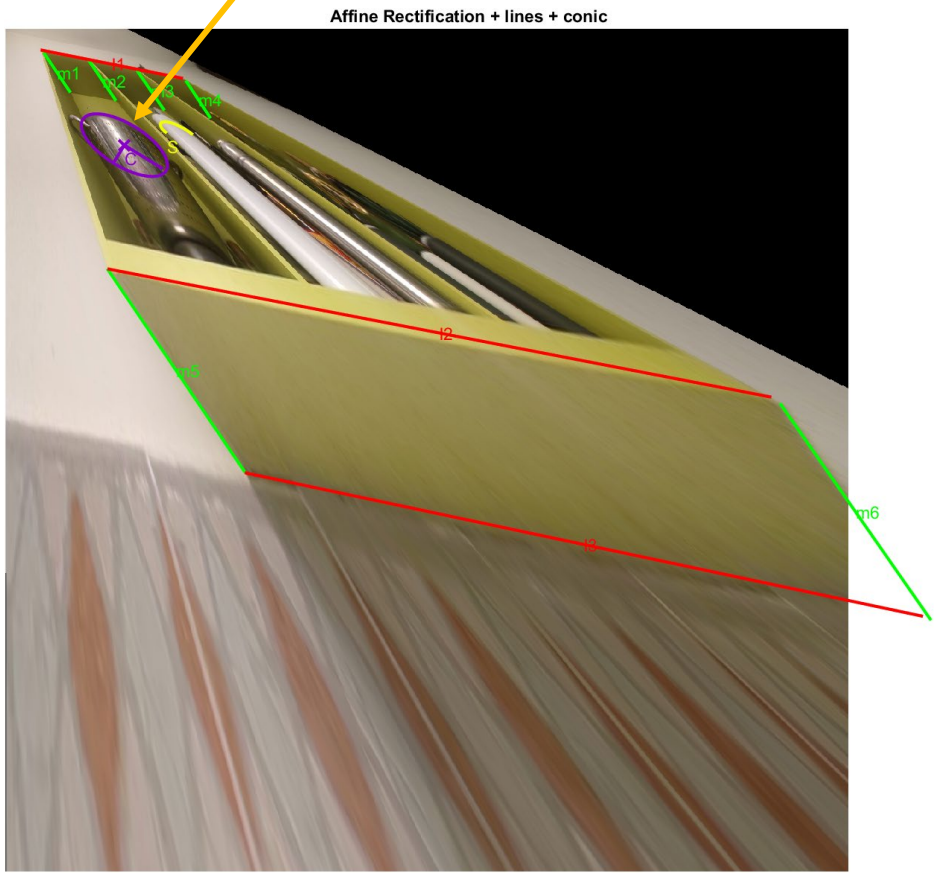


affine reconstruction:
mapping applied to whole image
but only horizontal faces are affinely rectified

After affine rectification:
2.2. Euclidean Reconstruction using the conic C

$$C_{aff} = |H_a^{-1T} * C * H_a^{-1}$$

is still an ellipse: however, its real shape is **circular**



Affine mapping from ellipse to circumference

$$R = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} \quad S = \begin{bmatrix} 1 & 0 \\ 0 & \frac{a_x}{a_y} \end{bmatrix}$$

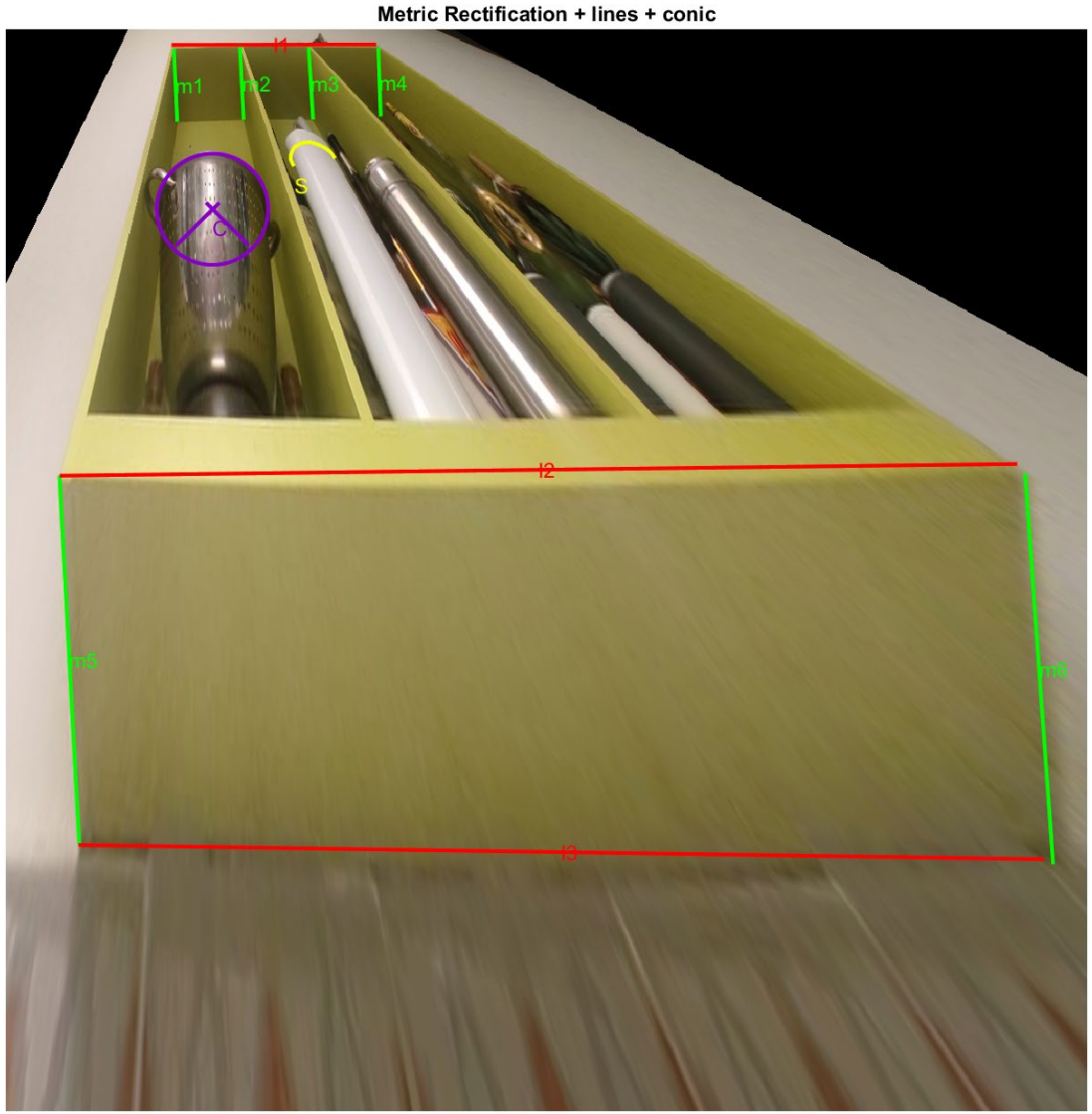
$$G = R * S * R^T$$

rotate according to axes, map to circumference, and rotate back

$$H_r = \begin{bmatrix} G_{11} & G_{12} & 0 \\ G_{21} & G_{22} & 0 \\ 0 & 0 & 1 \end{bmatrix} * H_a$$

complete rectification

Rectified horizontal faces



observe distortion

3. Intrinsic calibration

From plane-to-image homography $\mathbf{H} = [h_1 \quad h_2 \quad h_3]$ i.e., the inverse of the rectifying homography $\mathbf{H}_r$: $\mathbf{H} = \mathbf{H}_r^{-1}$

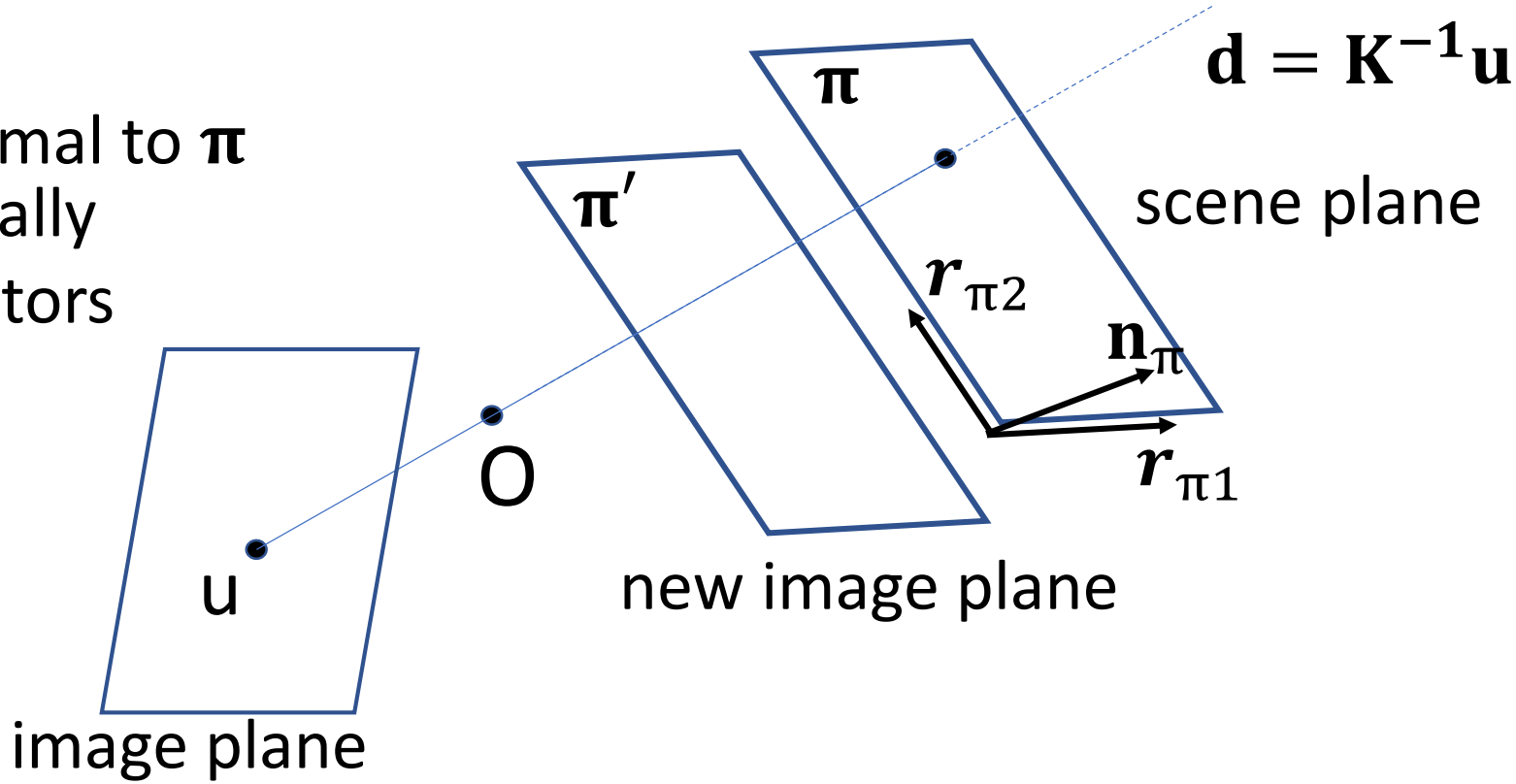
$$\begin{cases} h_1^T \omega h_2 = 0 \\ h_1^T \omega h_1 - h_2^T \omega h_2 = 0 \\ v_1^T \omega v_h = 0 \\ v_2^T \omega v_h = 0 \end{cases}$$

$$K = (\mathrm{chol}(\omega))^{-1} \longrightarrow K = \begin{bmatrix} 753.0127 & 0 & 805.7685 \\ 0 & 776.6877 & 557.2345 \\ 0 & 0 & 1 \end{bmatrix}$$

4. Rectification of the front face

reprojection of the image onto a new image plane $\boldsymbol{\pi}'$
parallel to the scene plane $\boldsymbol{\pi}$ by a **natural** camera:
Scene plane || image plane and natural camera
→ new (rectified) image **similar** to planar scene

$\mathbf{n}_\pi$: unit vector normal to $\boldsymbol{\pi}$
$\boldsymbol{r}_{\pi 1}$ and $\boldsymbol{r}_{\pi 2}$: mutually orthogonal unit vectors orthogonal to $\mathbf{n}_\pi$



4. Rectification of the front face

reprojection of the image onto a new image plane $\boldsymbol{\pi}'$

parallel to the scene plane $\boldsymbol{\pi}$ by a **natural** camera:

Scene plane || image plane AND natural camera

→ new (rectified) image **similar** to planar scene

coord.s of the image on the new plane:

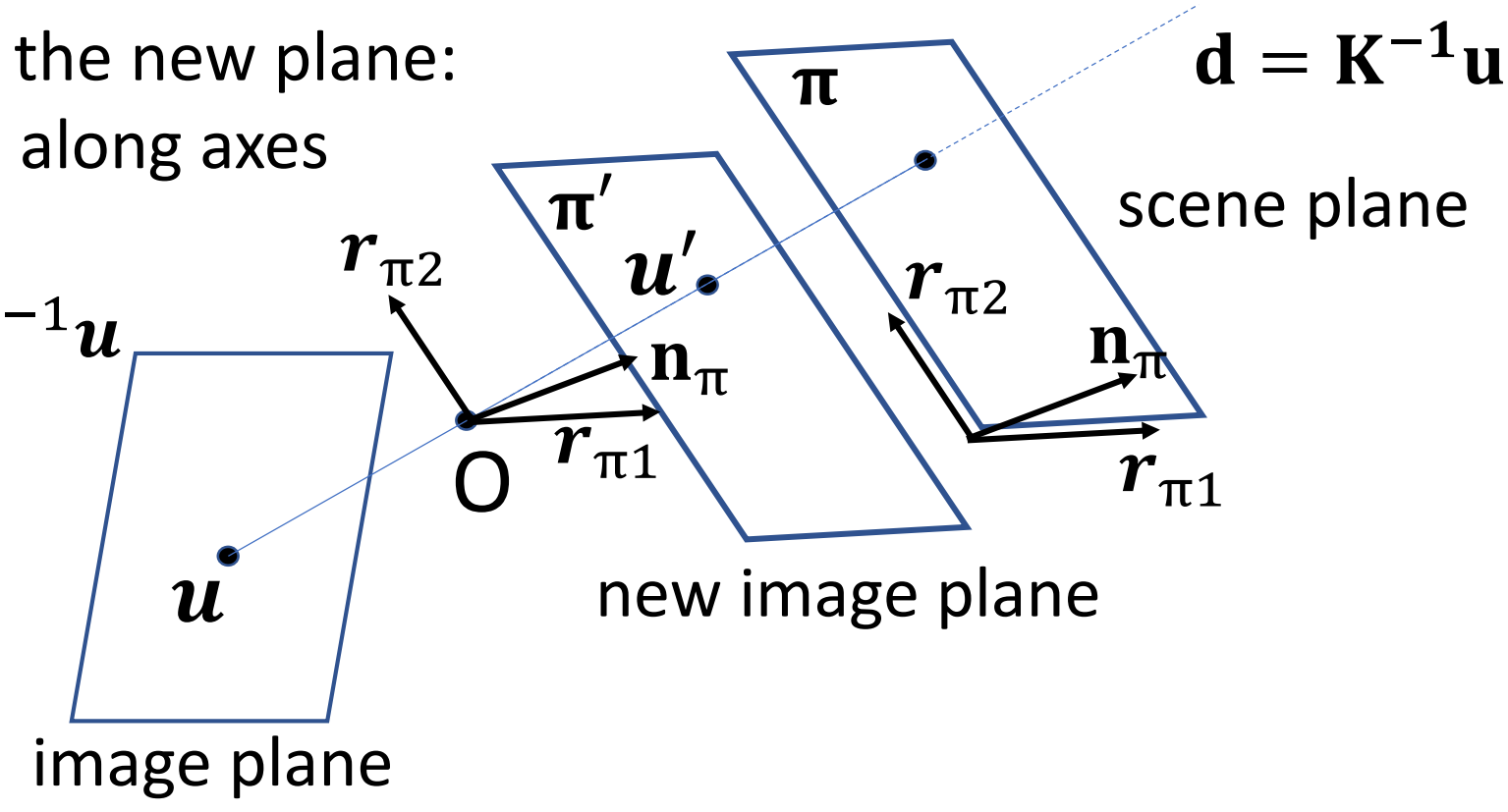
components of vector **d** along axes

$$\boldsymbol{u}' = \begin{bmatrix} \boldsymbol{r}_{\pi 1}{}^T \mathbf{d} \\ \boldsymbol{r}_{\pi 2}{}^T \mathbf{d} \\ \boldsymbol{n}_{\pi}{}^T \mathbf{d} \end{bmatrix} = \mathbf{R}_{\pi}{}^T \mathbf{K}^{-1} \boldsymbol{u}$$

→

$$\mathbf{H}_R = \mathbf{R}_{\pi}{}^T \mathbf{K}^{-1}$$

$$\mathbf{d} = \mathbf{K}^{-1}\mathbf{u}$$

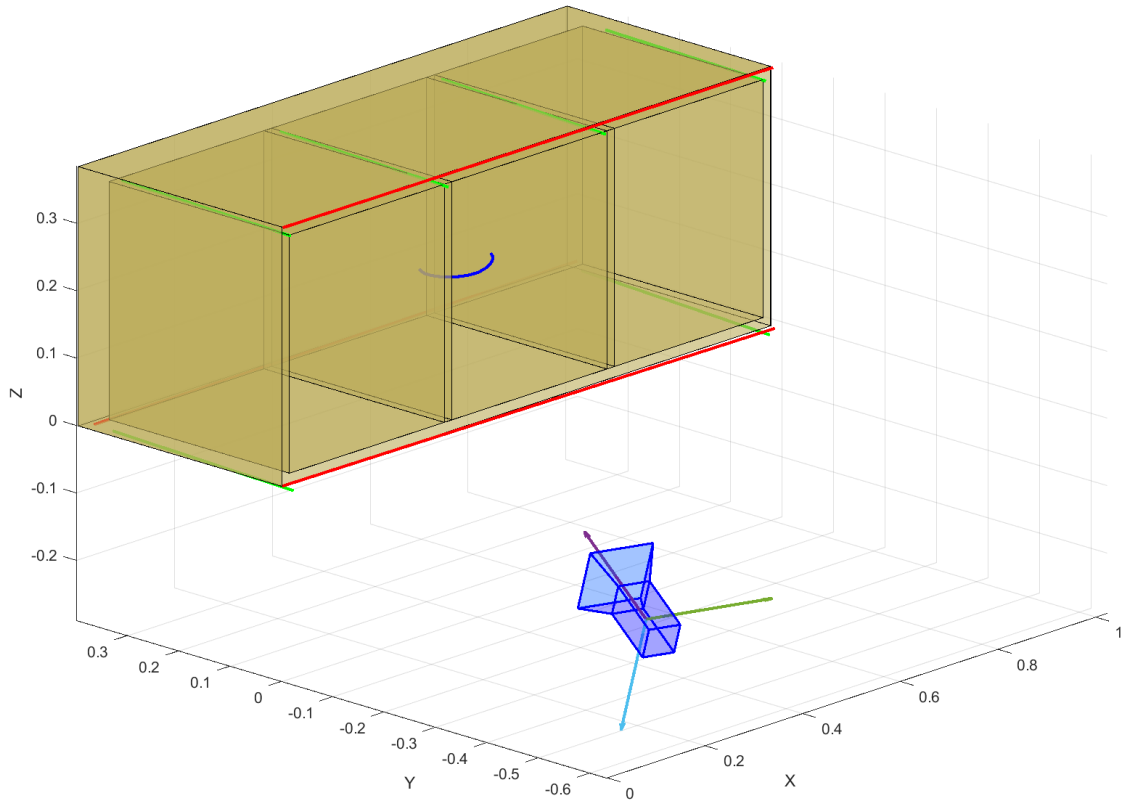


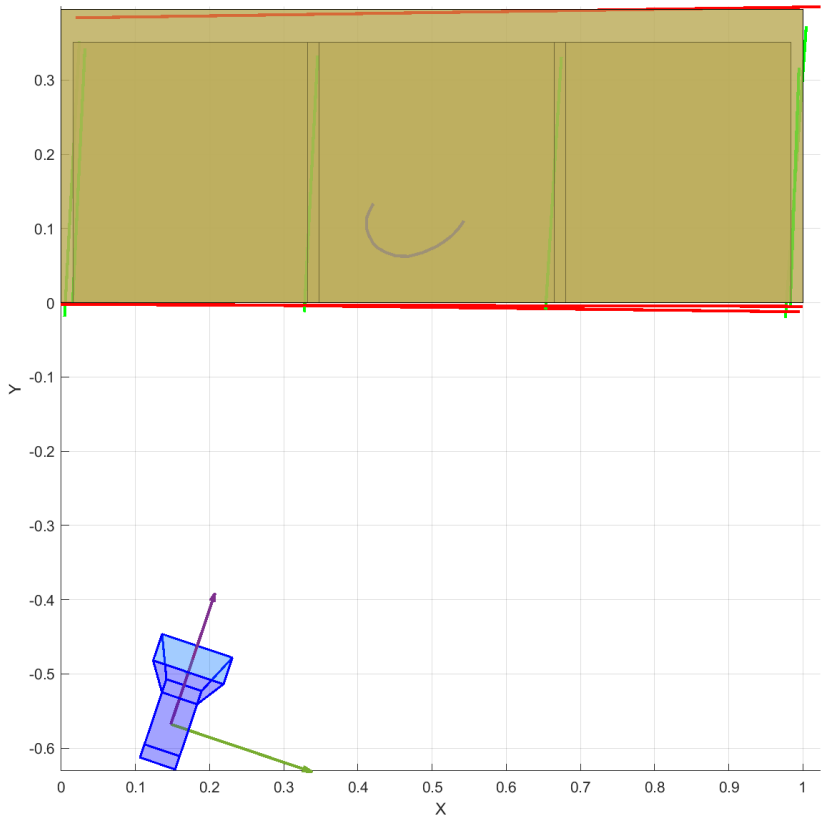
4. Rectification of the front face



5. Camera localization wrt the parallelepiped

$$[\boldsymbol{r}_1 \quad \boldsymbol{r}_2 \quad \mathbf{o}_\pi] = \mathbf{K}^{-1}\mathbf{H} = \mathbf{K}^{-1}\mathbf{H}_R{}^{-1} = \mathbf{K}^{-1}\,\mathbf{K}\mathbf{R}_\pi = \mathbf{R}_\pi$$





APPENDIX 2
The painter’s eye

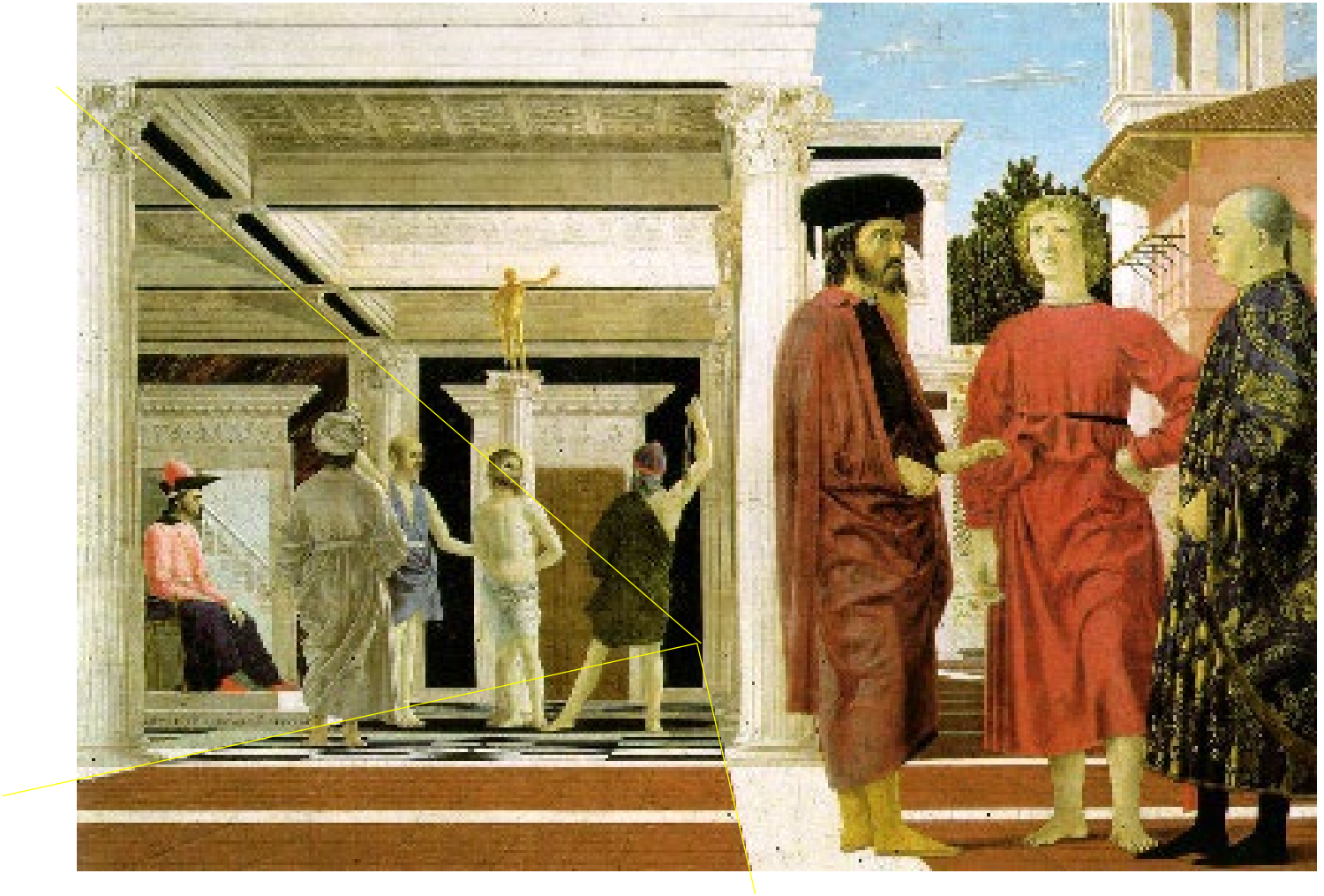
An Example of Camera Calibration

where was the painter’s eye (wrt to the painting)?



Assumptions: - the floor is horizontal
- floor tiles are square

Vanishing point of «x» direction: V_x

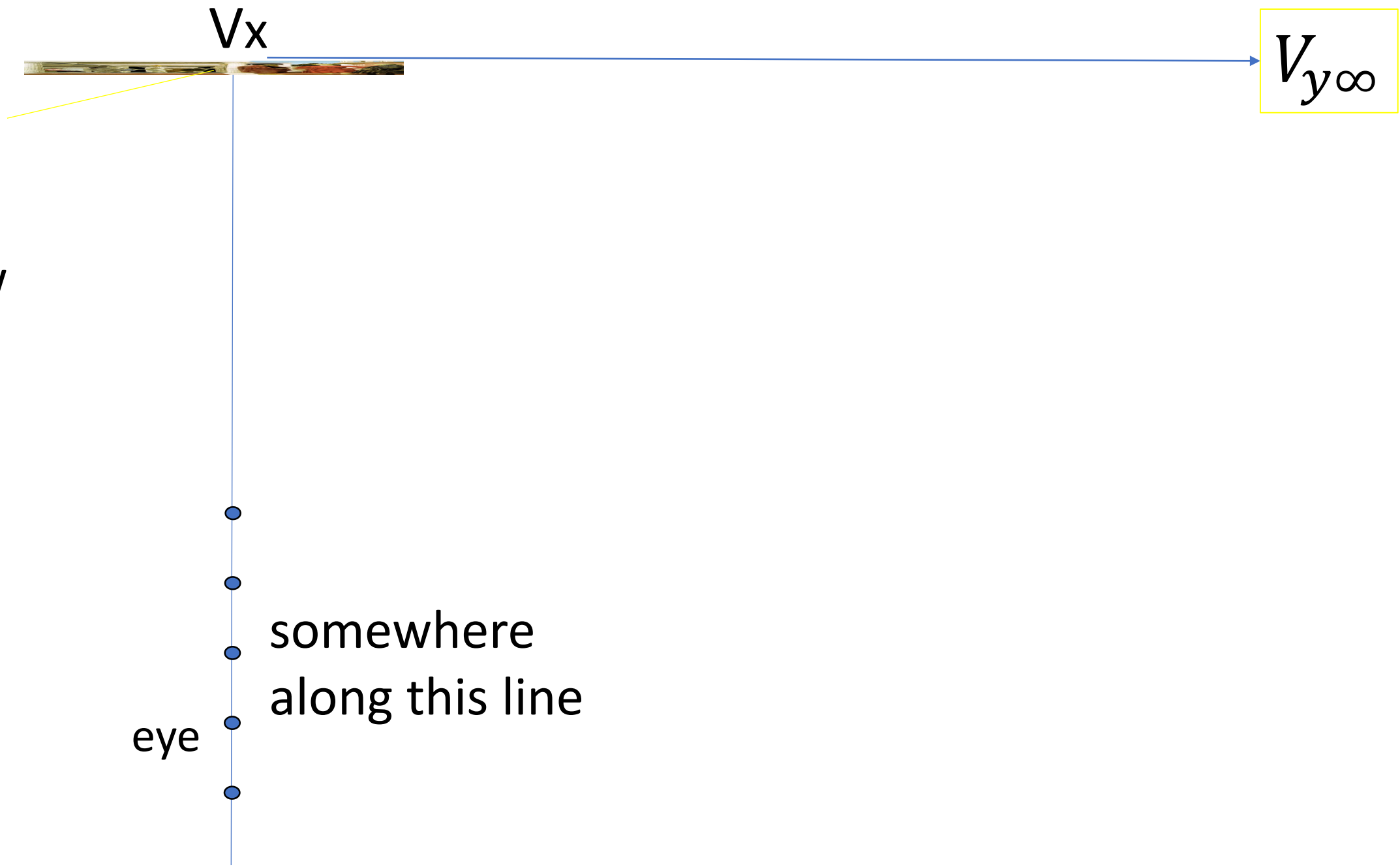


Vanishing point of «y» direction



Theorem: viewing ray associated to the vanishing point of a direction ***d*** is parallel to the direction ***d***

- Floor is horizontal → line (V_x, eye) is horizontal
- Vy is at the infinity → line (eye,$V_{y\infty}$) is parallel to the painting
- Theorem → line ($V_{y\infty}$, eye) is orthogonal to line (V_x, eye)
- → line (eye, V_x) is horizontal and perpendicular to y axis
- … eye is on the normal to the y-axis through the point V_x

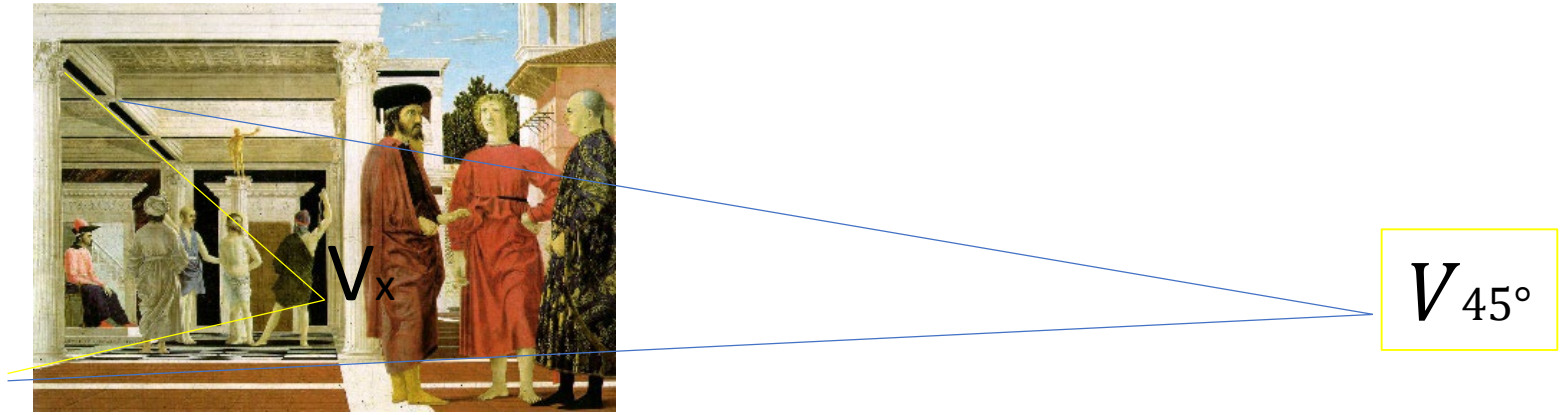

Vx
$V_{y\infty}$
this is a
top view
somewhere
along this line
eye

Theorem: viewing ray associated to the vanishing point of a direction d is parallel to the direction **d**

- Floor is horizontal → line (V_x, eye) is horizontal
- Vy is at the infinity → line (eye,$V_{y\infty}$) is parallel to the painting
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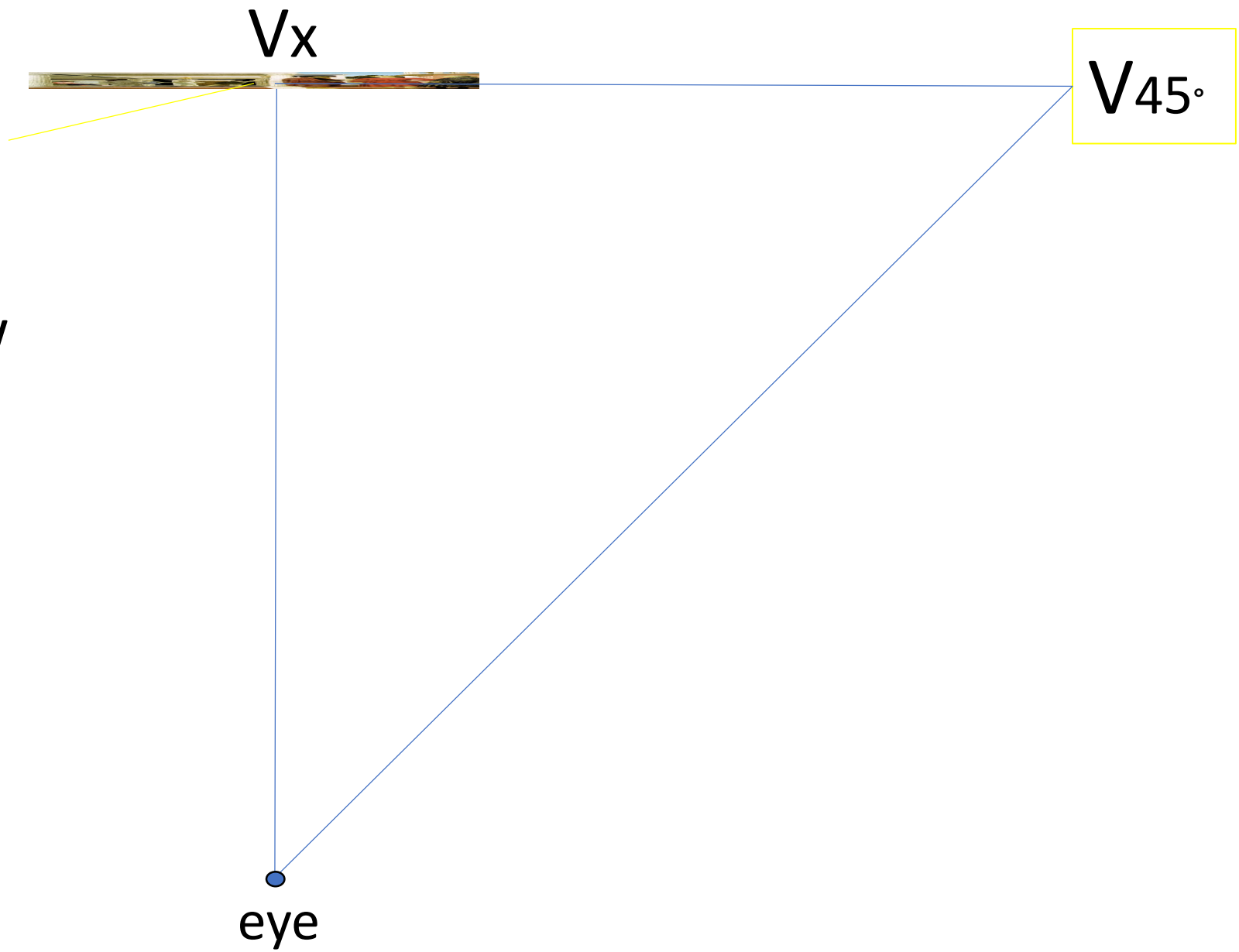
- How far?

Tiles are square → diagonals form a 45° angle with x-direction



Theorem → line (eye, $V_{45\circ}$) forms a 45° angle with line (eye, V_x)
floor horizontal → line (eye, $V_{45\circ}$) is also horizontal

this is a
top view



APPENDIX 3
Camera calibration from two images of an **unknown** planar scene

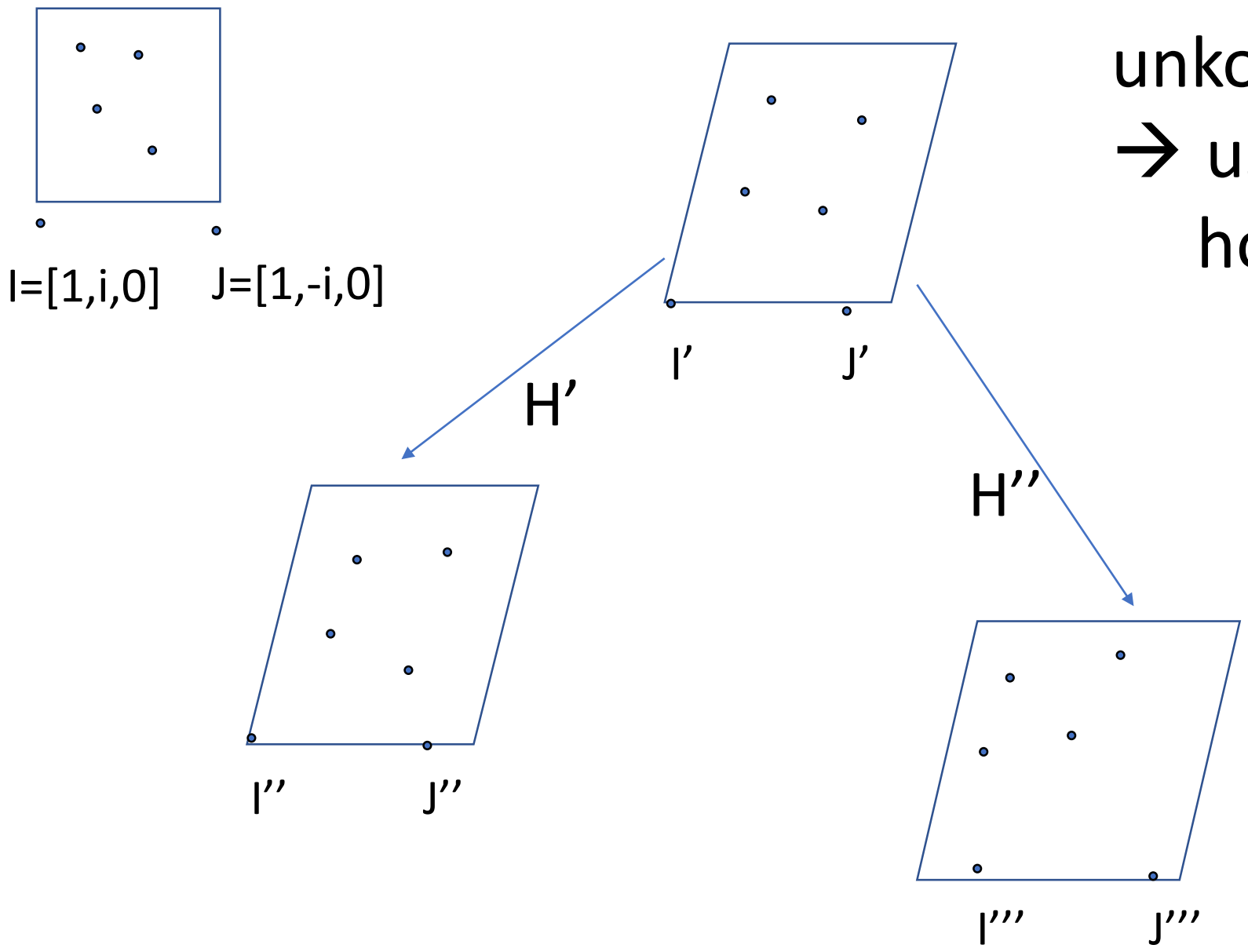
Theoretical result

Circular points of any plane π are self-orthogonal:

$$I_\pi{}^T I_\pi = 0 \text{ and } J_\pi{}^T J_\pi = 0$$

For any imaged circular point is $\boldsymbol{I'_\pi{}^T \omega I'_\pi = 0}$

In fact $I'_\pi = KRI_\pi \rightarrow I_\pi = (KR)^{-1} I'_\pi \rightarrow I_\pi{}^T I_\pi = I'_\pi{}^T (KR)^{-T} (KR)^{-1} I'_\pi = 0 \rightarrow I'_\pi{}^T \omega I'_\pi = 0$



unkown planar scene
→ use image-to-image homographies

$$I'' = H'I'$$

$$I'^{T}\omega I' = 0$$

$$I''^{T}\omega I'' = 0$$

$$I'^{T}H'^{T}\omega H'I' = 0$$

Unknowns: I' and J' and ω → at least 5 images
(each nonlinear eqn leads to 2 constraints: Re and Im part)

Camera calibration from two images of an **unknown** planar scene

- take images of a planar scene with camera with constant $\mathbf{K}$
- estimate image-image homographies
- formulate equations

$$I'^{T}\omega I' = 0$$

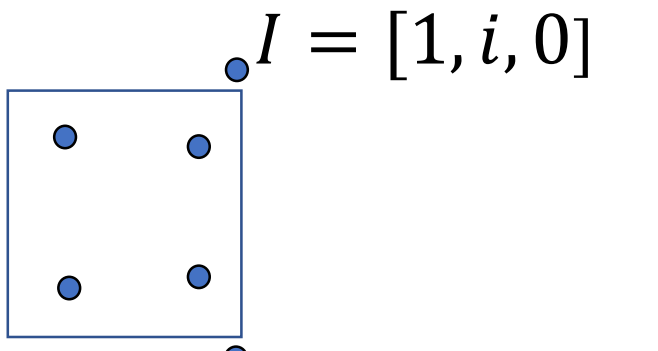
$$I'^{T}H'^{T}\omega H' I' = 0$$

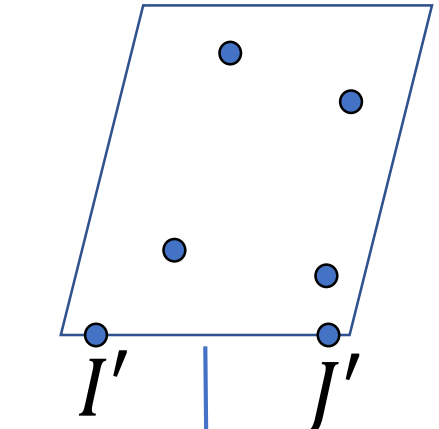
- solve them for ω and I'
- then take the inverse matrix ω^{-1}
- find $\mathbf{K}$ by Cholesky factorisation of $\omega^{-1} = \mathbf{K}\mathbf{K}^{\mathrm{T}}$

USUALLY LESS ACCURATE THAN ZHANG METHOD

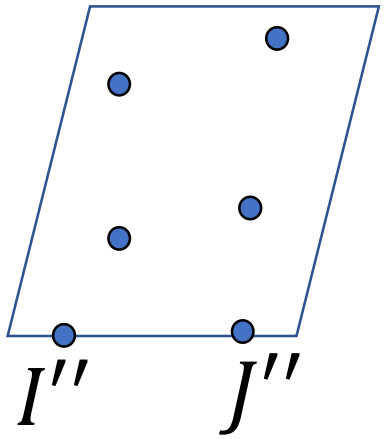
APPENDIX 4

Reconstruction of an **unknown** planar scene from two *calibrated* images (i.e. images acquired by calibrated cameras)





H



reconstruction of an **unkown** planar **scene**, from two **calibrated** images

→ use image-to-image homographies

remember: $\boldsymbol{I'}^T \omega \boldsymbol{I'} = 0$ for any imaged circular point $\boldsymbol{I'}$

$$\boldsymbol{I'}^T \omega \boldsymbol{I'} = 0$$
$$I''^T \omega I'' = 0$$
$$I'' = HI'$$
$$\boldsymbol{I'}^T H^T \omega H \boldsymbol{I'} = 0$$

same camera hypothesis
can easily be relaxed

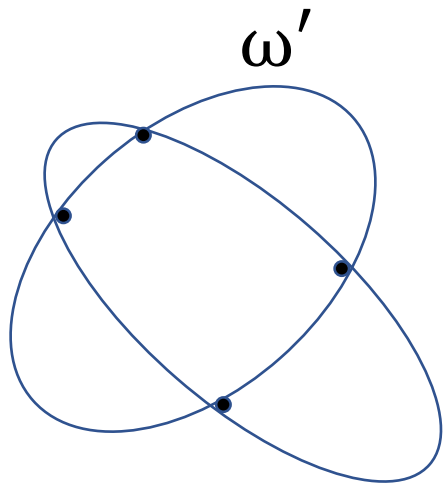
$\omega = \left(\mathbf{K}\mathbf{K}^{\mathrm{T}}\right)^{-1} = \mathbf{K}^{-\mathrm{T}}\mathbf{K}^{-1}$ is known after calibration

just 4 unknowns $\boldsymbol{I'}$ complex coordinates → at least 2 images
(each eqn leads to 2 constraints: Re and Im part)

$$\boldsymbol{I}'^{T}\omega'\boldsymbol{I}' = 0$$
$$\boldsymbol{I}'^{T}H'^{T}\omega''H'\boldsymbol{I}' = 0$$

two different cameras
ω' *and* ω''

Geometrically: intersection of two conics:
ω' and $H'^{T}\omega''H'$
→
two resulting pairs of
imaged circular points
→
selection based on reprojection
or on an additional (third) image



As usual: image rectification from the image $(\boldsymbol{I}', \boldsymbol{J}')$ of the circular points $(\boldsymbol{I}, \boldsymbol{J})$

- Image of the circular points → image of the conic dual to the circular points

$$C_\infty^{*\prime} = \boldsymbol{I}'\boldsymbol{J}'^{\boldsymbol{T}} + \boldsymbol{J}'\boldsymbol{I}'^{\boldsymbol{T}}$$

- Singular value decomposition

$$\text{svd}(C_\infty^{*\prime}) = U \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 0 \end{bmatrix} U^T = H_{SR}^{-1} C_\infty^* H_{SR}^{-T}$$

- Rectifying transformation (from svd output U)

$$H_{SR} = \begin{bmatrix} 1/\sqrt{a} & 0 & 0 \\ 0 & 1/\sqrt{b} & 0 \\ 0 & 0 & 1 \end{bmatrix} U^T$$

- Euclidean reconstructed model $M_S = H_{SR} \ * \text{Image}$

APPENDIX 5
Symmetry: a powerful tool

SYMMETRY

A SYSTEM IS SYMMETRIC WRT A CERTAIN TRANSFORMATION

IF

APPLYING THAT TRANSFORMATION TO THE SYSTEM,

NO MODIFICATION CAN BE MEASURED

Symmetry on the space (symmetry in 3D)

EXAMPLES

- EMPTY SPACE: is symmetric under any motion

IF WE ADD AN ELEMENT TO A SYMMETRIC SYSTEM, SYMMETRY CAN POSSIBLY STILL BE ENJOYED, BUT UNDER A RESTRICED SET OF TRANSFORMATIONS

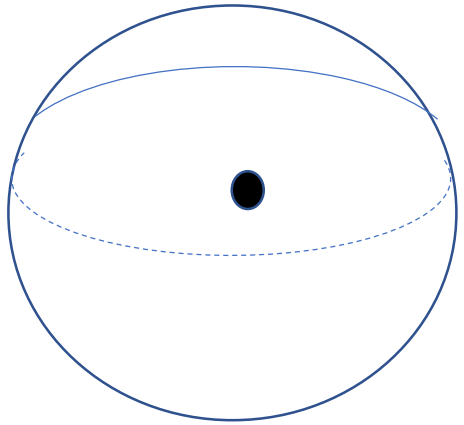
e.g. A SINGLE POINT: is symmetric under any motion preserving the distance from the point → SPHERICAL SYMMETRY

NOTE: the same symmetry is enjoyed by a SPHERE

EXAMPLES

e.g. A SINGLE POINT: is symmetric under any motion preserving the distance from the point → SPHERICAL SYMMETRY

NOTE: the same symmetry is enjoyed by a SPHERE



EXAMPLES

LET’S ADD AN ELEMENT TO THE SYSTEM: a second point



EXAMPLES

LET’S ADD AN ELEMENT TO THE SYSTEM: a second point

Let us call AXIS he straight line joining the two points



EXAMPLES



A TWO POINT system: is symmetric under any motion along a circumference orthogonal to the axis, centered on it

→ AXIAL SYMMETRY

EXAMPLES



A TWO POINT system: is symmetric under any motion along a circumference orthogonal to the axis, centered on it

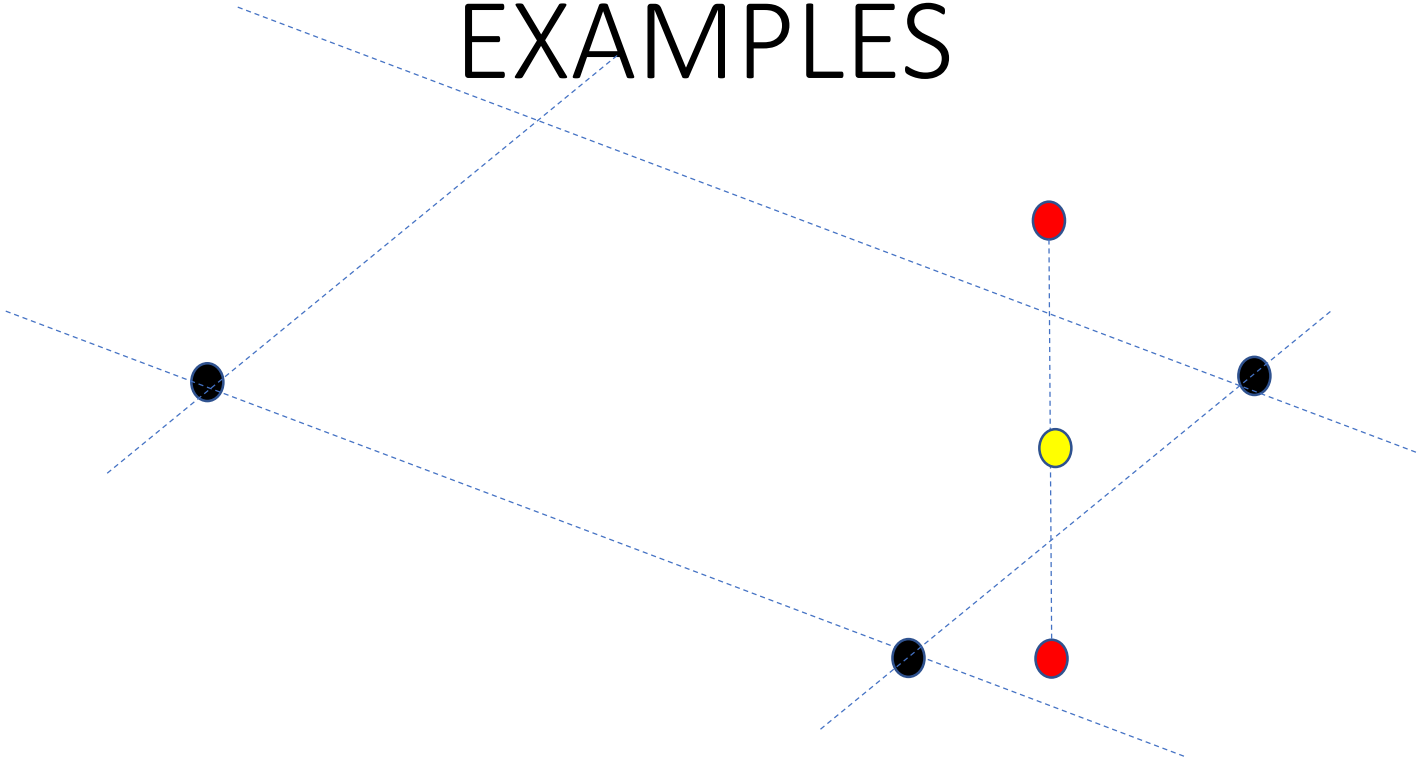
→AXIAL SYMMETRY

NOTE: the same symmetry is enjoyed by a TWO SHPERE system

EXAMPLES

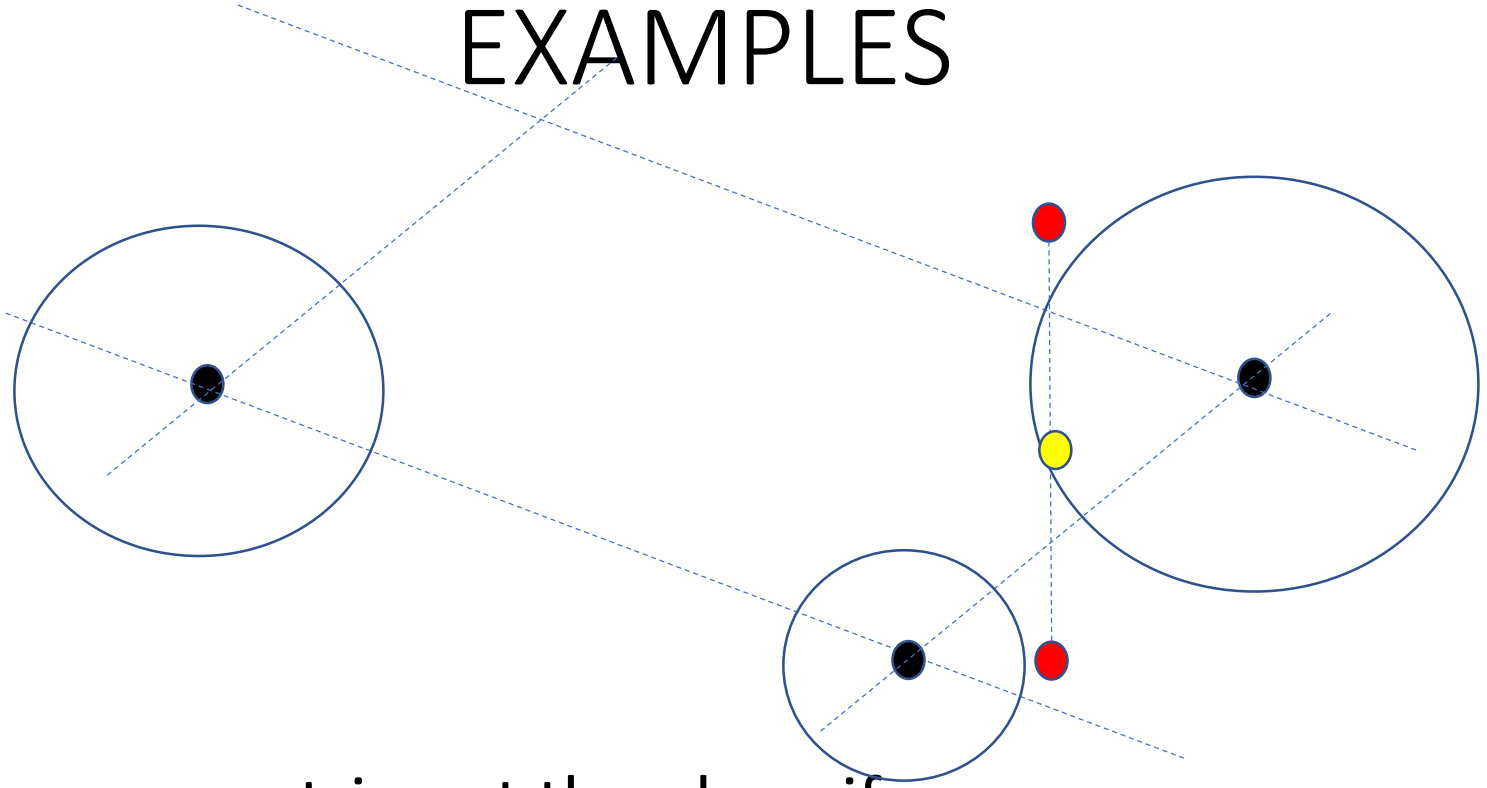
LET’S ADD ONE MORE ELEMENT TO THE SYSTEM: a third point

EXAMPLES



A THREE POINT system: is symmetric under «mirroring» wrt the plane passing through the three points → PLANAR SYMMETRY

EXAMPLES



Two points are symmetric wrt the plane if:

- The segment joining them is perpendicular to the plane
- The midpoint of the segment in on the plane

NOTE: the same symmetry is enjoyed by a THREE SHPERE system

Symmetry on the plane (symmetry in 2D)

SYMMETRY IN 2D (on the plane)

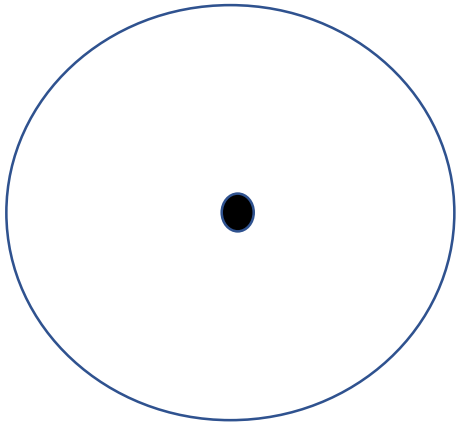
- EMPTY PLANE: is symmetric under any motion

IF WE ADD AN ELEMENT TO A SYMMETRIC SYSTEM, SYMMETRY CAN POSSIBLY STILL BE ENJOYED, BUT UNDER A RESTRICED SET OF TRANSFORMATIONS

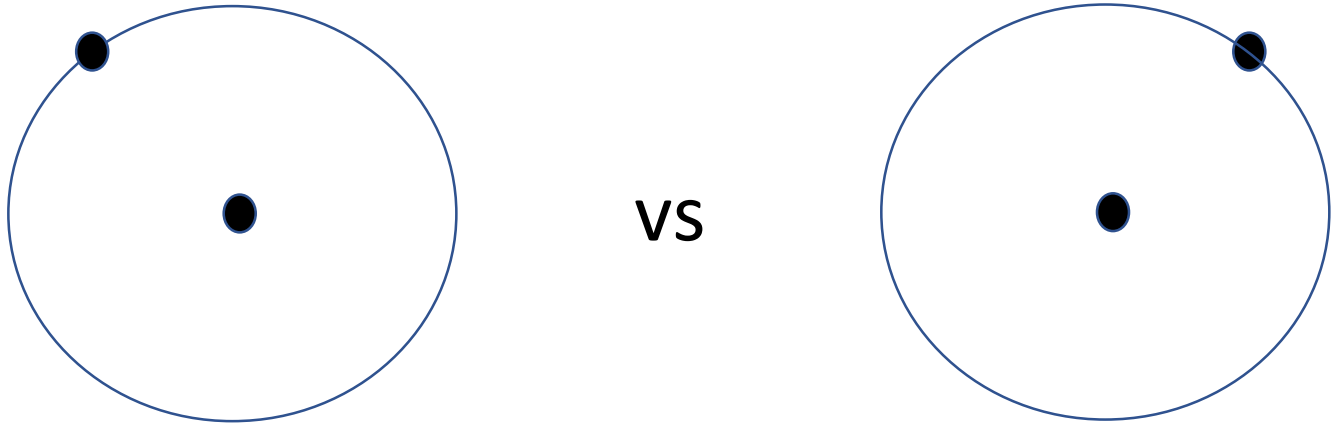
e.g. A SINGLE POINT: is symmetric under any motion preserving the distance from the point → CIRCULAR SYMMETRY OR ISOTROPY

NOTE: the same symmetry is enjoyed by a CIRCLE

Example of symmetry in 2D: circular symmetry



Example of symmetry in 2D: circular symmetry



The two systems are indistinguishable

EXAMPLES

LET’S ADD AN ELEMENT TO THE SYSTEM: a second point

Let us call AXIS he straight line joining the two points



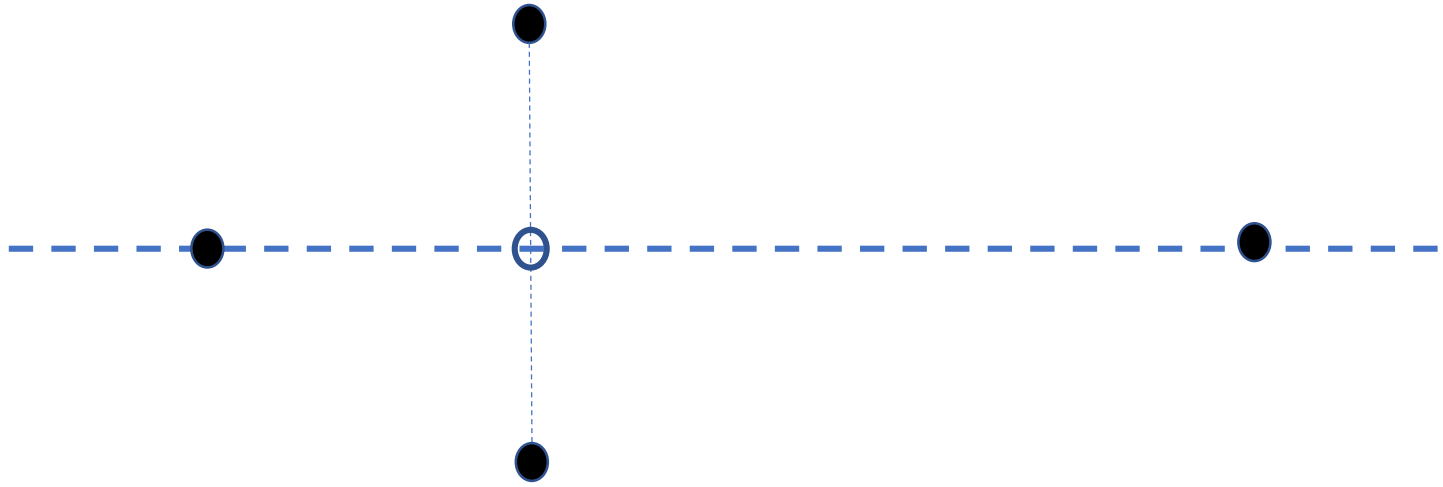
EXAMPLES

LET'S ADD AN ELEMENT TO THE SYSTEM: a second point

Let us call AXIS he straight line joining the two points

Two points are symmetric wrt the axis if:

- The segment joining them is perpendicular to the axis
- The midpoint of the segment in on the axis



EXAMPLES

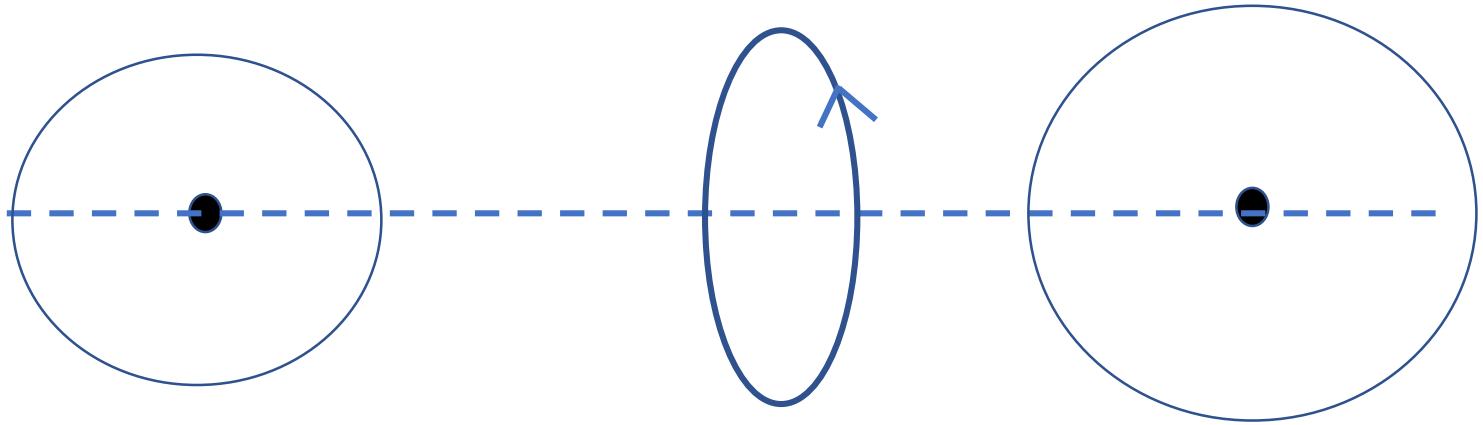


A TWO POINT system: is symmetric under any motion along a circumference orthogonal to the axis, centered on it

→AXIAL SYMMETRY

NOTE: the same symmetry is enjoyed by a TWO CIRCLE system

EXAMPLES



A TWO POINT system: is symmetric under any motion along a circumference orthogonal to the axis, centered on it

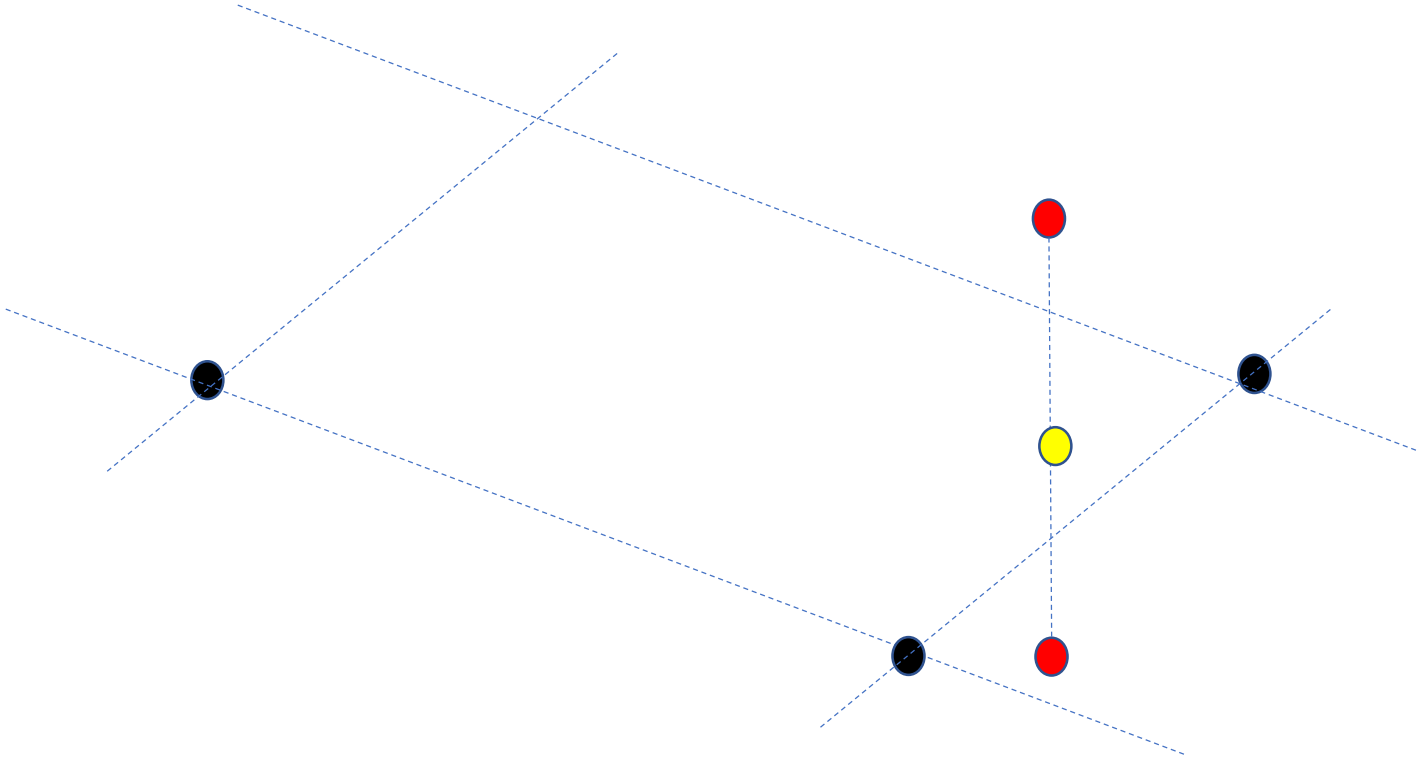
→AXIAL SYMMETRY

NOTE: the same symmetry is enjoyed by a TWO CIRCLE system

If we add a third point, there is no symmetry left



Exploiting symmetry: example on planar symmetry

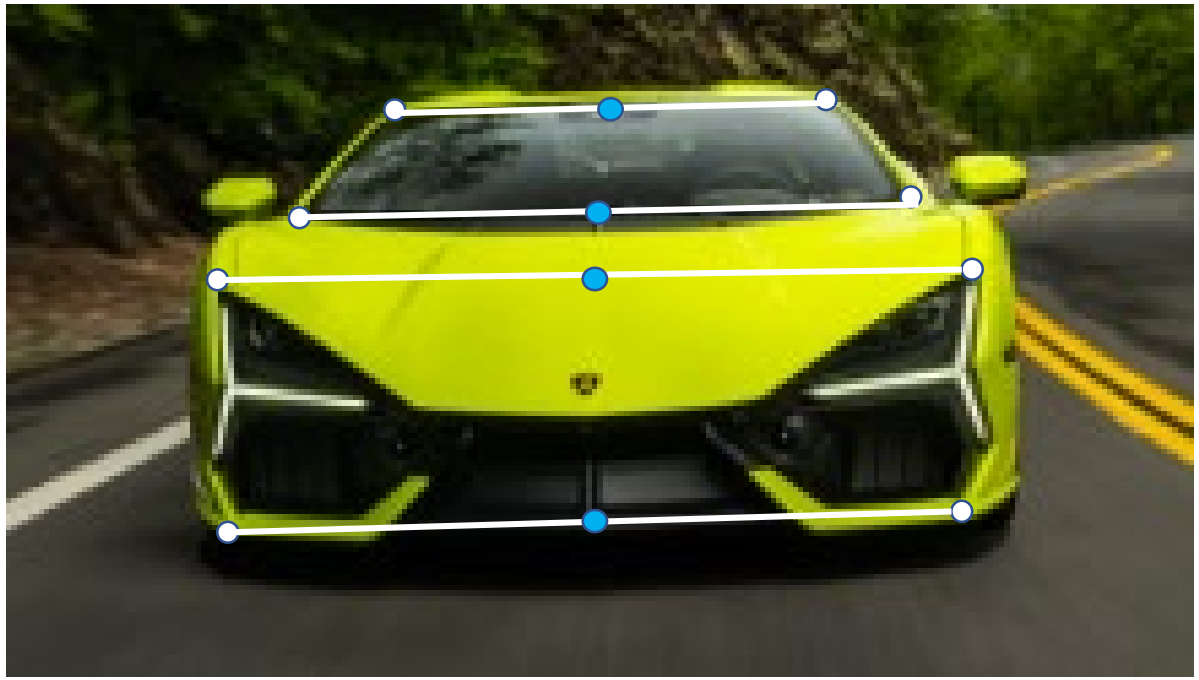


A THREE POINT system: is symmetric under «mirroring» wrt the plane passing through the three points → PLANAR SYMMETRY

Digression on CROSS RATIO and SYMMETRY

in planar symmetry

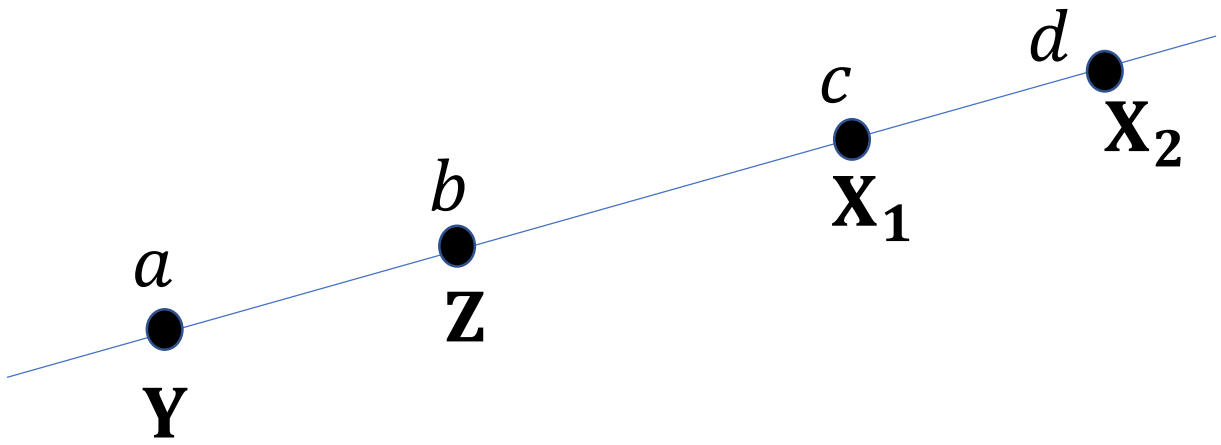
- pairs of symmetric points have their midpoints on the symmetry plane
- segments joining symmetric points are all perpendicular to the plane: there fore they are parallel to eachother



HOW DO THESE PROPERTIES TRANSLATE ONTO AN IMAGE?

REMEMBER: cross ratio of a 4-tuple of colinear points

$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{c-a}}{\boldsymbol{c-b}} \Big/ \frac{\boldsymbol{d-a}}{\boldsymbol{d-b}}$$ and linear combinations



THE CROSS RATIO
IS **INVARIANT**
FROM REAL WORLD
TO IMAGE

REMEMBER: images of parallel segment concur at the vanishing point V of their common direction

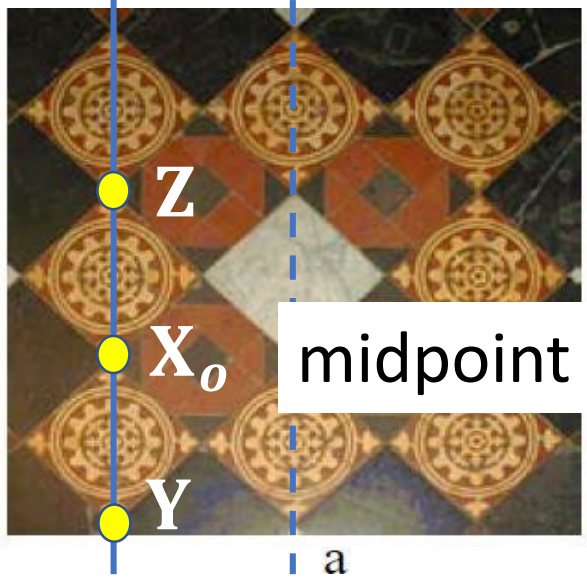
$\mathbf{X}_\infty$ point at the ∞

vanishing point

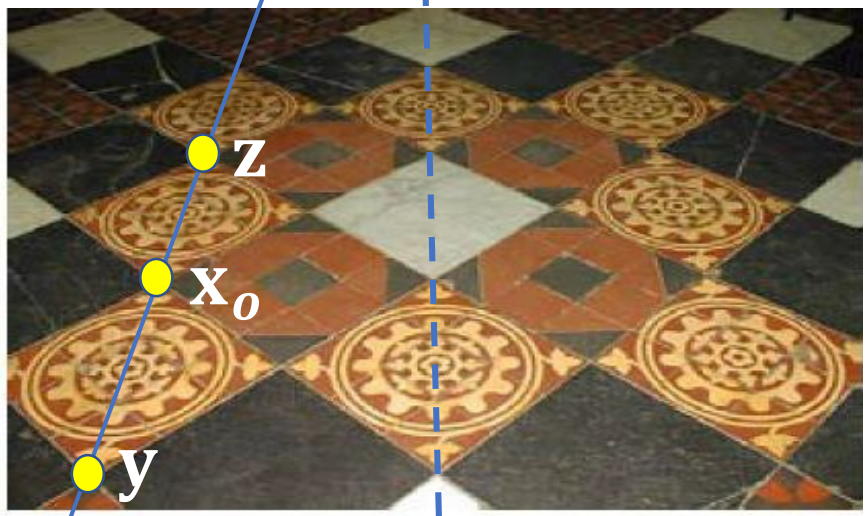
V

$CR_{Y,Z,X_o,X_\infty} = -1 \quad \rightarrow \quad CR_{y,z,x_o,V} = -1$

$CR_{y,z,x_o,V} = -1$



a

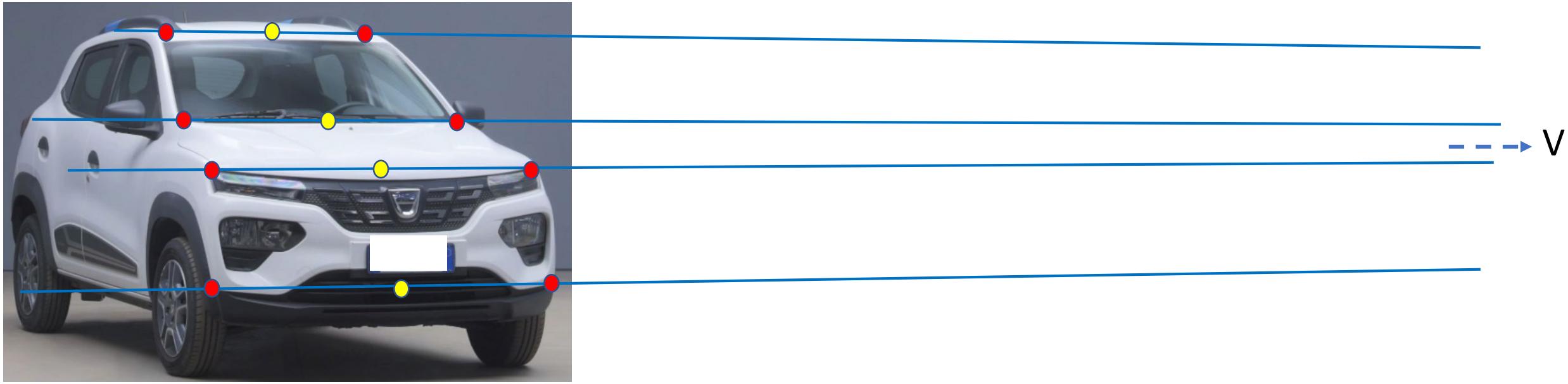


c

CROSS RATIO SPECIAL VALUE: -1 (*harmonic* 4-tuple)

$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$, is the value of the CROSS RATIO of a 4-tuple consisting of, two endpoint, their midpoint, and the point at the ∞

Since the CR is invariant under projective mapping, it keeps the same value if the image of the above «symmetric» 4-tuple is considered



Example: exploiting symmetry

3D position of symmetric features in planar symmetric objects (e.g., cars, airplanes etc.)

Hp:

- calibrated camera: $\boldsymbol{K}$ known
- one known size, e.g. the real distance AB between two symmetric features A, B



Vanishing point

Find line segments joining symmetric features

Find vanishing point V: concurrency point of those lines



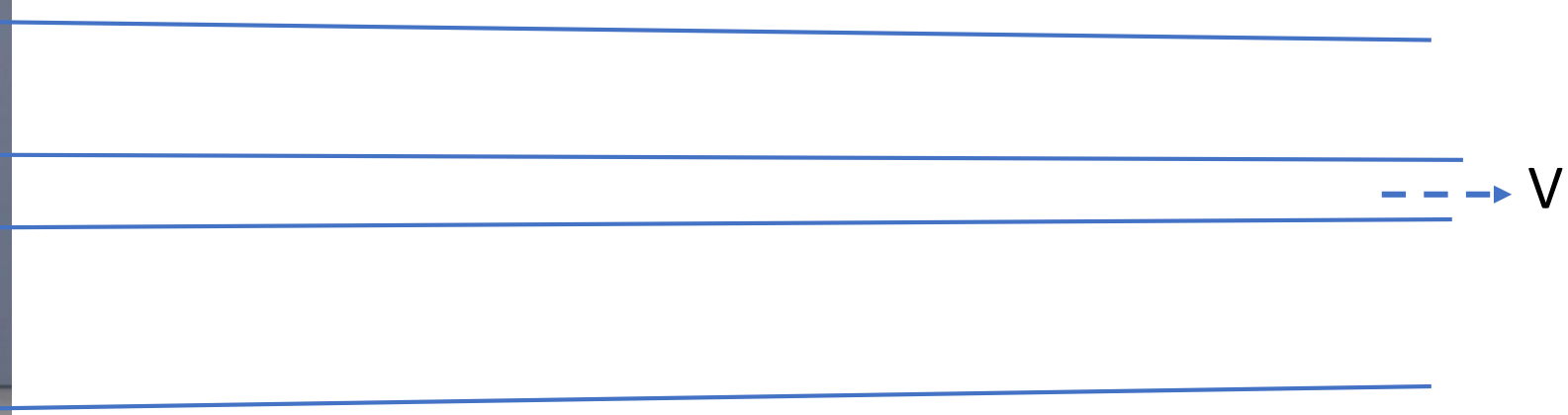


Image of midpoints

Find image of the midpoints of the parallel segments:

→ e.g., for A'B', point C' such that CR(A', B', C', V) = -1

Symmetry plane in 3D

Find symmetry plane π (through C): normal $\boldsymbol{n} = \boldsymbol{K}^{-1}\boldsymbol{V}$

Localize C in 3D by solving the **triangle** OAB (size AB known):

OAB directions: $\mathrm{AB} = \boldsymbol{n} = \boldsymbol{K}^{-1}\boldsymbol{V}$, $\mathrm{OA} = \boldsymbol{K}^{-1}\boldsymbol{A}'$, $\mathrm{OB} = \boldsymbol{K}^{-1}\boldsymbol{B}'$

→ A and B found in 3D, C = midpoint(AB) = (A+B)/2

→ symmetry plane $\pi = [\boldsymbol{n}^T \quad d]^T$ with $d = -[X_C \quad Y_C \quad Z_C]\boldsymbol{n}$

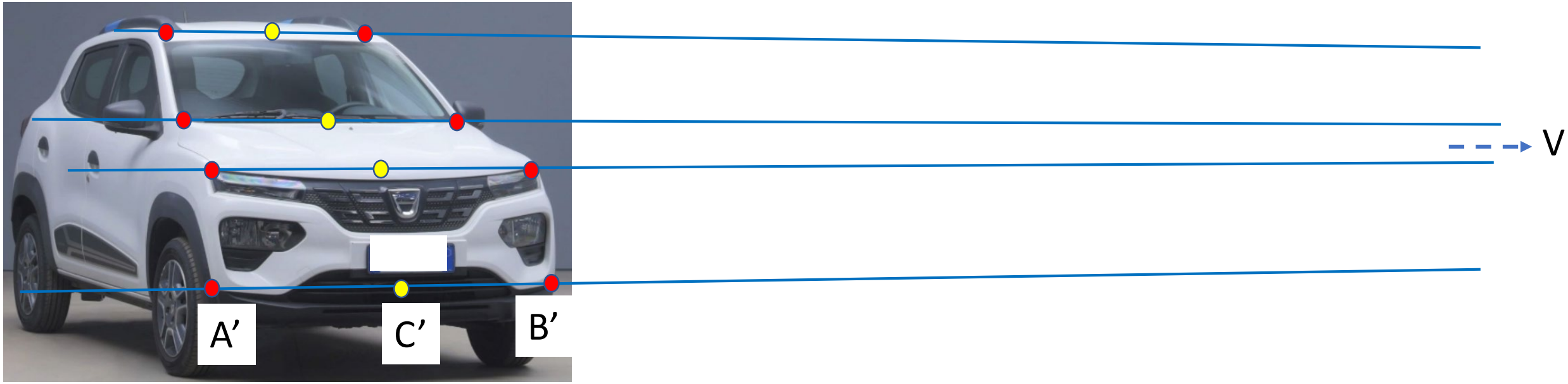


Image of midpoints

Symmetry plane in 3D

Backproject other midpoints onto π

Solve triangle OAB → A B in 3D → C = (A+B)/2

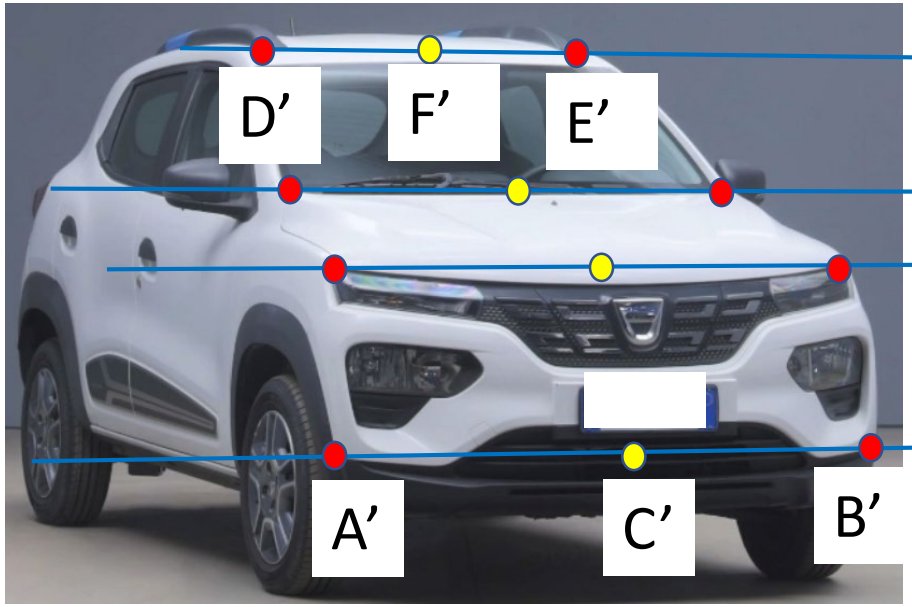
$\boldsymbol{n} = \boldsymbol{K^{-1}V}$ $\quad d = -[X_C \quad Y_C \quad Z_C]\boldsymbol{n}$ $\quad \pi = [\boldsymbol{n}^T \quad d]^T$

$w = -\boldsymbol{n}^T\boldsymbol{K^{-1}F'}/d$ → cartesian coord.s of F $= -\boldsymbol{K^{-1}F'}/w$

3D position of symmetric features

Find 3D positions of symmetric features, e.g. D (or E)

Solve triangle OFD using two known vertices O=(0, 0, 0) and F and known directions OD $= -\boldsymbol{K^{-1}D'}$ and FD $= -\boldsymbol{K^{-1}V}$



V

symmetric features: reconstructed!!

Homework 2025-26:

single uncalibrated image → reconstruct 3D points



APPENDIX 6
Polarity: the projective extension of **symmetry**

POLARITY

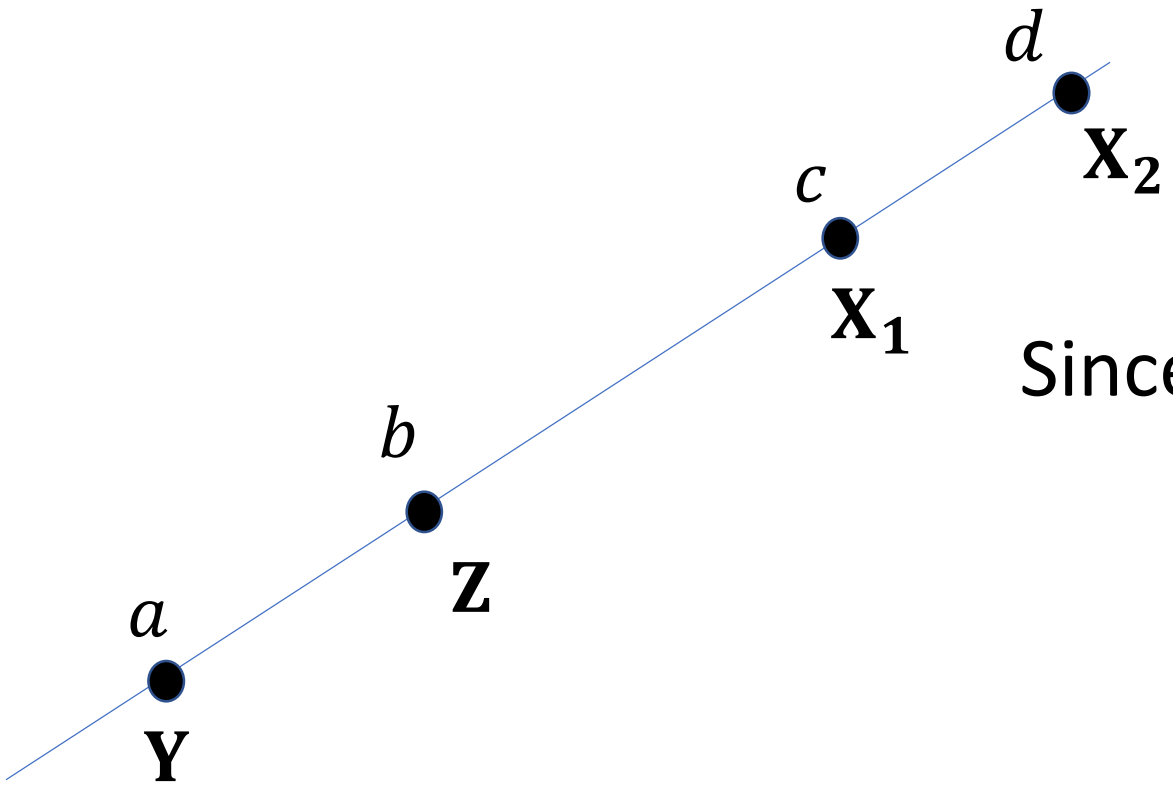
Polarity (conjugacy)
is the projective extension of symmetry

→

useful in the analysis of images of scenes containing symmetric objects (circles, spheres, right cones, right cylinders, planar mirrors)

Cross ratio of a 4-tuple of colinear points

$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{c-a}}{\boldsymbol{c-b}} \Big/ \frac{\boldsymbol{d-a}}{\boldsymbol{d-b}}$$ and linear combinations



Since $\mathbf{X_1}$ and $\mathbf{X_2}$ are colinear with $\mathbf{Y}, \mathbf{Z}$

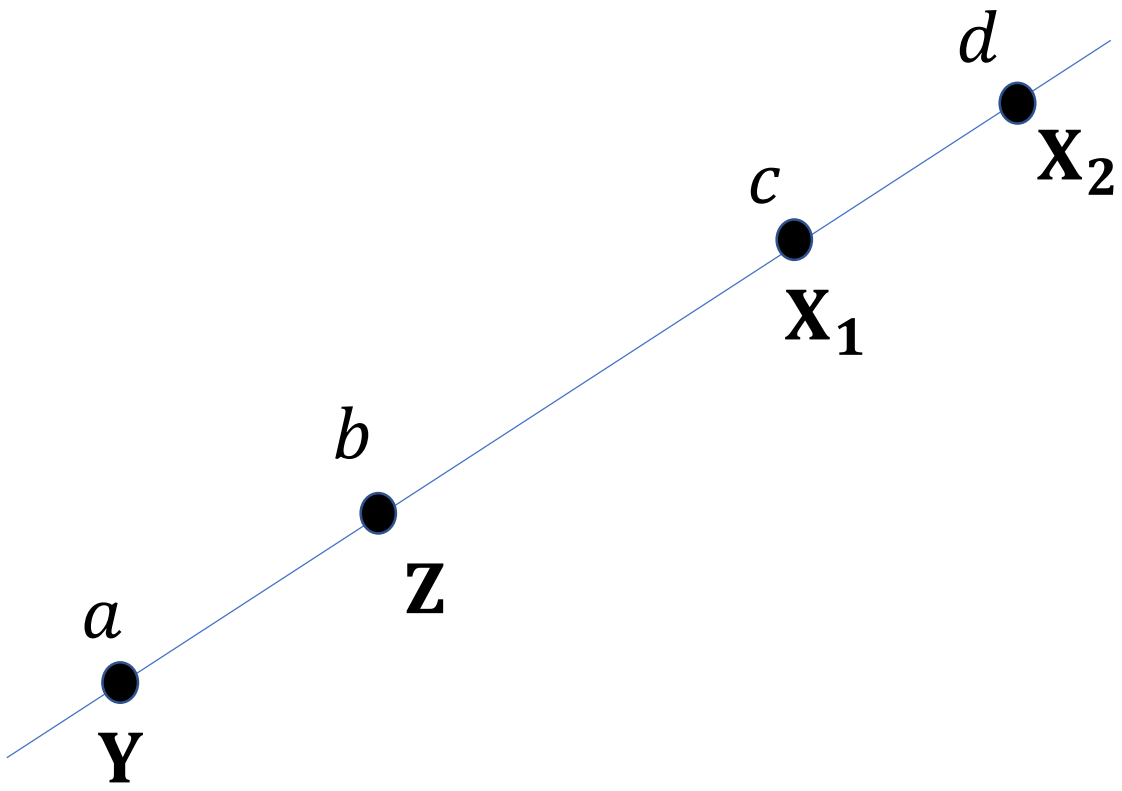
$$\mathbf{X_1} = \alpha_1 \mathbf{Y} + \beta_1 \mathbf{Z}$$

and

$$\mathbf{X_2} = \alpha_2 \mathbf{Y} + \beta_2 \mathbf{Z}$$

Cross ratio and linear combination coefficients: A result

$$\boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{c-a}}{\boldsymbol{c-b}} \Big/ \frac{\boldsymbol{d-a}}{\boldsymbol{d-b}} = \frac{\boldsymbol{\beta_1/\alpha_1}}{\boldsymbol{\beta_2/\alpha_2}}$$



Proof: since the abscissae are proportional to, e.g., the X cartesian coordinates, we can replace the abscissae by the X coordinates (let us the same names …)

Let $\mathbf{Y} = \begin{bmatrix} \boldsymbol{y} \\ * \\ \boldsymbol{v} \end{bmatrix}, \mathbf{Z} = \begin{bmatrix} \boldsymbol{z} \\ * \\ \boldsymbol{w} \end{bmatrix}$: then $\boldsymbol{X_1} = \begin{bmatrix} \boldsymbol{\alpha_1 y + \beta_1 z} \\ * \\ \boldsymbol{\alpha_1 v + \beta_1 w} \end{bmatrix}$ and $\boldsymbol{X_2} = \begin{bmatrix} \boldsymbol{\alpha_2 y + \beta_2 z} \\ * \\ \boldsymbol{\alpha_2 v + \beta_2 w} \end{bmatrix}$

The difference between the X coordinates of $\boldsymbol{X_1}$ and $\mathbf{Y}$ is

$$c - a = \frac{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v} - (\boldsymbol{\alpha_1 v + \beta_1 w})\boldsymbol{y}}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v}} = \frac{\boldsymbol{\beta_1}(\boldsymbol{zv - yw})}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{v}}$$

The difference between the X coordinates of $\boldsymbol{X_1}$ and $\mathbf{Z}$ is

$$c - b = \frac{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w} - (\boldsymbol{\alpha_1 v + \beta_1 w})\boldsymbol{z}}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w}} = \frac{-\boldsymbol{\alpha_1}(\boldsymbol{zv - yw})}{(\boldsymbol{\alpha_1 y + \beta_1 z})\boldsymbol{w}}$$

The difference between the X coordinates of $\boldsymbol{X_1}$ and $\mathbf{Y}$ is

$$c - a = \frac{(\boldsymbol{\alpha_1 y + \beta_1 z})v - (\boldsymbol{\alpha_1 v + \beta_1 w})y}{(\boldsymbol{\alpha_1 y + \beta_1 z})v} = \frac{\boldsymbol{\beta_1}(zv - yw)}{(\boldsymbol{\alpha_1 y + \beta_1 z})v}$$

The difference between the X coordinates of $\boldsymbol{X_1}$ and $\mathbf{Z}$ is

$$c - b = \frac{(\boldsymbol{\alpha_1 y + \beta_1 z})w - (\boldsymbol{\alpha_1 v + \beta_1 w})z}{(\boldsymbol{\alpha_1 y + \beta_1 z})w} = \frac{-\boldsymbol{\alpha_1}(zv - yw)}{(\boldsymbol{\alpha_1 y + \beta_1 z})w}$$

The ratio between these differences is $\frac{c-a}{c-b} = -\frac{\boldsymbol{\beta_1}}{\boldsymbol{\alpha_1}}\frac{w}{v}$ and, similarly,

$$\frac{d-a}{d-b} = -\frac{\boldsymbol{\beta_2}}{\boldsymbol{\alpha_2}}\frac{w}{v}$$

Thus the cross ratio of the four colinear points $\boldsymbol{Y, Z, \mathbf{X}_1, \mathbf{X}_2}$ is

$$\boldsymbol{CR_{Y,Z,\mathbf{X}_1,\mathbf{X}_2}} = \frac{\boldsymbol{c-a}}{\boldsymbol{c-b}} \Big/ \frac{\boldsymbol{d-a}}{\boldsymbol{d-b}} = \frac{\boldsymbol{\beta_1}}{\boldsymbol{\alpha_1}} \Big/ \frac{\boldsymbol{\beta_2}}{\boldsymbol{\alpha_2}}$$

Q.E.D.

CROSS RATIO SPECIAL VALUE: -1 (*harmonic* 4-tuple)

When $\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$, $\mathbf{y}$ and $\mathbf{z}$ are said to be **conjugate** wrt $\mathbf{x}_1$ and $\mathbf{x}_2$

Given the segment $\boldsymbol{Y, Z}$, an instance of two conjugate points wrt $\boldsymbol{Y, Z}$ is pair ($\mathbf{X_0}$ = midpoint of $\boldsymbol{Y, Z}$, $\mathbf{X}_\infty$ = point at the infinity along $\boldsymbol{Y, Z}$)

Other instances: since the cross ratio of 4 colinear points is invariant under image projection, other instances of conjugate points are images $\mathbf{y}, \mathbf{z}$ of segment endpoints, image $\mathbf{x}_o$ of midpoint and vanishing point $\mathbf{x}_\infty$ (image of point at the infinity)

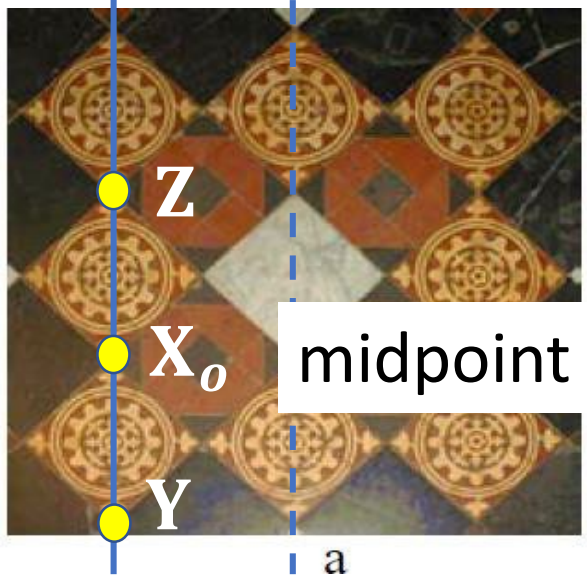
$\mathbf{X}_\infty$ point at the ∞

vanishing point

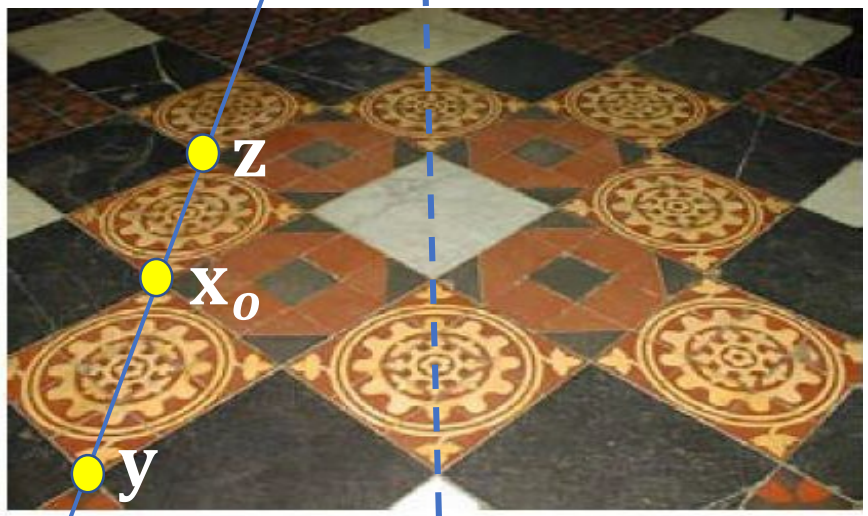
V

$CR_{Y,Z,X_o,X_\infty} = -1 \quad \rightarrow \quad CR_{y,z,x_o,V} = -1$

$CR_{y,z,x_o,V} = -1$



a



c

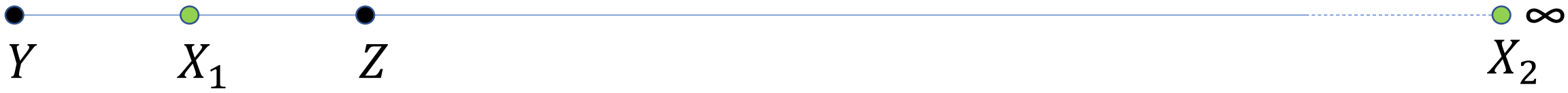
Harmonic 4-tuples and conjugate points

Harmonic 4-tuples and conjugate points

$$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$$

$\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are conjugate wrt $(\boldsymbol{Y}, \boldsymbol{Z})$

$\boldsymbol{Y}$ and $\boldsymbol{Z}$ are conjugate wrt $(\boldsymbol{X_1}, \boldsymbol{X_2})$



A representative case:

- two points Y and Z are the endpoints of a segment

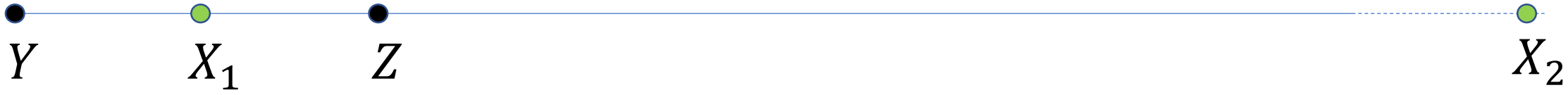
$-$ X_1 and X_2 are the midpoint of (Y, Z) and the point at the ∞

Harmonic 4-tuples and conjugate points

$$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$$

$\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are conjugate wrt $(\boldsymbol{Y}, \boldsymbol{Z})$

$\boldsymbol{Y}$ and $\boldsymbol{Z}$ are conjugate wrt $(\boldsymbol{X_1}, \boldsymbol{X_2})$



Other case: to preserve the value of the cross ratio = -1:
the «far» point X_2 approaches a little bit
and the other point X_1 also moves to get a bit closer to X_2

Harmonic 4-tuples and conjugate points

$$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$$

$\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are conjugate wrt $(\boldsymbol{Y}, \boldsymbol{Z})$

$\boldsymbol{Y}$ and $\boldsymbol{Z}$ are conjugate wrt $(\boldsymbol{X_1}, \boldsymbol{X_2})$



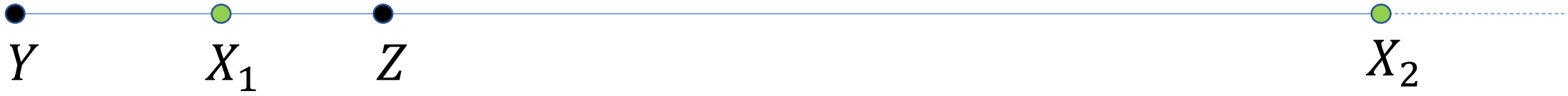
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Harmonic 4-tuples and conjugate points

$$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$$

$\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are conjugate wrt $(\boldsymbol{Y}, \boldsymbol{Z})$

$\boldsymbol{Y}$ and $\boldsymbol{Z}$ are conjugate wrt $(\boldsymbol{X_1}, \boldsymbol{X_2})$



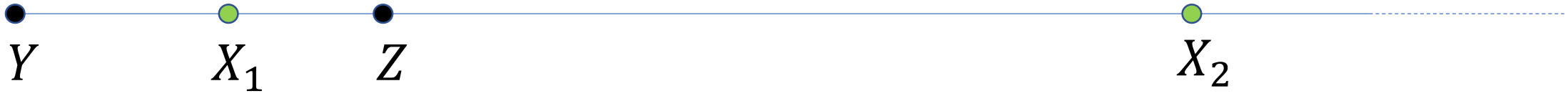
Other case: to preserve the value of the cross ratio = -1:
the «far» point X_2 approaches a little bit
and the other point X_1 also moves to get a bit closer to X_2

Harmonic 4-tuples and conjugate points

$$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$$

$\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are conjugate wrt $(\boldsymbol{Y}, \boldsymbol{Z})$

$\boldsymbol{Y}$ and $\boldsymbol{Z}$ are conjugate wrt $(\boldsymbol{X_1}, \boldsymbol{X_2})$



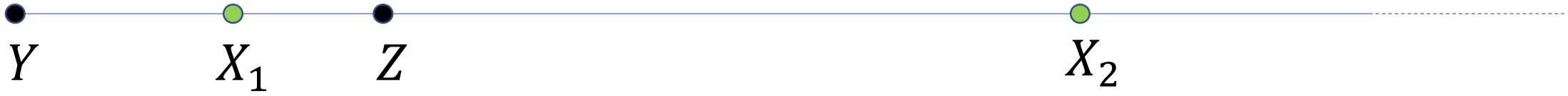
Other case: to preserve the value of the cross ratio = -1:
the «far» point X_2 approaches a little bit
and the other point X_1 also moves to get a bit closer to X_2

Harmonic 4-tuples and conjugate points

$$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$$

$\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are conjugate wrt $(\boldsymbol{Y}, \boldsymbol{Z})$

$\boldsymbol{Y}$ and $\boldsymbol{Z}$ are conjugate wrt $(\boldsymbol{X_1}, \boldsymbol{X_2})$



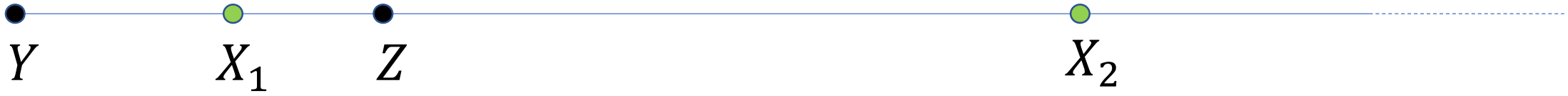
Other case: to preserve the value of the cross ratio = -1:
the «far» point X_2 approaches a little bit
and the other point X_1 also moves to get a bit closer to X_2

Harmonic 4-tuples and conjugate points

$$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$$

$\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are conjugate wrt $(\boldsymbol{Y}, \boldsymbol{Z})$

$\boldsymbol{Y}$ and $\boldsymbol{Z}$ are conjugate wrt $(\boldsymbol{X_1}, \boldsymbol{X_2})$



If we revert X_1 and X_2 , the cross ratio inverts its value:
but the inverse of -1 is itself!

→

In a harmonic tuple we can revert the roles of X_1 and X_2

→

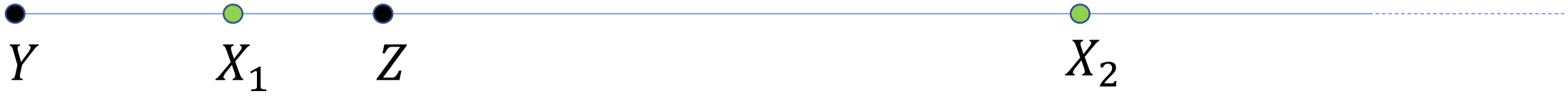
X_1 and X_2 are **conjugate** points wrt Y and Z, namely both
X_1 is **conjugate** to X_2 wrt Y and Z, and X_2 is **conjugate** to X_1 wrt Y and Z

Harmonic 4-tuples and conjugate points

$$\boldsymbol{CR_{Y,Z,X_1,X_2} = -1}$$

$\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are conjugate wrt $(\boldsymbol{Y}, \boldsymbol{Z})$

$\boldsymbol{Y}$ and $\boldsymbol{Z}$ are conjugate wrt $(\boldsymbol{X_1}, \boldsymbol{X_2})$



NOTE:

if X_1 and X_2 are **conjugate** points wrt Y and Z

since the cross ratio = -1 has a negative value, the simple ratios

$\frac{\boldsymbol{c-a}}{\boldsymbol{c-b}}$ and $\frac{\boldsymbol{d-a}}{\boldsymbol{d-b}}$ have opposite signs: i.e. one is positive and the other is negative

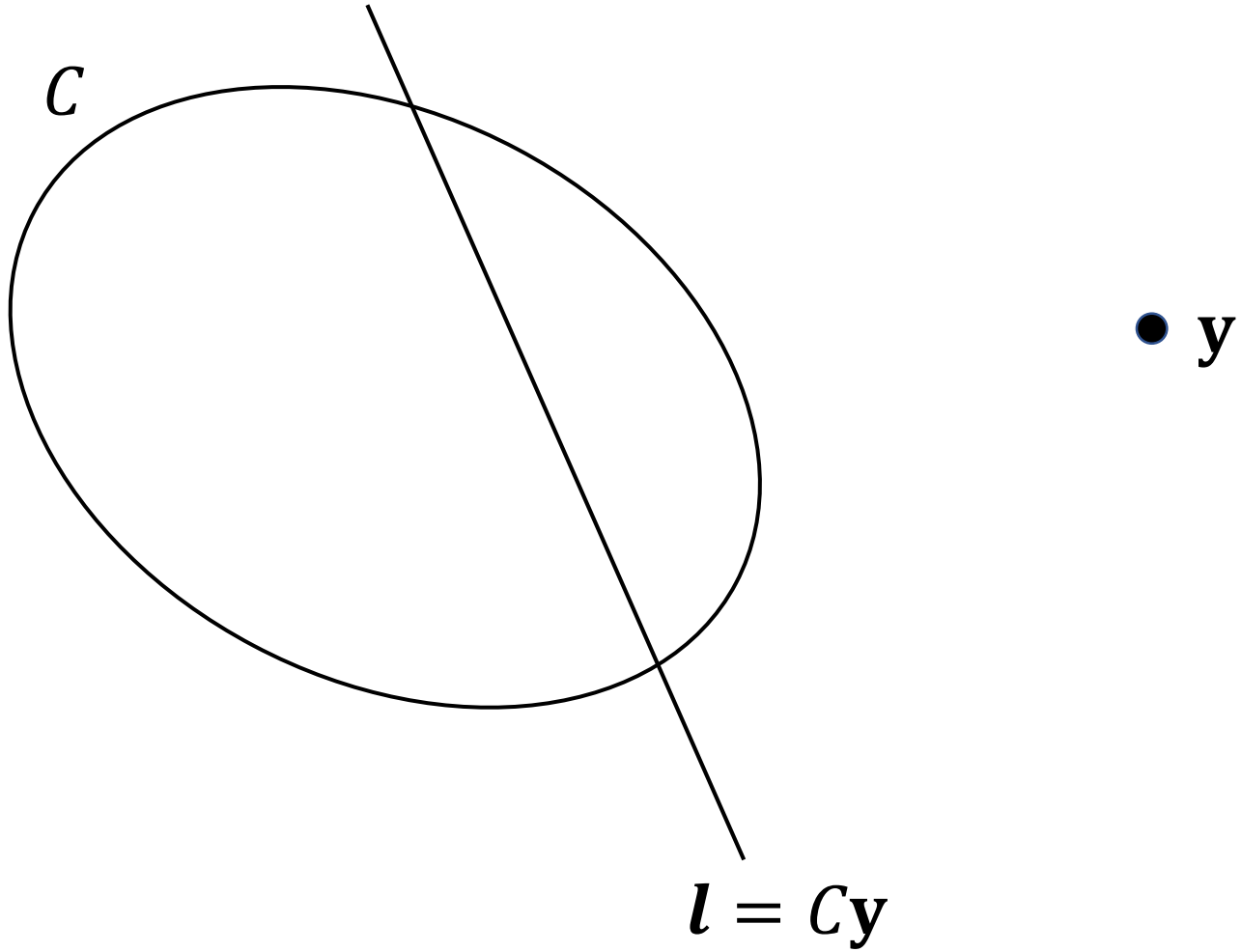
→

one among X_1 and X_2 is **internal** to the segment YZ, while the other is **external**

The polar line of a point wrt a conic

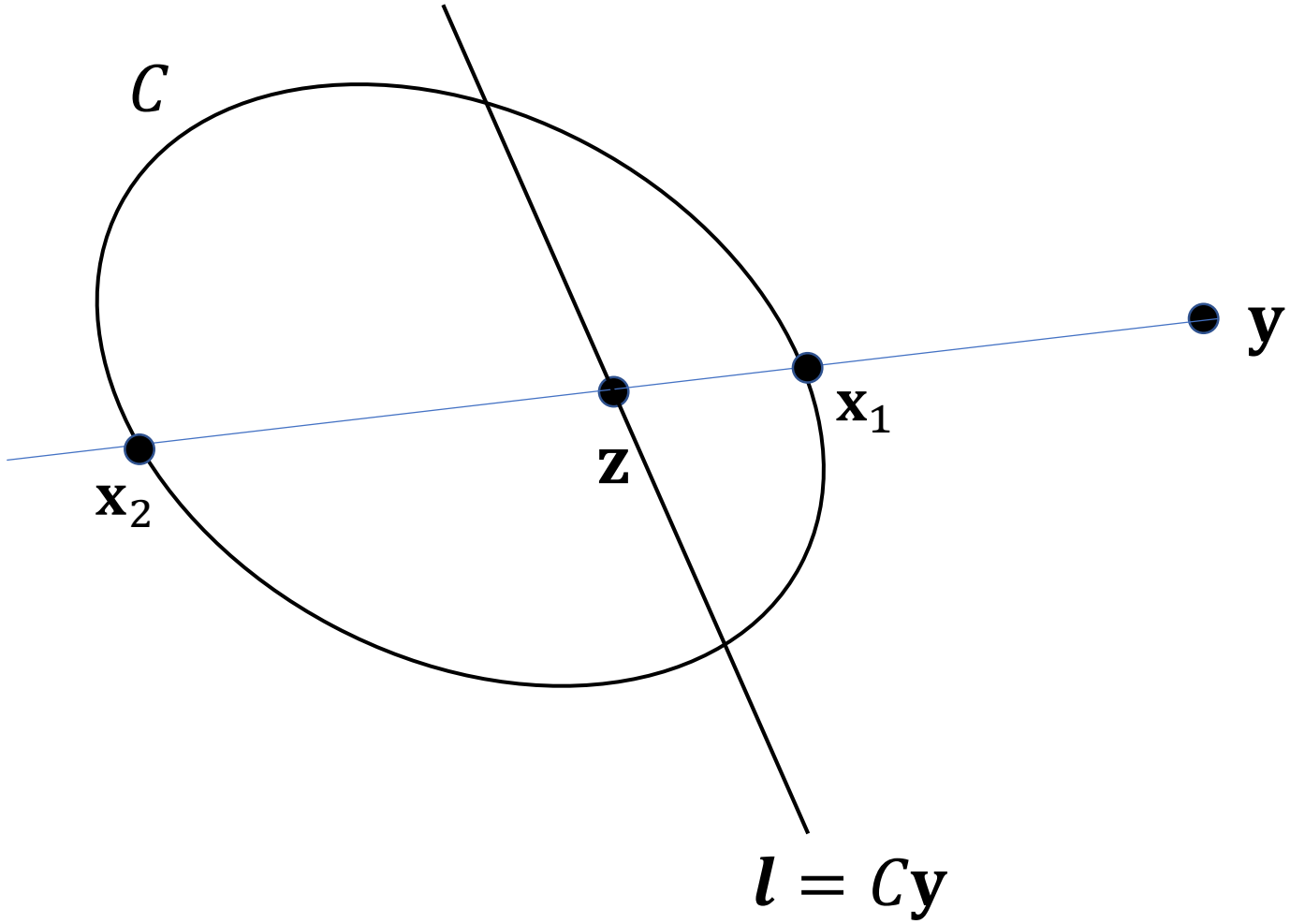
Polar line of a point wrt a conic

Given a point $\mathbf{y}$ and a conic C in the plane, the line $\boldsymbol{l} = C\mathbf{y}$ is called the *polar line* of point $\mathbf{y}$ with respect to the conic C.



Polar line and cross ratio (= -1)

Theorem. Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be the points where the line through $\mathbf{y}$ and $\mathbf{z}$ crosses C: cross ratio $\boldsymbol{CR}_{Y,Z,\mathbf{X_1},\mathbf{X_2}} = \mathbf{-1}$ ($\mathbf{y}$ and $\mathbf{z}$ are said to be **conjugate** wrt $\mathbf{x}_1$ and $\mathbf{x}_2$)



Exercise: prove it!

(or see next page)

Theorem. Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be the points where the line through $\mathbf{y}$ and $\mathbf{z}$ crosses C:

$\mathbf{y}$ and $\mathbf{z}$ are **conjugate** with respect to $\mathbf{x}_1$ and $\mathbf{x}_2$, i.e. their cross ratio = -1.

Proof sketch. $\mathbf{x}_1$ and $\mathbf{x}_2$ are the two solutions (for $\mathbf{x}$) to the intersection problem:

$\mathbf{x} = \alpha\mathbf{y} + \beta\mathbf{z}$ (since $\mathbf{x}$ is on line ($\mathbf{y}$, $\mathbf{z}$)) AND $\mathbf{x}^T C\,\mathbf{x} = \mathbf{0}$ (since $\mathbf{x}$ is on conic C). Namely, $(\alpha\mathbf{y} + \beta\mathbf{z})^T C(\alpha\mathbf{y} + \beta\mathbf{z}) = 0$, with $\mathbf{z}$ on line $\boldsymbol{l} = C\mathbf{y}$, i.e. $\mathbf{z}^T C\,\mathbf{y} = 0 = \mathbf{y}^T C\mathbf{z}$ (remember C is symmetric). The remaining terms are therefore $\alpha^2\mathbf{y}^T C\mathbf{y} + \beta^2\mathbf{z}^T C\mathbf{z} = \mathbf{0}$. Thus there are two opposite solutions for the ratio β/α:

$$\beta/\alpha = \pm\sqrt{-\mathbf{y}^T C\mathbf{y}/\mathbf{z}^T C\mathbf{z}} \qquad \text{i.e. } \frac{\beta_2}{\alpha_2} = -\frac{\beta_1}{\alpha_1}$$

Remember we are solving the intersection problem:

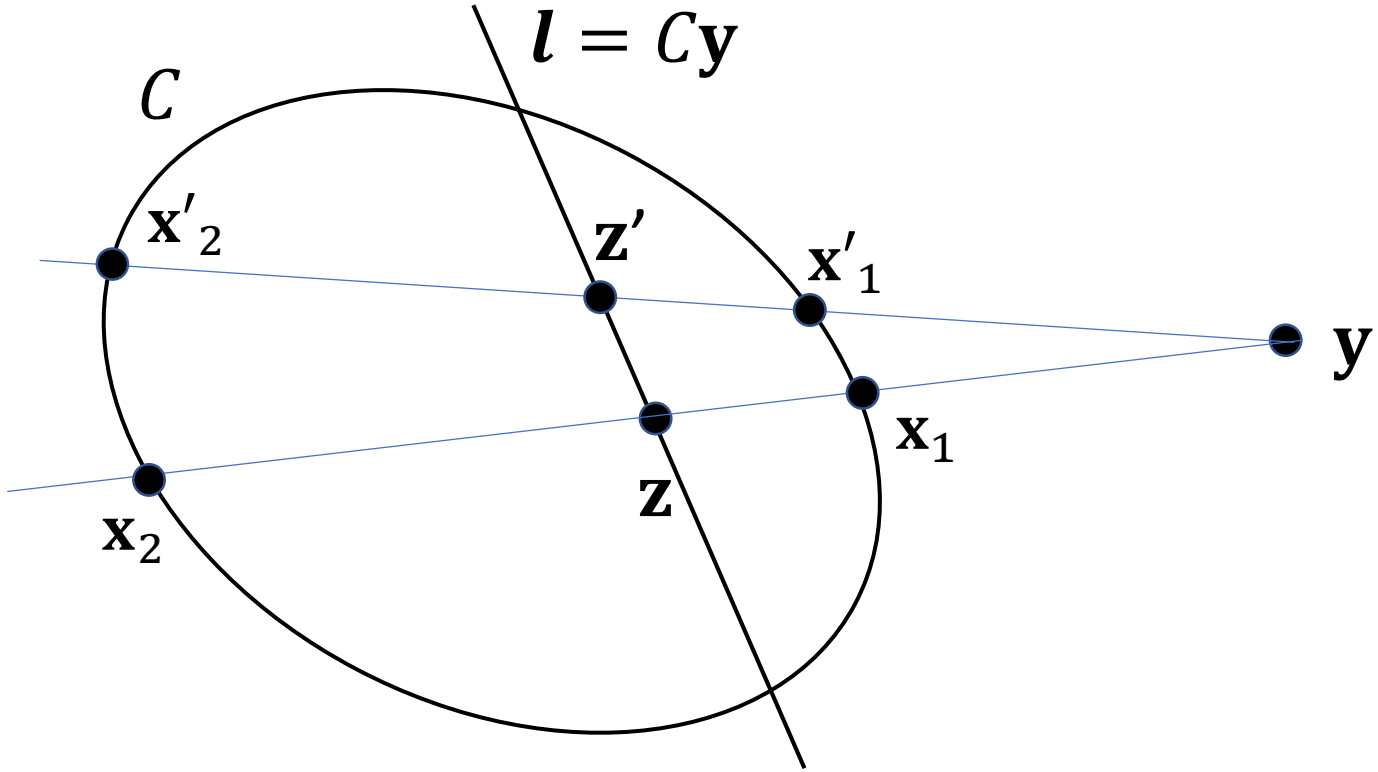
→ these two solutions correspond to the points $\mathbf{x}_1$ and $\mathbf{x}_2$

$$\text{Hence } \boldsymbol{CR_{Y,Z,X_1,X_2}} = \frac{\boldsymbol{\beta_1}}{\boldsymbol{\alpha_1}}\Big/\frac{\boldsymbol{\beta_2}}{\boldsymbol{\alpha_2}} = \mathbf{-1}$$

The polar line $\boldsymbol{l} = C\mathbf{y}$ is the locus of points $\mathbf{z}$ conjugate of $\mathbf{y}$ wrt conic C (more precisely, conjugate wrt to the intersection points of C with any line through $\mathbf{y}$)

Polar line and cross ratio (= -1)

Theorem. Let $\mathbf{x}_1$ and $\mathbf{x}_2$ be the points where a line through $\mathbf{y}$ and $\mathbf{z}$ crosses C: cross ratio $\boldsymbol{CR}_{Y,Z,\mathbf{X_1},\mathbf{X_2}} = \mathbf{-1}$ ($\mathbf{y}$ and $\mathbf{z}$ are said to be **conjugate** wrt $\mathbf{x}_1$ and $\mathbf{x}_2$)



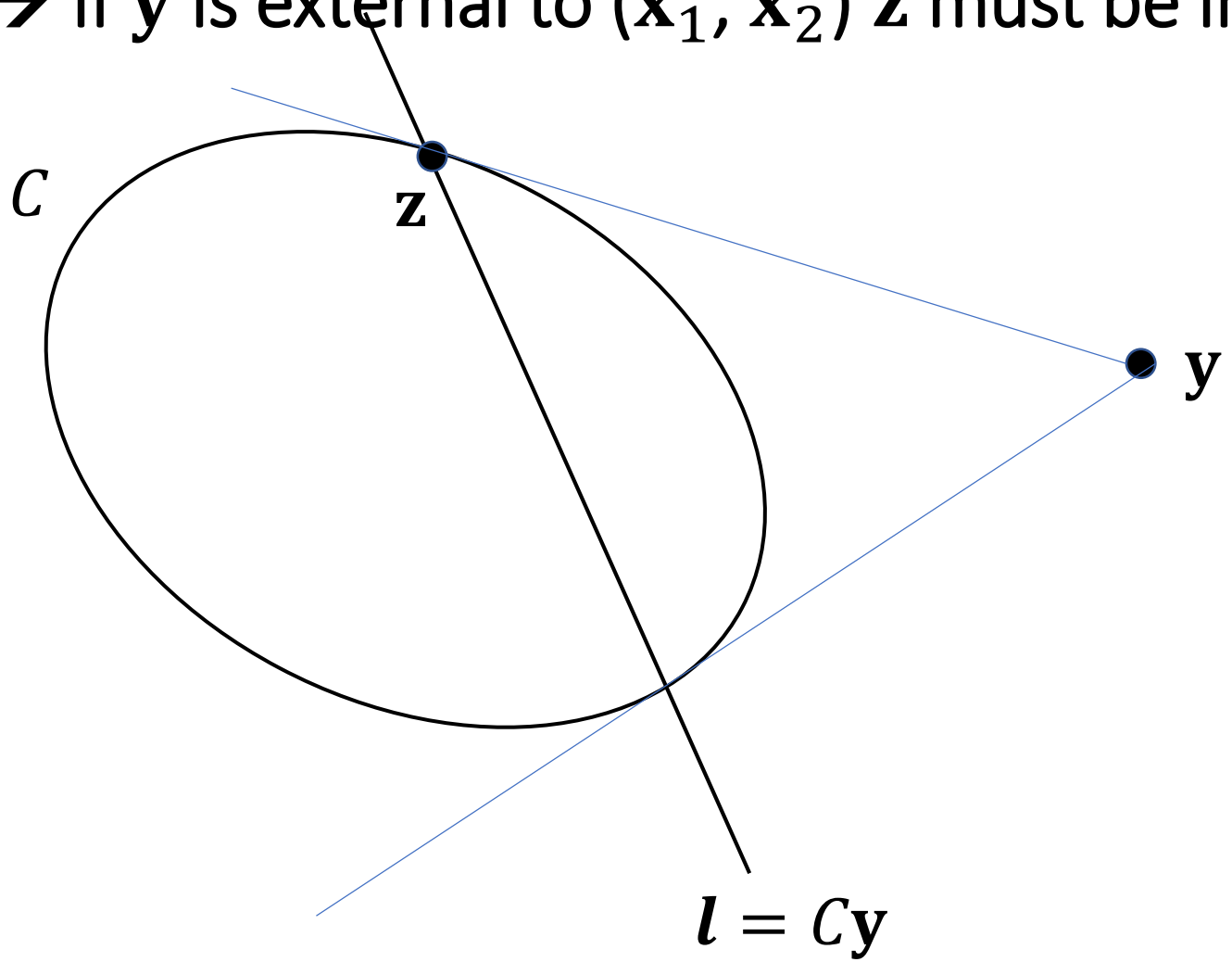
Therefore, the polar line $\boldsymbol{l}$ of $\mathbf{y}$ wrt C is the set of points $\mathbf{z}$, that are conjugate to $\mathbf{y}$ wrt the intersection points $\mathbf{x}_1$ and $\mathbf{x}_2$ between C and any line through $\mathbf{y}$.

Polar line and tangency points

Property:

The polar line $\boldsymbol{l} = C\mathbf{y}$ goes through the tangency points from $\mathbf{y}$ to C

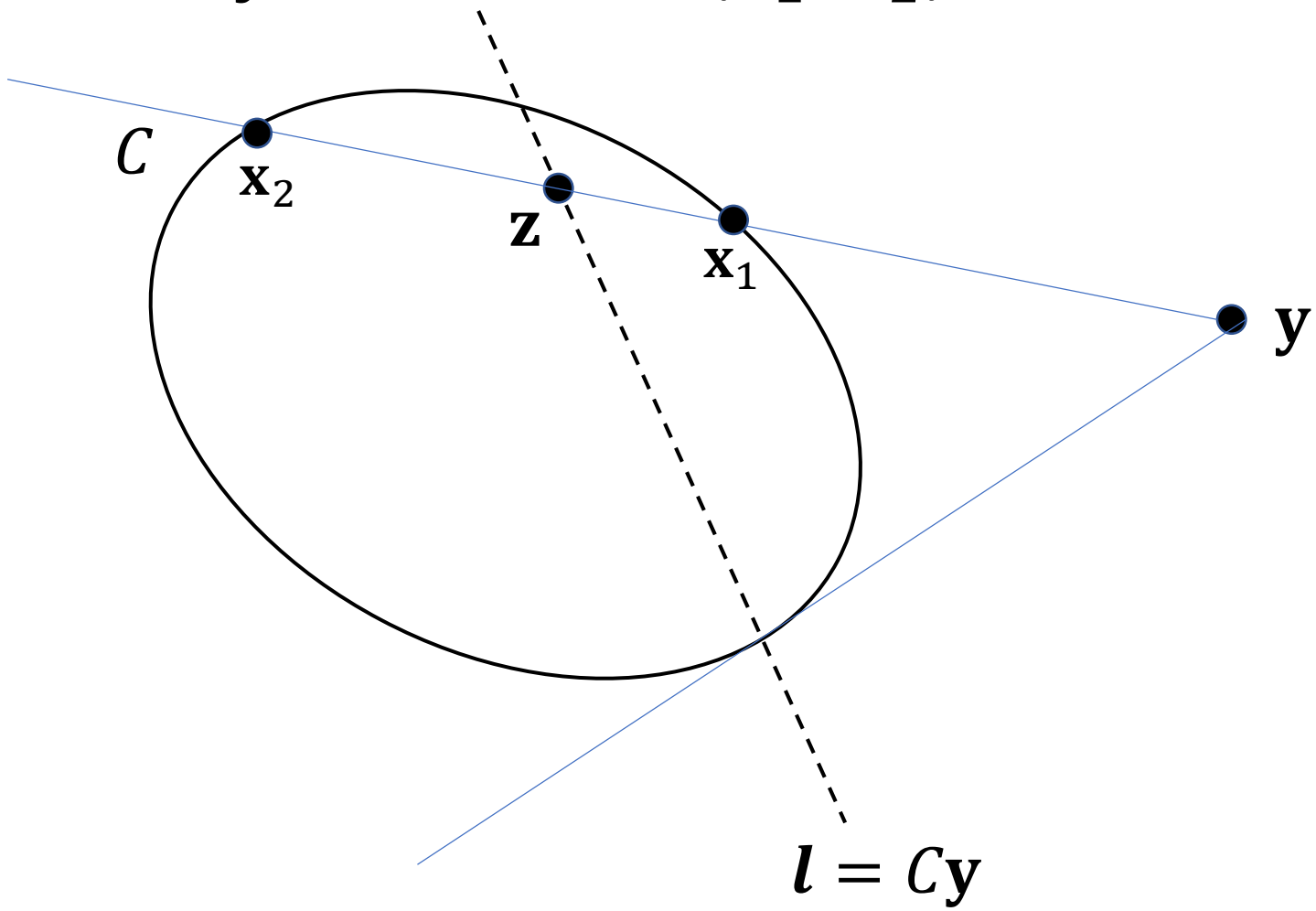
Geometric/ Qualitative Proof: $\mathbf{y}$ and $\mathbf{z}$ conjugate wrt $\mathbf{x}_1$ and $\mathbf{x}_2$ → negative cross ratio → if $\mathbf{y}$ is external to $(\mathbf{x}_1, \mathbf{x}_2)$ $\mathbf{z}$ must be internal



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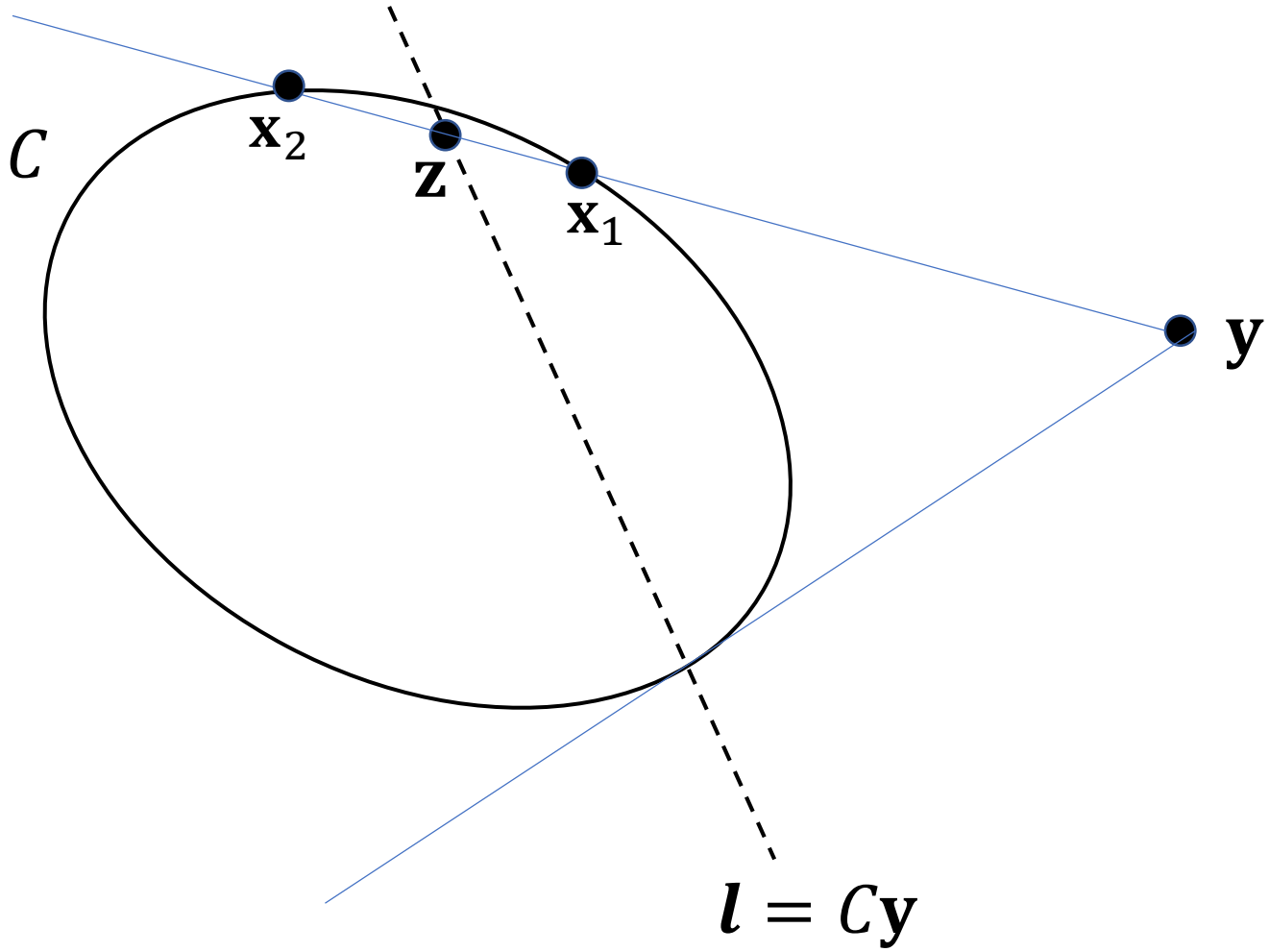
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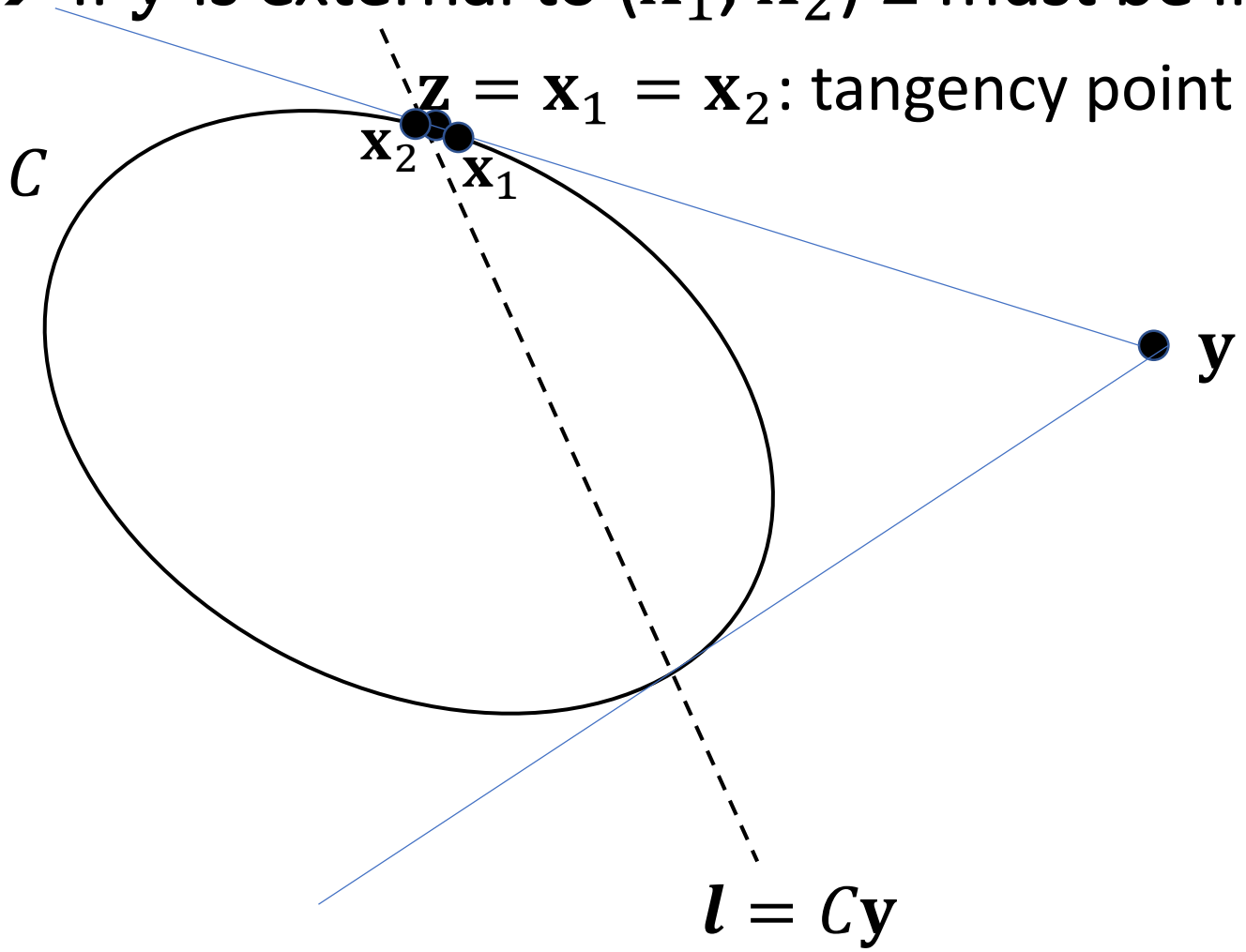
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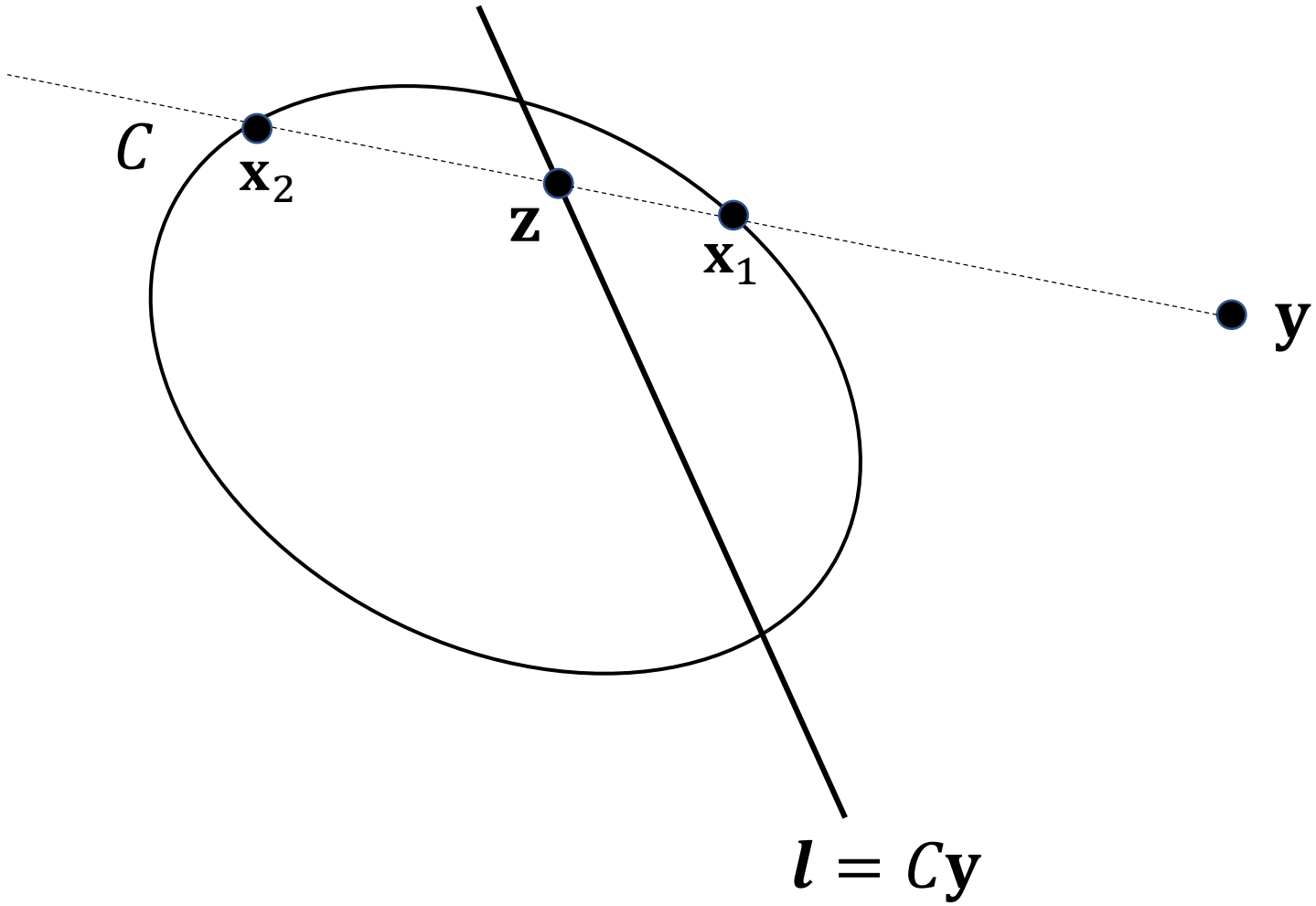
Algebraic proof

Property:

The polar line $\boldsymbol{l} = C\mathbf{y}$ goes through the tangency points from $\mathbf{y}$ to C

Algebraic Proof:

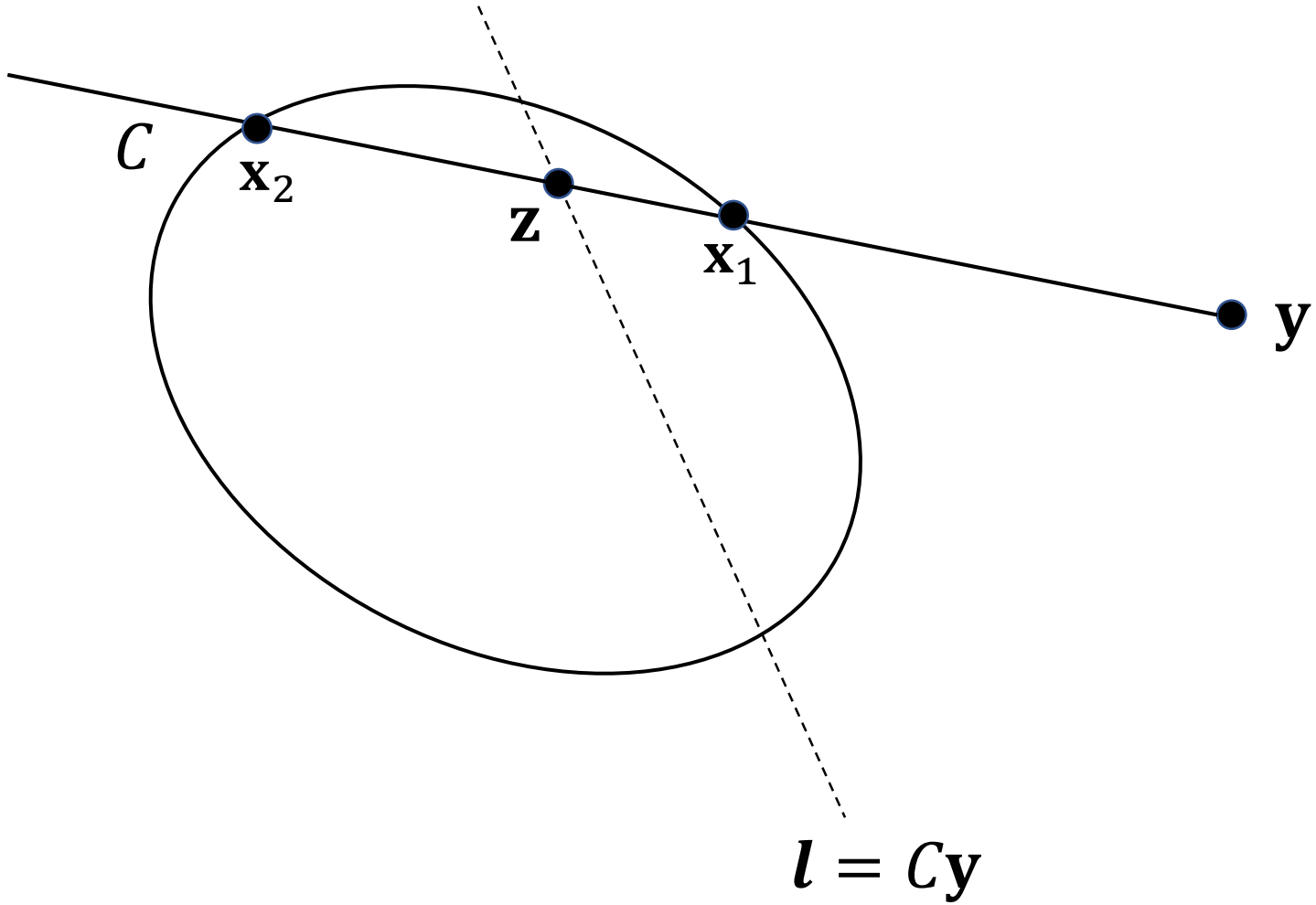
$\mathbf{z} \in \boldsymbol{l} = C\mathbf{y} \rightarrow \boldsymbol{l}^T\mathbf{z} = \mathbf{y}^T\boldsymbol{C}\mathbf{z} = \mathbf{0}$



Algebraic Proof: Intersect polar line $\boldsymbol{l}$ with line joining $\mathbf{y}$ and $\mathbf{z}$

$\mathbf{z} \in \boldsymbol{l} = C\mathbf{y} \rightarrow \boldsymbol{l}^T\mathbf{z} = \mathbf{y}^T\boldsymbol{C}\mathbf{z} = \mathbf{0}$ and $\mathbf{z} + \alpha\mathbf{y} \in \boldsymbol{C} \rightarrow (\mathbf{z} + \alpha\mathbf{y})^T\boldsymbol{C}(\mathbf{z} + \alpha\boldsymbol{y}) = \mathbf{0}$

$\rightarrow \mathbf{z}^T\boldsymbol{C}\mathbf{z} + \alpha^2\,\mathbf{y}^T\boldsymbol{C}\mathbf{y} = \mathbf{0}$ two solutions $\alpha = \pm\sqrt{-\mathbf{z}^T\boldsymbol{C}\mathbf{z}/\mathbf{y}^T\boldsymbol{C}\mathbf{y}}$

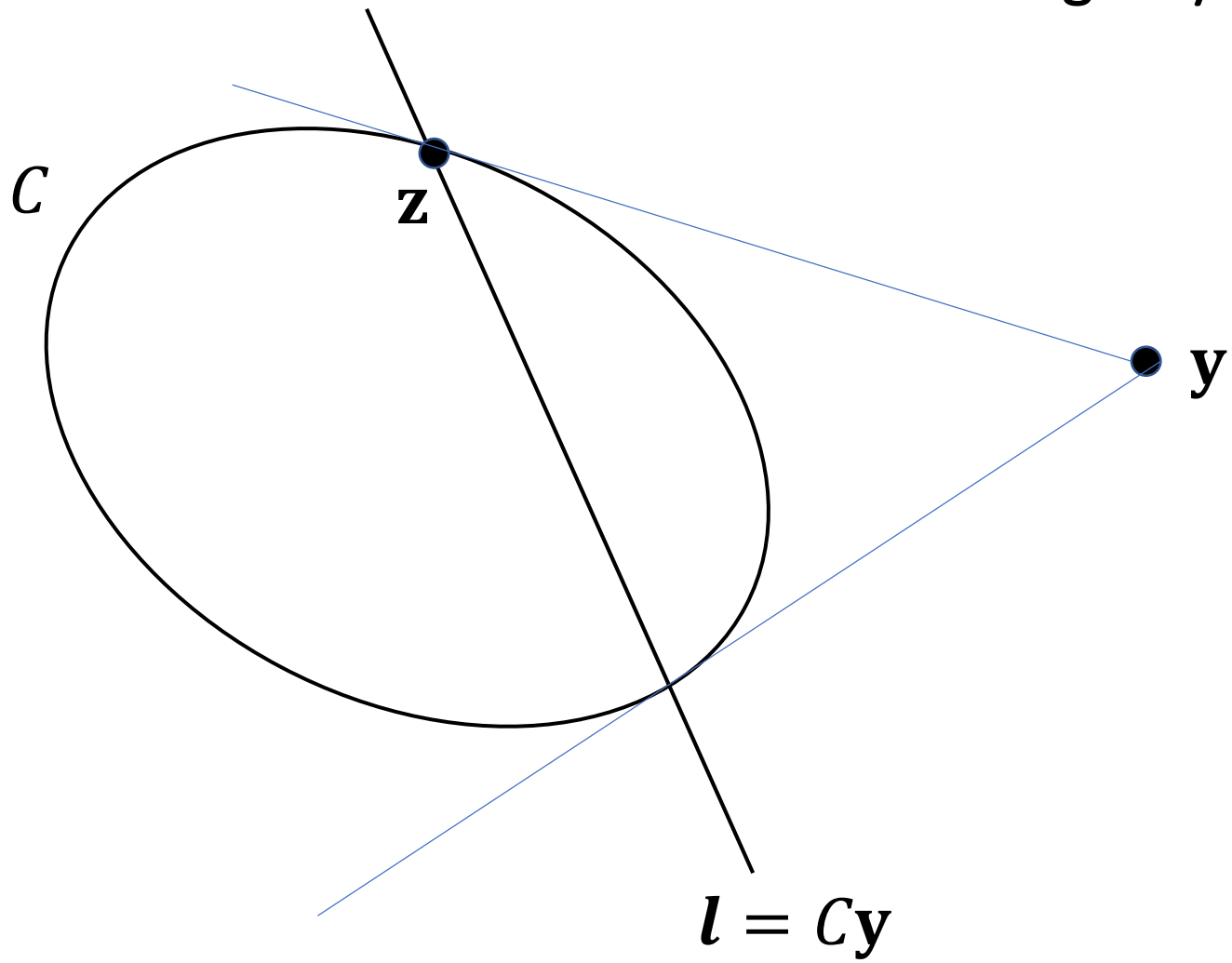


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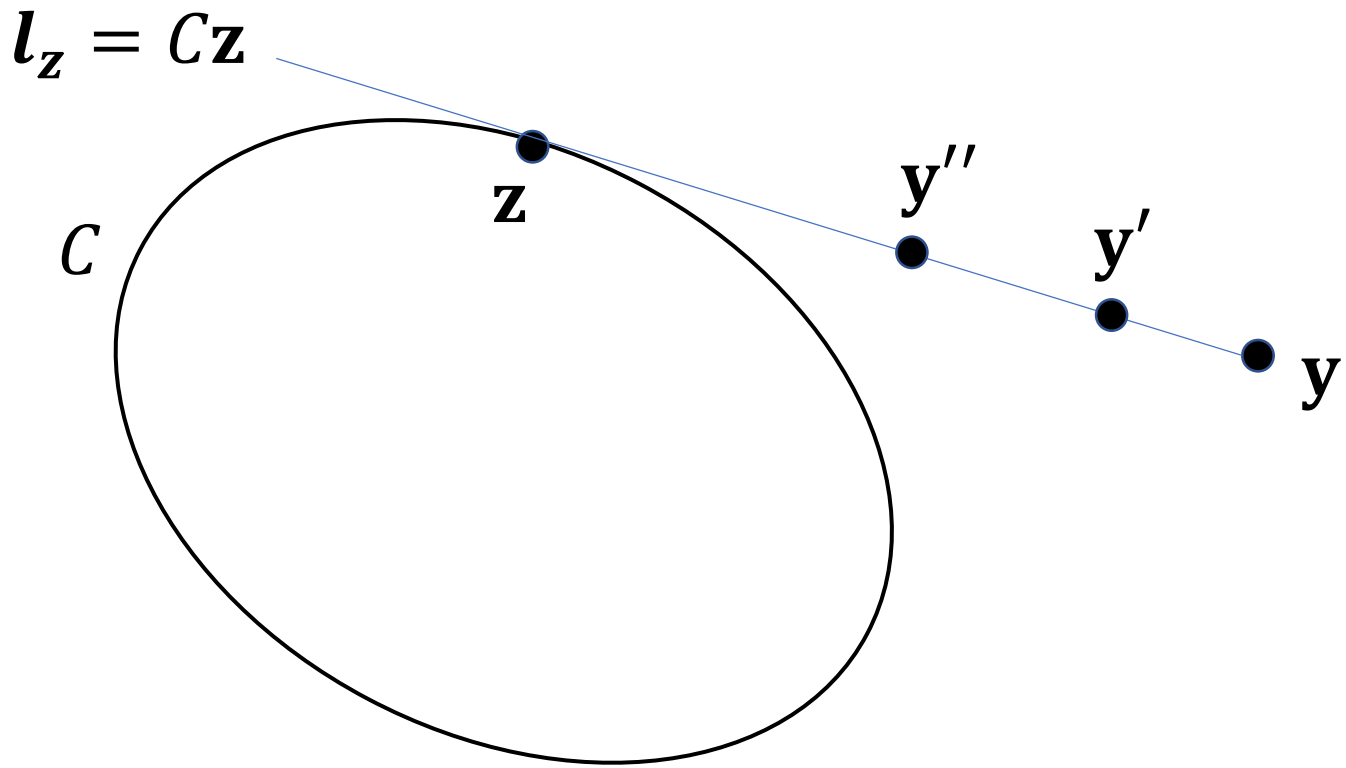
$\rightarrow \mathbf{z}^T\boldsymbol{C}\mathbf{z} + \alpha^2\,\mathbf{y}^T\boldsymbol{C}\mathbf{y} = \mathbf{0}$ two solutions $\alpha = \pm\sqrt{-\mathbf{z}^T\boldsymbol{C}\mathbf{z}/\mathbf{y}^T\boldsymbol{C}\mathbf{y}}$

tangency: one double solution when $-\mathbf{z}^T\boldsymbol{C}\mathbf{z} = \mathbf{0}$ i.e. $\mathbf{z} \in \boldsymbol{C}$ → tangency point from y



The polar line of a point **ON** the conic C

The polar line $\boldsymbol{l_z} = C\mathbf{z}$ of a point the conic C is the tangent line to C through $\mathbf{z}$

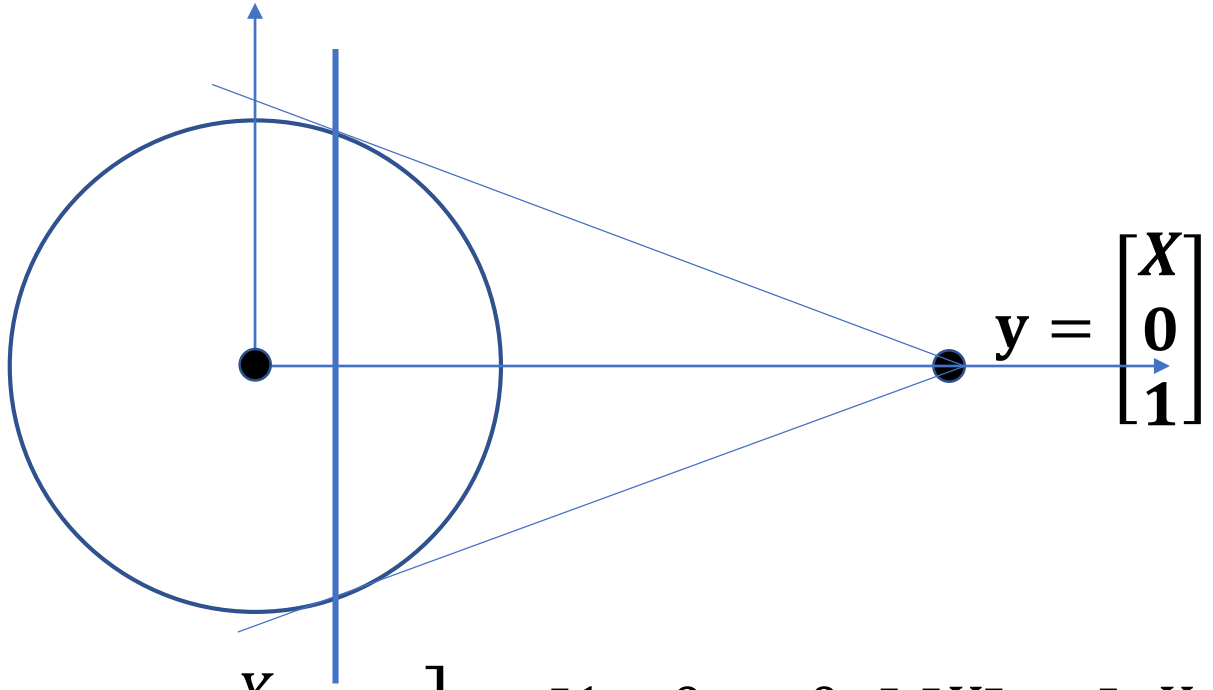


In fact, $\mathbf{z}$ on C is conjugate to any point $\mathbf{y}$ on the tangent to C through $\mathbf{z}$

Examples: the polar line with respect to a circumference

the polar of a point $\mathbf{y}$ wrt a circumference is a line perpendicular to the segment (*center*, $\mathbf{y}$)

Simple case: center on the origin, and point $\mathbf{y}$ on the X-axis

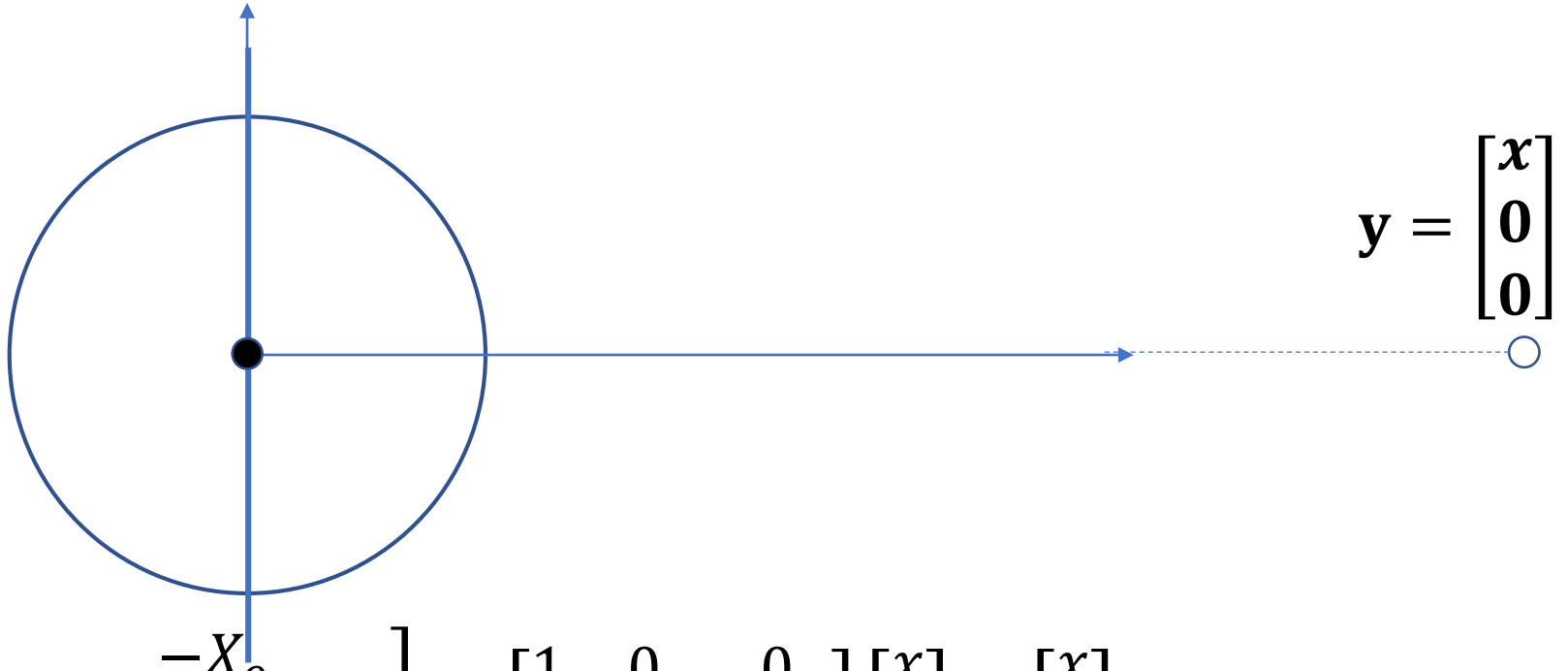


polar line: $\boldsymbol{l} = C\mathbf{y} = \begin{bmatrix} 1 & 0 & -X_o \\ 0 & 1 & -Y_o \\ -X_o & -Y_o & X_o^2 + Y_o^2 - r^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -r^2 \end{bmatrix} \begin{bmatrix} X \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} X \\ 0 \\ -r^2 \end{bmatrix}$

the polar line (cartesian) equation X X $-r^2 = 0 \rightarrow$ X $= r^2/X$: a vertical line

the polar of a point *at the* ∞ wrt a circumference

Particular case: point $\mathbf{y}$ at the ∞



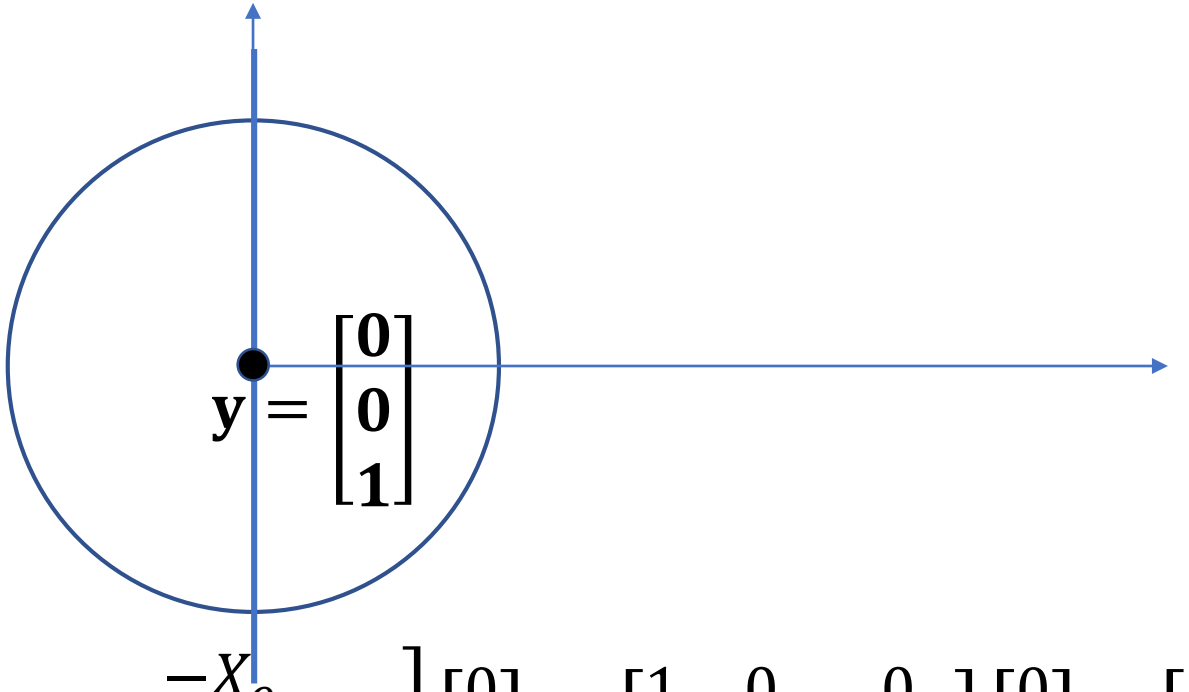
$$\mathbf{y} = \begin{bmatrix} \boldsymbol{x} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}$$

polar line: $\boldsymbol{l} = C\mathbf{y} = \begin{bmatrix} 1 & 0 & -X_o \\ 0 & 1 & -Y_o \\ -X_o & -Y_o & X_o^2 + Y_o^2 - r^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -r^2 \end{bmatrix} \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix}$

the polar line (cartesian) equation $X = 0$ is the **diameter** ⊥ direction of the point $\mathbf{y}$

the polar line of the center of a circumference

Particular case: point $\mathbf{y}$ is the origin



$\boldsymbol{l}_\infty$

polar line: $\boldsymbol{l} = C\mathbf{y} = \begin{bmatrix} 1 & 0 & -X_o \\ 0 & 1 & -Y_o \\ -X_o & -Y_o & X_o^2 + Y_o^2 - r^2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -r^2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -r^2 \end{bmatrix}$

the polar line equation is $-r^2 w = 0$ i.e. $w = 0$ i.e. the line at the infinity $\boldsymbol{l}_\infty$